Statistical Computing

Lecture notes, Lectures 1–11

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Lecture 1. Computer Arithmetic

1.1 Storage units and characters

A **bit** (binary digit, a word coined by Tukey) takes the values 0 and 1; a **byte** is 8 bits. Decimal prefixes (KB $= 10^3$, MB $= 10^6$, GB $= 10^9$, TB $= 10^{12}$ bytes, ...) are distinguished from binary ones (KiB $= 2^{10}$, MiB $= 2^{20}$ bytes, ...).

Text is stored as character codes. ASCII uses 7 bits (128 characters) and extended ASCII 8 bits (256). Unicode encodings (UTF-8, UTF-16, UTF-32) cover all scripts; UTF-8, whose one-byte codes agree with ASCII, is dominant on the internet.

Memory is finite, so *every* representation of $\mathbb{Z}$ or $\mathbb{R}$ in a computer is an approximation.

1.2 Integers: fixed-point numbers

A fixed-point system with M bits models integers.

- **Unsigned** integers represent $\{0, 1, \ldots, 2^M - 1\}$ (e.g. 0–255 for 8 bits).
- **Signed** integers use the most significant bit as a sign bit and **two's complement** for negatives: to represent $-x$, flip all bits of x and add 1. Equivalently, the bit pattern of $-x$ is that of the unsigned integer $2^M - x$. The range is $[-2^{M-1}, 2^{M-1} - 1]$; for $M = 32$ the maximum is $2^{31} - 1 = 2147483647$.

Two's complement turns subtraction into addition modulo 2^M, correct whenever the result is in range: with 8 bits, $57 - 35$ is $00111001_2 + 11011101_2 \equiv 00010110_2 = 22$.

Out-of-range results **overflow**: C and Julia wrap around ($255 + 1 = 0$ for unsigned 8-bit integers), R returns `NA`. Arbitrary-precision integers avoid overflow at a cost in speed.

1.3 Floating-point numbers

A real number is written in scientific notation with base $b = 2$:

$$\pm d_1.d_2 \cdots d_p \times 2^e, \qquad d_i \in \{0, 1\}.$$

In **normalized** form $d_1 = 1$ (e.g. $18 = 1.0010_2 \times 2^4$), so the leading bit is not stored. Exactly representable numbers are **floating-point numbers**; other reals are rounded to a nearby one. IEEE 754 (1985) stores

- a sign bit,
- the fraction (mantissa, significand) of the normalized representation, $p - 1$ bits,
- the exponent plus a **bias**.

format	bits	fraction bits	p	$e_{\min}, e_{\max}$	range	decimal digits
half (Float16)	16	10	11	$-14, 15$	$10^{\pm 4}$	3.3
single (float)	32	23	24	$-126, 127$	$10^{\pm 38}$	7.2
double	64	52	53	$-1022, 1023$	$10^{\pm 308}$	15.9

The bias is 15, 127 and 1023 respectively; the precision in decimal digits is $\log_{10} 2^p$. For half precision, the value of the bit string $b_{15}b_{14}\cdots b_0$ is

$$(-1)^{b_{15}} \times 2^{(b_{14}\cdots b_{10})_2 - 15} \times \Big(1 + \sum_{i=1}^{10} b_{10-i} 2^{-i}\Big).$$

Double precision is the default for numerical computing (R has no native single precision).

Special values. The exponent codes $e_{\min} - 1$ (all zeros) and $e_{\max} + 1$ (all ones) are reserved.

- Exponent $e_{\max} + 1$ with zero fraction: $\pm\infty$ (e.g. $10^{300} \times 10^{300}$).
- Exponent $e_{\max}+1$ with nonzero fraction: `NaN` (e.g. $0/0$, $0 \cdot \infty$). A `NaN` is not equal to anything, including itself.
- Exponent $e_{\min} - 1$ with zero fraction: an exact zero (which cannot be written in normalized form).
- Exponent $e_{\min} - 1$ with nonzero fraction: **denormalized** (subnormal) numbers $0.d_2 \cdots d_p \times 2^{e_{\min}}$ filling $(0, 2^{e_{\min}})$. This **gradual underflow** guarantees $x - y = 0 \Rightarrow x = y$; e.g. $2^{-149} = 0.0\cdots01_2 \times 2^{-126}$ in single precision.

1.4 Rounding and machine epsilon

A real number with more than p significant bits is rounded. IEEE 754's default is **round to nearest, ties to even** (at a tie, the neighbour whose last bit is 0). With $p = 4$:

exact	rounded	reason
1.000011×2^1	1.000×2^1	remainder $011 < 100$: down
1.000110×2^1	1.001×2^1	remainder $110 > 100$: up
1.011100×2^1	1.100×2^1	tie: to even (up)
1.010100×2^1	1.010×2^1	tie: to even (down)

If x is represented by $[x]$, the **absolute error** is $|[x] - x|$ and the **relative error** is $|[x] - x|/|x|$.

Floating-point numbers are not evenly spaced: each $(2^i, 2^{i+1}]$ holds 2^{p-1} equally spaced ones, so the spacing doubles per binade.

Definition (Machine epsilons)

$\epsilon_{\max}$ is the smallest positive floating-point number whose addition to 1 gives a result different from 1; $\epsilon_{\min}$ is the smallest positive floating-point number whose subtraction from 1 gives a result different from 1.

Proposition 1.1

With precision p and ties to even,

$$\epsilon_{\max} = 2^{-p} + 2^{-(2p-1)}, \qquad \epsilon_{\min} = 2^{-(p+1)} + 2^{-2p}.$$

For double precision, $\epsilon_{\max} = 2^{-53} + 2^{-105} \approx 1.1 \times 10^{-16}$.

Proof. The neighbours of 1 are $1 + 2^{-(p-1)}$ and $1 - 2^{-p}$. The midpoint $1 + 2^{-p}$ ties to the even value 1, so $\epsilon_{\max} > 2^{-p}$. The next floating-point number after 2^{-p} is $2^{-p} + 2^{-p-(p-1)}$; adding it to 1 passes the midpoint and rounds to $1 + 2^{-(p-1)}$, so $\epsilon_{\max} = 2^{-p} + 2^{-(2p-1)}$. Likewise $1 - 2^{-(p+1)}$, the midpoint

of $1-2^{-p}$ and 1, rounds to 1, while the next number $2^{-(p+1)}+2^{-(p+1)-(p-1)}=2^{-(p+1)}+2^{-2p}$ does move 1. ■

Theorem 1.2 (Relative error of representation)

If a real number $x>0$ in the normal range is represented by $[x]$, then

$$\frac{|[x]-x|}{|x|}\le \epsilon_{\max}.$$

Proof. Let $[x]=(1+\sum_{i=1}^{p-1} b_{i+1}2^{-i})\times 2^e$, so $[x]\ge 2^e$ and the spacing of floating-point numbers near $[x]$ is $2^{e-(p-1)}$. Rounding to nearest gives $|x-[x]|\le 2^e/2^p$. Then

$$\frac{|x-[x]|}{|x|}\le \frac{2^e}{2^p}\cdot\frac{1}{[x]-2^e/2^p}\le \frac{2^e}{2^p}\cdot\frac{1}{2^e(1-2^{-p})}\le \frac{1}{2^p}\Big(1+\frac{1}{2^{p-1}}\Big)=\epsilon_{\max},$$

using $1/(1-u)\le 1+2u$ for $0\le u\le \frac{1}{2}$. ■

So $[x]=x(1+\delta)$ with $|\delta|\le\epsilon_{\max}$: the machine epsilon is the **unit roundoff** or machine precision.

Example 1.1 (A toy decimal system)

Take $b=10$, $p=5$, $-9\le e\le 9$, numbers $\pm 0.d_1\cdots d_5\times 10^e$, normalized except that subnormals and zero are allowed at $e=-9$. Let $X\sim\text{Unif}(0,1)$ and round to nearest.

- The smallest positive number is $0.00001\times 10^{-9}=10^{-14}$, so $[X]=0$ iff $X\le\frac{1}{2}\cdot 10^{-14}$ (ties to the even neighbour 0): $P([X]=0)=5\times 10^{-15}$.
- $1=0.10000\times 10^1$ and its predecessor is 0.99999×10^0, so $[X]=1$ iff $X\ge 0.999995$: $P([X]=1)=5\times 10^{-6}$.
- $[X]$ is not uniform over the floating-point numbers in $(0,1)$: within $[10^{e-1},10^e)$ the spacing is 10^{e-5}, so each number there has probability 10^{e-5}. For instance $P([X]=0.5)=10^{-5}$ but $P([X]=0.05)=10^{-6}$.

1.5 Floating-point arithmetic and its pitfalls

IEEE 754 computes $+,-,\times,\div,\sqrt{\ }$, remainder and conversions *as if exactly* and then rounds: the computed sum of $[x]$ and $[y]$ is $[[x]+[y]]$.

Overflow and underflow. Magnitudes beyond $\approx 10^{308}$ overflow to $\pm\infty$; below $\approx 10^{-308}$ they underflow gradually to subnormals, then 0. Underflow is the lesser evil, so arrange formulas to avoid overflow, e.g.

$$p=\frac{1}{1+e^{-x^\top\beta}}\qquad\text{rather than}\qquad\frac{e^{x^\top\beta}}{1+e^{x^\top\beta}},$$

since the latter gives $\infty/\infty=$ `NaN` for large $x^\top\beta$ while the former only underflows. Likewise $\log(e^{710})=\infty$: work with likelihoods on the log scale.

Adding numbers of different magnitudes. If $a=x.xx\cdots\times 2^{30}$ and $b=y.yy\cdots\times 2^{-30}$, then b lies below the last bit of a and $a+b=a$ in double precision.

Proposition 1.3 (Error of a sum)

For $x, y > 0$, the computed sum satisfies

$$\frac{|[[x] + [y]] - (x + y)|}{x + y} \leq 2\epsilon_{\max} + \epsilon_{\max}^2 \approx 2\epsilon_{\max}.$$

Proof. Write $[x] = x(1+\delta_x)$, $[y] = y(1+\delta_y)$ and $[[x]+[y]] = ([x]+[y])(1+\delta_s)$, all $|\delta| \leq \epsilon_{\max}$. Then $[x] + [y] \leq (x + y)(1 + \epsilon_{\max})$ and

$$|[[x] + [y]] - (x + y)| \leq |\delta_s|([x] + [y]) + |\delta_x|x + |\delta_y|y \leq \epsilon_{\max}(1 + \epsilon_{\max})(x + y) + \epsilon_{\max}(x + y).$$

■

Adding positive numbers is benign; subtraction is not.

Catastrophic cancellation. Subtracting nearly equal numbers that carry rounding errors destroys significant digits. With 3 digits, $b = 3.34$, $a = 1.22$, $c = 2.28$: exactly $b^2 - 4ac = 0.0292$, but $[b^2] = 11.2$, $[4ac] = 11.1$ give 0.1; the subtraction is exact and magnifies earlier errors. In binary, $[x] = 1.d_2 \cdots d_{p-1}1$ and $[y] = 1.d_2 \cdots d_{p-1}0$ give $[x] - [y] = 2^{-(p-1)}$, while $x - y$ may be anywhere in $(0, 2^{-(p-2)} + 2^{-2p}]$: the relative error is unbounded. Likewise $1 - \Phi(z)$ returns 0 for $z \geq 9$; compute tail probabilities directly.

Rules of thumb.

(1) Add small numbers before large ones.

(2) Add numbers of like magnitude together (pairwise summation).

(3) Avoid subtracting nearly equal numbers. For example, for $ax^2 + bx + c = 0$ with $b > 0$, compute $x_1 = \frac{-b-\sqrt{b^2-4ac}}{2a}$ (no cancellation) and then $x_2 = c/(ax_1)$ from $x_1x_2 = c/a$, instead of $\frac{-b+\sqrt{b^2-4ac}}{2a}$.

Floating-point arithmetic breaks algebraic laws: addition is not associative ($(a + b) - a \neq b$ for $a \gg b$), multiplication does not distribute exactly, zero has two signs ± 0 (equal under comparison), $x + 0 = x$ fails for `NaN`, and $x - x = 0$ fails for $\pm\infty$ and `NaN`. Decimal constants such as $0.1, 0.3$ are inexact, so `0.1+0.2==0.3` and `0.1==0.3/3` are false; test equality with a tolerance (R's `all.equal`).

Example 1.2 (Cancellation in the quadratic formula)

For $a = 10^{-30}$, $b = -10^{30}$, $c = 10^{30}$ the roots are about 10^{60} and 1. In double precision $4ac = 4$ is below the last bit of $b^2 = 10^{60}$, so $\sqrt{b^2 - 4ac} = |b|$ and the textbook formula gives the small root $(-b - |b|)/(2a) = 0$. The stable recipe

$$q = -\tfrac{1}{2}(b + \text{sign}(b)\sqrt{b^2 - 4ac}), \qquad x_1 = q/a, \qquad x_2 = c/q,$$

never subtracts nearly equal numbers and gives $x_2 = 1$ to full precision (for $b = +10^{30}$, reverse the signs).

Example 1.3 (The harmonic series "converges")

The partial sums $s_n \approx \log n$ diverge, but in floating point they freeze once $1/n$ is below half the spacing near s_n: in single precision at $n = 2^{21} = 2097152$ with $s = 15.4037$; in double precision (spacing 2^{-47} on $[32, 64)$) near 34 after about $2^{48} \approx 3 \times 10^{14}$ terms.

For Euler's constant $\gamma = \lim_n(\sum_{i\le n} 1/i - \log n) \approx 0.5772$: take $n = 10^6$, add $1/n + \cdots + 1/1$ (small first), subtract $\log n$ and the leading error term $1/(2n)$. This gives about 13 digits; without the correction the error is 5×10^{-7}.

1.6 Selected exercises

Motivation. Three facts used above: the $1/(2n)$ correction for γ, exact floating-point subtraction, and error growth in long sums.

Exercise 1.1 (Euler's constant as an integral)

Let $\lfloor x \rfloor$ be the largest integer not exceeding x, and $H_n = \sum_{i=1}^{n} 1/i$.
(a) Show that for every integer $n \ge 1$,

$$H_n - \log n = \frac{1}{n} + \int_1^n \Big(\frac{1}{\lfloor x \rfloor} - \frac{1}{x}\Big) dx.$$

(b) Deduce that $\gamma = \lim_n(H_n - \log n)$ exists, that $\gamma = \int_1^\infty (1/\lfloor x \rfloor - 1/x)\, dx$, that $0 < \gamma < 1$, and that $0 < H_n - \log n - \gamma < 1/n$.
(c) Show that $H_n - \log n - \gamma = \frac{1}{2n} + O(n^{-2})$, which justifies the correction used in Example 1.3.

Solution. Write $r(x) = 1/\lfloor x \rfloor - 1/x$, which is ≥ 0 because $\lfloor x \rfloor \le x$, and $c_i = \int_i^{i+1} r(x)\, dx$ for integers $i \ge 1$.

(a) On $[i, i+1)$ we have $\lfloor x \rfloor = i$, so

$$c_i = \frac{1}{i} - \log \frac{i+1}{i}.$$

Summing over $i = 1, \ldots, n-1$, the logarithms telescope: $\int_1^n r = H_{n-1} - \log n = H_n - \frac{1}{n} - \log n$.

(b) On $[i, i+1]$, $1/x \ge 1/(i+1)$, so $0 < c_i \le \frac{1}{i} - \frac{1}{i+1}$ ($r > 0$ on $(i, i+1)$). Hence

$$0 < \int_n^\infty r = \sum_{i \ge n} c_i \le \sum_{i \ge n} \Big(\frac{1}{i} - \frac{1}{i+1}\Big) = \frac{1}{n}.$$

So $\int_1^\infty r < \infty$, and by (a) $H_n - \log n = \frac{1}{n} + \int_1^\infty r - \int_n^\infty r \to \int_1^\infty r = \gamma$. Also $H_n - \log n - \gamma = \frac{1}{n} - \int_n^\infty r \in (0, \frac{1}{n})$, positive because $c_i < \frac{1}{i} - \frac{1}{i+1}$ strictly ($1/x > 1/(i+1)$ on $(i, i+1)$). Finally $\gamma > 0$ as $r > 0$ on $(1, 2)$, and $\gamma < 1$ by the display with $n = 1$ and strictness.

(c) For $0 < t \le 1$ the series $\log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \ldots$ is alternating with decreasing terms, so $t - \frac{t^2}{2} \le \log(1+t) \le t - \frac{t^2}{2} + \frac{t^3}{3}$. With $t = 1/i$,

$$\frac{1}{2i^2} - \frac{1}{3i^3} \le c_i \le \frac{1}{2i^2}.$$

Comparing sums with integrals, $\frac{1}{n} = \int_n^\infty x^{-2} dx \le \sum_{i \ge n} i^{-2} \le \frac{1}{n^2} + \frac{1}{n}$, and $\sum_{i \ge n} i^{-3} \le \frac{1}{n^3} + \frac{1}{2n^2}$. Therefore $\int_n^\infty r = \sum_{i \ge n} c_i = \frac{1}{2n} + O(n^{-2})$, and by (b)

$$H_n - \log n - \gamma = \frac{1}{n} - \int_n^\infty r = \frac{1}{2n} + O(n^{-2}).$$

Subtracting $1/(2n)$ reduces the error from $O(n^{-1})$ to $O(n^{-2})$. ■

Exercise 1.2 (Sterbenz's lemma: exact subtraction)

Consider binary floating-point numbers with precision p, exponent range $[e_{\min}, e_{\max}]$ and gradual underflow, and ignore overflow.
(a) Show that if x, y are positive floating-point numbers with $y/2 \le x \le 2y$, then $x - y$ is a floating-point number; hence the computed difference $[x - y]$ is exact.
(b) Show by an example that the hypothesis cannot be dropped.

Solution. [Supplement]

Step 1. (Representability.) Every $D \cdot 2^{f-p+1}$ with integers $0 \le D < 2^p$ and $f \in [e_{\min}, e_{\max}]$ is a floating-point number. $D = 0$ is the exact zero. Otherwise let $2^k \le D < 2^{k+1}$ $(k \le p-1)$ and write $D \cdot 2^{f-p+1} = (D/2^k) \cdot 2^{f'}$, $f' = f - (p-1-k) \le e_{\max}$, where $D/2^k \in [1, 2)$ has at most p significant bits. If $f' \ge e_{\min}$ it is normalized; if $f' < e_{\min}$ it is a multiple of $2^{e_{\min}-p+1}$ (as $f \ge e_{\min}$) in $(0, 2^{e_{\min}})$, i.e. subnormal.

Step 2. (Common grid.) By symmetry $(x - y = -(y - x))$ assume $x \ge y$. With e_y the exponent of y ($e_y = e_{\min}$ if y is subnormal), $y = M_y 2^{e_y-p+1}$ with an integer $0 < M_y < 2^p$. As $e_x \ge e_y$, $x = M_x 2^{e_x-e_y} \cdot 2^{e_y-p+1}$ is also an integer multiple of 2^{e_y-p+1}.

Step 3. (Conclusion.) So $x - y = D \cdot 2^{e_y-p+1}$ with an integer $D \ge 0$, and $x \le 2y$ gives $x - y \le y$, i.e. $D \le M_y < 2^p$. By Step 1 ($f = e_y$), $x - y$ is a floating-point number; rounding returns it unchanged, so $[x - y] = x - y$.

(b) $x = 2^p$, $y = \frac{1}{2}$ (so $x > 2y$): $x - y = (2^{p+1} - 1)/2$ is $p + 1$ ones in binary, needs $p + 1$ significant bits, and is not representable. ■

Exercise 1.3 (Rounding error of recursive summation)

Let $x_1, \ldots, x_n$ be floating-point numbers, $u = \epsilon_{\max}$, and suppose no overflow or underflow occurs. The loop $\hat{s}_1 = x_1$, $\hat{s}_k = [\hat{s}_{k-1} + x_k]$ computes $\hat{s}_n$.
(a) Show that

$$\Big|\hat{s}_n - \sum_{i=1}^{n} x_i\Big| \le \gamma_{n-1} \sum_{i=1}^{n} |x_i|, \qquad \gamma_m = (1+u)^m - 1 \le \frac{mu}{1-mu} \quad (mu < 1).$$

(b) Show the sharper bound $|\hat{s}_n - \sum_i x_i| \le \sum_{i=1}^{n} \gamma_{n+1-\max(i,2)} |x_i|$, and explain why it supports adding numbers of small magnitude first.

Solution. [Supplement]

Step 1. (One addition.) By Theorem 1.2, $\hat{s}_k = (\hat{s}_{k-1} + x_k)(1 + \delta_k)$ with $|\delta_k| \le u$ (take $\delta_k = 0$ if the exact sum is 0, which is representable).

Step 2. (Unrolling.) By induction on n,

$$\hat{s}_n = \sum_{i=1}^{n} x_i \prod_{k=\max(i,2)}^{n} (1 + \delta_k),$$

since multiplying the formula for $\hat{s}_{n-1}$ by $(1+\delta_n)$ and adding $x_n(1+\delta_n)$ gives it for $\hat{s}_n$. So x_i carries $m_i = n + 1 - \max(i, 2) \le n - 1$ factors.

Step 3. (Products of rounding factors.) For $|\delta_k| \le u$, expanding the product,

$$\Big|\prod_{k=1}^{m}(1+\delta_k) - 1\Big| = \Big|\sum_{\emptyset \ne S \subseteq \{1,\ldots,m\}} \prod_{k \in S} \delta_k\Big| \le \sum_{\emptyset \ne S} u^{|S|} = (1+u)^m - 1 = \gamma_m.$$

Step 4. (Bounds.) Subtracting $\sum_i x_i$ in Step 2 and using Step 3 gives (b): $|\hat{s}_n - \sum_i x_i| \le \sum_i \gamma_{m_i} |x_i|$.

Since γ_m increases in m and $m_i \le n-1$, (a) follows. For the bound on γ_m: $(1+u)^m \le e^{mu}$, and $e^{-mu} \ge 1-mu > 0$ gives $e^{mu} \le 1/(1-mu)$, so $\gamma_m \le \frac{1}{1-mu} - 1 = \frac{mu}{1-mu}$.

In (b), x_1, x_2 carry γ_{n-1} and x_n only γ_1; by the rearrangement inequality the bound is smallest when the $|x_i|$ increase, i.e. small terms first (rule (1)). For positive x_i the relative error is at most $\gamma_{n-1} \approx (n-1)u$, consistent with Proposition 1.3 for $n = 2$. ■

Lecture 2. Linear Systems: LU Decomposition

2.1 Algorithms and their cost

We study algorithms for $Ax = b$. A guiding principle (Gentle):

Mantra

The form of a mathematical expression and the way the expression should be evaluated in practice may be quite different.

An **algorithm** is a finite set of instructions mapping input to output. Its cost is counted in **flops** (one floating-point $+, -, \times, \div$ or comparison, with fetch and store); it has order $O(f(n))$ if its leading flop count is $c\, f(n)$. E.g. Ab with $A \in \mathbb{R}^{m\times n}$ costs $2mn$ flops and AB with $B \in \mathbb{R}^{n\times p}$ costs $2mnp$.

At 10^{12} flops per second, $n = 10^{12}$ takes about 27 seconds with an $O(n \log n)$ algorithm and 31700 years with an $O(n^2)$ one; quicksort and Tukey's FFT turn $O(n^2)$ into $O(n \log n)$.

2.2 Triangular systems

General systems are reduced to easy ones. If A is lower triangular, **forward substitution** solves $Ax = b$:

$$x_1 = \frac{b_1}{a_{11}}, \qquad x_i = \frac{1}{a_{ii}}\Big(b_i - \sum_{j<i} a_{ij}x_j\Big), \quad i = 2, \dots, n,$$

and for upper triangular A, **back substitution** solves from x_n upwards. Each costs $1 + 3 + \cdots + (2n-1) = n^2$ flops, in place (b overwritten by x); BLAS routine `trsm`.

Triangular matrices have their diagonal entries as eigenvalues and their product as determinant; products and inverses of (unit) lower triangular matrices are (unit) lower triangular, and likewise upper. Products: for A, B upper triangular, $(AB)_{ij} = \sum_k a_{ik}b_{kj}$ needs $i \le k \le j$, so it vanishes for $i > j$ and equals $a_{ii}b_{ii}$ for $i = j$. Inverses: induct on n with $A = \left(\begin{smallmatrix} A_{11} & a_{12} \\ 0 & a_{22} \end{smallmatrix}\right)$; the block B_{21} of $B = A^{-1}$ satisfies $a_{22}B_{21} = 0$, so $B_{21} = 0$, $A_{11}B_{11} = I$, and $B_{11} = A_{11}^{-1}$ is upper triangular by induction (unit diagonal if A has one). Lower triangular cases follow by transposition. Hence an orthogonal triangular Q is diagonal, since Q^{-1} is upper and $Q^\top$ lower triangular.

2.3 Gaussian elimination and the LU decomposition

The **elementary operator matrix** $E_{jk}(c) = I + c\, e_j e_k^\top$ $(j \ne k)$ is unit triangular, $E_{jk}(c)^{-1} = E_{jk}(-c)$, and left-multiplication by it adds $c\times$row k to row j ($2m$ flops for m columns). Gaussian elimination reduces A to upper triangular form by such operations.

Those on column k combine into the **Gauss transformation**

$$M_k = E_{nk}(c_{nk}) \cdots E_{k+1,k}(c_{k+1,k}) = I + \mathbf{c}_k e_k^\top, \qquad \mathbf{c}_k = (0, \dots, 0, c_{k+1,k}, \dots, c_{nk})^\top,$$

with $c_{ik} = -a_{ik}^{(k)}/a_{kk}^{(k)}$, which zeros column k below the diagonal; it is unit lower triangular with $M_k^{-1} = I - \mathbf{c}_k e_k^\top$. Elimination computes

$$M_{n-1} \cdots M_1 A = U \quad \text{(upper triangular)}, \qquad \text{so} \qquad A = LU, \quad L = M_1^{-1} \cdots M_{n-1}^{-1}.$$

Proposition 2.1

$L = I - \sum_{k=1}^{n-1} \mathbf{c}_k e_k^\top$: it is unit lower triangular, and its (i,k) entry below the diagonal is the multiplier $\ell_{ik} = a_{ik}^{(k)}/a_{kk}^{(k)}$.

Proof. $M_k^{-1}M_{k+1}^{-1} = (I - \mathbf{c}_k e_k^\top)(I - \mathbf{c}_{k+1}e_{k+1}^\top) = I - \mathbf{c}_k e_k^\top - \mathbf{c}_{k+1}e_{k+1}^\top + \mathbf{c}_k(e_k^\top \mathbf{c}_{k+1})e_{k+1}^\top$, and $e_k^\top \mathbf{c}_{k+1} = 0$ because $\mathbf{c}_{k+1}$ vanishes in positions $\le k+1$. Inductively the product is $I - \sum_k \mathbf{c}_k e_k^\top$. ■

Theorem 2.2 (Existence and uniqueness of LU)

If the leading principal submatrices $A[1{:}k, 1{:}k]$, $k = 1, \dots, n-1$, are nonsingular, then $A = LU$ with L unit lower triangular and U upper triangular. If moreover A is nonsingular, the decomposition is unique and $\det A = \prod_k u_{kk}$.

Proof. *Existence.* Unit lower triangular operations preserve leading principal minors, so the step-k pivot is $a_{kk}^{(k)} = \det A[1{:}k, 1{:}k] / \det A[1{:}k-1, 1{:}k-1] \neq 0$ and elimination completes. *Uniqueness.* If $L_1U_1 = L_2U_2$ with A nonsingular, all factors are invertible and $L_2^{-1}L_1 = U_2U_1^{-1}$ is both unit lower and upper triangular, hence I. Finally $\det A = \det L \det U = \prod u_{kk}$ (how software computes determinants). ■

For nonsingular A the condition is also necessary: A has an LU decomposition without pivoting *iff* all leading principal submatrices are nonsingular. Indeed, partitioning conformally, $A[1{:}k, 1{:}k] = L_kU_k$ with L_k, U_k the leading blocks, and $0 \neq \det A = \prod_i u_{ii}$ gives $\det A[1{:}k, 1{:}k] = \prod_{i \le k} u_{ii} \neq 0$.

Cost. Step k updates an $(n-k) \times (n-k)$ block at 2 flops per entry, so LU costs

$$\sum_{k=1}^{n-1} 2(n-k)^2 \approx \tfrac{2}{3}n^3 \text{ flops},$$

in place (L below the diagonal, U on and above). Solving $LUx = b$ adds $2n^2$ (forward and back substitution), $\frac{2}{3}n^3 + 2n^2$ in total; several right-hand sides reuse the factorization.

No inversion. Inverting A via LU solves n systems with right-hand sides $e_1, \dots, e_n$, costing $\frac{4}{3}n^3$ (exploiting zeros), $2n^3$ in total: three times one solve. Rule: *for $A^{-1}b$, solve $Ax = b$.* Explicit inverses are needed only for standard errors, very many right-hand sides, or small n.

2.4 Pivoting, conditioning and stability

For $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ elimination fails at once (zero pivot); small pivots are just as harmful.

Example 2.1

Solve $0.001x_1 + x_2 = 1$, $x_1 + x_2 = 2$ (exact solution $\approx (1.001, 0.999)$) with two significant digits. Pivot 0.001 gives multiplier 1000 and $(1 - 1000)x_2 = 2 - 1000$, rounded to $-1000x_2 = -1000$: $x_2 = 1$, $x_1 = 0$. The computed $L = \begin{pmatrix} 1 & 0 \\ 1000 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 0.001 & 1 \\ 0 & -1000 \end{pmatrix}$ give $LU = \begin{pmatrix} 0.001 & 1 \\ 1 & 0 \end{pmatrix} \neq A$: the large multiplier wiped out a_{22} and b_2. Swapping the equations first gives multiplier 0.001, $(1 - 0.001)x_2 = 1 - 0.002$, so $x_2 = 1$, $x_1 = 1$, correct to two digits.

Conditioning. The *problem* $A \mapsto A^{-1}b$ is **well conditioned** if small relative perturbations of A cause small relative changes of the solution, as measured by the **condition number**

$$\kappa(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}, \qquad \sigma_{\max}(A) = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}, \quad \sigma_{\min}(A) = \min_{x \neq 0} \frac{\|Ax\|}{\|x\|},$$

(∞ for singular A). In Example 2.1, $\kappa(A) = 2.62$: well conditioned.

Example 2.2 (An ill-conditioned matrix)

The symmetric matrix

$$A = \begin{pmatrix} 5 & 7 & 6 & 5 \\ 7 & 10 & 8 & 7 \\ 6 & 8 & 10 & 9 \\ 5 & 7 & 9 & 10 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 68 & -41 & -17 & 10 \\ -41 & 25 & 10 & -6 \\ -17 & 10 & 5 & -3 \\ 10 & -6 & -3 & 2 \end{pmatrix},$$

looks harmless. Changing a_{33} from 10 to 9.99 (relative change 10^{-3}) gives the inverse (two decimals) $\begin{pmatrix} 71.04 & -42.79 & -17.89 & 10.54 \\ -42.79 & 26.05 & 10.53 & -6.32 \\ -17.89 & 10.53 & 5.26 & -3.16 \\ 10.54 & -6.32 & -3.16 & 2.09 \end{pmatrix}$, entries moving by up to 5.4%: an amplification of about 54. The worst direction is much worse: $\kappa_2(A) = \sigma_{\max}/\sigma_{\min} \approx 2984$, so about 3.5 decimal digits can be lost in solving $Ax = b$.

Stability. An *algorithm* $\tilde{f}$ is **stable** if for every $\tilde{A}$ within relative distance ϵ of A, the computed $\hat{x} = \tilde{f}(A)$ is within relative distance $O(\epsilon)$ of $\tilde{A}^{-1}b$. By Example 2.1, elimination without pivoting is unstable: a well-conditioned problem gives $\|x - \hat{x}\|/\|x\| \geq 1/\sqrt{2}$.

Partial pivoting. Before eliminating column k, swap into row k the row with the largest $|a_{ik}|$, $i \geq k$; at $O(n^2)$ comparisons this gives

$$PA = LU, \qquad P \text{ a permutation matrix}, \quad |\ell_{ij}| \leq 1.$$

Complete pivoting also swaps columns, moving the largest remaining entry to the pivot: $PAQ = LU$, more stable but costlier. Gaussian elimination with partial pivoting (GEPP) is the standard dense solver and is stable in practice, partly because $|\ell_{ij}| \leq 1$.

Software. BLAS (basic operations) and LAPACK (decompositions): `getrf` computes $PA = LU$ in place; R's `solve` calls `dgesv`.

2.5 Selected exercises

Motivation. One LDU factorization by hand, then three facts used without proof: the condition-number error bound, that partial pivoting never breaks down, and when no pivoting is needed.

Exercise 2.1 (An LDU decomposition by hand)

Let

$$A = \begin{pmatrix} 2 & 4 & 2 \\ 4 & 7 & 7 \\ -2 & -7 & 5 \end{pmatrix}, \qquad b = \begin{pmatrix} 4 \\ 13 \\ 7 \end{pmatrix}.$$

(a) Compute $A = LU$ by Gaussian elimination, and write it as $A = LDU_1$ with D diagonal and U_1 unit upper triangular.
(b) Solve $Ax = b$ by forward and back substitution, and compute $\det A$.

Solution.

(a) The leading principal minors are 2, $2 \cdot 7 - 4 \cdot 4 = -2$ and 4 (below), so by Theorem 2.2 the factorization exists and is unique. Column 1 ($\ell_{21} = 2$, $\ell_{31} = -1$):

$$\begin{pmatrix} 2 & 4 & 2 \\ 0 & -1 & 3 \\ 0 & -3 & 7 \end{pmatrix}.$$

Column 2 ($\ell_{32} = 3$) leaves $(0, 0, -2)$ in row 3. By Proposition 2.1,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 4 & 2 \\ 0 & -1 & 3 \\ 0 & 0 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \quad U_1 = D^{-1}U = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}.$$

Check: rows 2, 3 of LU are $2(2,4,2) + (0,-1,3) = (4,7,7)$ and $-(2,4,2) + 3(0,-1,3) + (0,0,-2) = (-2,-7,5)$.

(b) $Ly = b$: $y_1 = 4$, $y_2 = 13 - 2 \cdot 4 = 5$, $y_3 = 7 + 4 - 3 \cdot 5 = -4$. $Ux = y$: $x_3 = 2$, $x_2 = 3x_3 - 5 = 1$, $x_1 = (4 - 4x_2 - 2x_3)/2 = -2$. So $x = (-2, 1, 2)^\top$, and $Ax = (-4+4+4,\ -8+7+14,\ 4-7+10)^\top = b$. Finally $\det A = \prod_k u_{kk} = 2 \cdot (-1) \cdot (-2) = 4$. ■

Exercise 2.2 (The condition number bounds the error)

Let A be nonsingular, $b \neq 0$ and $Ax = b$. Write $\|\cdot\|$ for the Euclidean norm and the induced matrix norm, $\|M\| = \sigma_{\max}(M)$.
(a) Show that $\kappa(A) = \|A\|\,\|A^{-1}\|$.
(b) If $A(x + \delta x) = b + \delta b$, show that $\dfrac{\|\delta x\|}{\|x\|} \leq \kappa(A) \dfrac{\|\delta b\|}{\|b\|}$.
(c) If $(A + \delta A)(x + \delta x) = b$ and $t = \kappa(A)\|\delta A\|/\|A\| < 1$, show that $\dfrac{\|\delta x\|}{\|x\|} \leq \dfrac{t}{1-t}$.

Solution. [Supplement]

(a) $y = Ax$ ranges over nonzero vectors with x, so

$$\sigma_{\min}(A) = \min_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \min_{y \neq 0} \frac{\|y\|}{\|A^{-1}y\|} = \Big(\max_{y \neq 0} \frac{\|A^{-1}y\|}{\|y\|} \Big)^{-1} = \frac{1}{\|A^{-1}\|},$$

and $\kappa(A) = \sigma_{\max}(A)/\sigma_{\min}(A) = \|A\|\|A^{-1}\|$.

(b) $\delta x = A^{-1}\delta b$ gives $\|\delta x\| \leq \|A^{-1}\|\|\delta b\|$, and $\|b\| \leq \|A\|\|x\|$ ($x \neq 0$ as $b \neq 0$) gives $1/\|x\| \leq \|A\|/\|b\|$. Multiply and use (a).

(c) $A\delta x = -\delta A(x + \delta x)$, so $\delta x = -A^{-1}\delta A(x + \delta x)$ and

$$\|\delta x\| \leq \|A^{-1}\|\|\delta A\|(\|x\| + \|\delta x\|) = t(\|x\| + \|\delta x\|),$$

as $\|A^{-1}\|\|\delta A\| = t$ by (a). Rearranging, $(1 - t)\|\delta x\| \leq t\|x\|$ with $1 - t > 0$.

So relative data perturbations ϵ (rounding: $\epsilon \approx 10^{-16}$) give relative errors about $\kappa(A)\epsilon$: $\kappa(A) \approx 10^k$ loses about k digits, as in Example 2.2. ■

Exercise 2.3 (Partial pivoting never breaks down)

Show that for every nonsingular $A \in \mathbb{R}^{n\times n}$, Gaussian elimination with partial pivoting runs to completion and yields $PA = LU$ with P a permutation matrix, L unit lower triangular with $|\ell_{ij}| \le 1$, and U upper triangular and nonsingular.

Solution. [Supplement]

Step 1. (The pivots are nonzero.) Suppose steps $1, \dots, k-1$ have been carried out, giving

$$M_{k-1}P_{k-1}\cdots M_1P_1A = A^{(k)} = \begin{pmatrix} U_{11} & U_{12} \\ 0 & S_k \end{pmatrix},$$

with P_j swapping rows j and some $j' \ge j$, $M_j = I + \mathbf{c}_j e_j^\top$, and U_{11} upper triangular of size $k-1$. All factors are nonsingular, so $0 \ne \det A^{(k)} = \det U_{11} \det S_k$; the first column of S_k is nonzero, its largest-modulus entry is a nonzero pivot, and each multiplier is an entry over one of larger or equal modulus: $|\ell_{ik}| \le 1$. By induction elimination completes with $U = A^{(n)}$ upper triangular and nonsingular.

Step 2. (Moving permutations right.) Put $Q_k = P_{n-1}\cdots P_{k+1}$ ($Q_{n-1} = I$) and $N_k = Q_k M_k Q_k^{-1}$. Each P_j is a symmetric involution, so

$$M_{n-1}P_{n-1}M_{n-2}P_{n-2}\cdots M_1P_1 = N_{n-1}N_{n-2}\cdots N_1\,P, \qquad P = P_{n-1}\cdots P_1,$$

by inserting $P_jP_j = I$; e.g. for $n = 4$,

$$M_3P_3M_2P_2M_1P_1 = M_3\,(P_3M_2P_3)\,(P_3P_2M_1P_2P_3)\,(P_3P_2P_1).$$

Step 3. (Structure of N_k.) P_j, $j > k$, swaps rows $j, j' > k$, so $Q_k e_k = e_k$, $e_k^\top Q_k^{-1} = e_k^\top$ and $N_k = I + \tilde{\mathbf{c}}_k e_k^\top$ with $\tilde{\mathbf{c}}_k = Q_k\mathbf{c}_k$ a permutation of $\mathbf{c}_k$ within positions $k+1, \dots, n$: zero in positions $\le k$, entries of modulus ≤ 1.

Step 4. (Conclusion.) $N_{n-1}\cdots N_1PA = U$, and the proof of Proposition 2.1 applies verbatim: $L = N_1^{-1}\cdots N_{n-1}^{-1} = I - \sum_k \tilde{\mathbf{c}}_k e_k^\top$ is unit lower triangular, $|\ell_{ij}| \le 1$, and $PA = LU$. ■

Exercise 2.4 (Diagonal dominance makes pivoting unnecessary)

Suppose A is strictly diagonally dominant by columns: $|a_{jj}| > \sum_{i\ne j}|a_{ij}|$ for every j. Show that Gaussian elimination without pivoting runs to completion with all multipliers $|\ell_{ik}| < 1$, so that partial pivoting would never swap rows.

Solution. [Supplement]

Step 1. (One step.) Partition $A = \left(\begin{smallmatrix} \alpha & w^\top \\ v & B \end{smallmatrix}\right)$. Dominance of column 1 gives $\sum_i |v_i| < |\alpha|$, so $\alpha \ne 0$ and $|v_i/\alpha| < 1$. The remaining block is the Schur complement $S = B - vw^\top/\alpha$, $s_{ij} = b_{ij} - v_iw_j/\alpha$.

Step 2. (S inherits dominance.) For column j: $|b_{jj}| > \sum_{i\ne j}|b_{ij}| + |w_j|$, and Step 1 gives $\sum_{i\ne j}|v_i| < |\alpha| - |v_j|$. Hence

$$\begin{aligned}\sum_{i\ne j}|s_{ij}| &\le \sum_{i\ne j}|b_{ij}| + \frac{|w_j|}{|\alpha|}\sum_{i\ne j}|v_i| < |b_{jj}| - |w_j| + \frac{|w_j|}{|\alpha|}(|\alpha| - |v_j|) \\ &= |b_{jj}| - \frac{|w_j||v_j|}{|\alpha|} \le \left|b_{jj} - \frac{v_jw_j}{\alpha}\right| = |s_{jj}|.\end{aligned}$$

(Strictness comes from the bound on $|b_{jj}|$; the bound on $\sum|v_i|$ enters with the factor $|w_j| \ge 0$.)

Step 3. (Induction.) S is strictly column dominant of size $n-1$, so every pivot is nonzero and every multiplier has modulus < 1. Each pivot exceeds in modulus every other entry of its column below it, so partial pivoting always picks the diagonal entry. ■

Lecture 3. Cholesky Decomposition

Mantra 2

The structure should be exploited whenever possible in solving a problem.

Common structures: symmetry, positive (semi)definiteness, sparsity, Kronecker products, low rank. Statisticians mostly meet positive (semi)definite matrices (covariances, Gram matrices $X^\top X$): $x^\top Mx > 0$ (≥ 0) for $x \neq 0$, and since $x^\top Mx = x^\top \frac{M+M^\top}{2} x$ we may assume M symmetric.

3.1 The Cholesky decomposition

Motivation. Cholesky factors a symmetric positive definite A as $LL^\top$ at half the cost of LU, stably and without pivoting.

Theorem 3.1 (Cholesky)

If $A \in \mathbb{R}^{n\times n}$ is symmetric positive definite, then $A = LL^\top$ for a unique lower triangular L with positive diagonal entries.

Proof. Induction on n. For $n = 1$, $a = \sqrt{a}\sqrt{a}$ with $a > 0$. For $n > 1$, partition and match blocks:

$$\begin{pmatrix} a_{11} & \mathbf{a}^\top \\ \mathbf{a} & A_{22} \end{pmatrix} = \begin{pmatrix} \ell_{11} & \mathbf{0}^\top \\ \mathbf{b} & L_{22} \end{pmatrix} \begin{pmatrix} \ell_{11} & \mathbf{b}^\top \\ \mathbf{0} & L_{22}^\top \end{pmatrix} \iff a_{11} = \ell_{11}^2, \quad \mathbf{a} = \ell_{11}\mathbf{b}, \quad A_{22} = \mathbf{b}\mathbf{b}^\top + L_{22}L_{22}^\top.$$

As $a_{11} = e_1^\top Ae_1 > 0$, $\ell_{11} = \sqrt{a_{11}}$ and $\mathbf{b} = \mathbf{a}/\sqrt{a_{11}}$ are forced. The **Schur complement** $S = A_{22} - \mathbf{b}\mathbf{b}^\top = A_{22} - \mathbf{a}\mathbf{a}^\top/a_{11}$ is positive definite: for $\mathbf{y} \neq \mathbf{0}$ put $x = (-\mathbf{a}^\top\mathbf{y}/a_{11}, \mathbf{y}^\top)^\top \neq 0$; then

$$0 < x^\top Ax = \frac{(\mathbf{a}^\top\mathbf{y})^2}{a_{11}} - 2\frac{(\mathbf{a}^\top\mathbf{y})^2}{a_{11}} + \mathbf{y}^\top A_{22}\mathbf{y} = \mathbf{y}^\top S\mathbf{y}.$$

By induction $S = L_{22}L_{22}^\top$ for a unique such L_{22}. ■

The proof is the algorithm: first column of L, Schur complement update, recurse, in place in the lower triangle of A.

Cost. By symmetry only half the update entries are computed:

$$\tfrac{1}{2}\{2(n-1)^2 + 2(n-2)^2 + \cdots + 2\} \approx \tfrac{1}{3}n^3 \text{ flops}$$

plus n square roots: *half* the cost of LU. It is very stable without pivoting and fails (a nonpositive number under a root) exactly when A is not positive definite, so it is the cheapest *test* of positive definiteness.

Example 3.1

For

$$A = \begin{pmatrix} 4 & 6 & 4 & 4 \\ 6 & 37/4 & 11/2 & 6 \\ 4 & 11/2 & 29/4 & 7 \\ 4 & 6 & 7 & 33/4 \end{pmatrix}$$

the first column of L is $(4,6,4,4)^\top/2 = (2,3,2,2)^\top$. The Schur complement has first column $(37/4-9,\ 11/2-6,\ 6-6)^\top = (1/4,-1/2,0)^\top$, so $\ell_{22}=1/2$ and the second column of L is $(1/2,-1,0)^\top$. The pivots are $4, \frac{1}{4}, \frac{9}{4}, \frac{1}{4}$ and

$$L = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 3 & 1/2 & 0 & 0 \\ 2 & -1 & 3/2 & 0 \\ 2 & 0 & 2 & 1/2 \end{pmatrix}.$$

All pivots are positive, so A is positive definite, and $\det A = (2\cdot\frac{1}{2}\cdot\frac{3}{2}\cdot\frac{1}{2})^2 = \frac{9}{16}$.

Pivoting for semidefinite matrices. If A is only positive semidefinite (e.g. $X^\top X$, X rank deficient), a zero pivot can occur. **Symmetric pivoting** moves the largest remaining diagonal entry to the pivot position by a symmetric row and column swap. If it is 0, the remaining block is 0 (since $|a_{ij}| \le \sqrt{a_{ii}a_{jj}}$) and the algorithm stops:

$$PAP^\top = LL^\top, \qquad L \in \mathbb{R}^{n\times r}, \quad r = \operatorname{rank} A.$$

LAPACK provides `potrf` (no pivoting) and `pstrf` (pivoting); R's `chol` returns the upper factor $R = L^\top$, $A = R^\top R$.

3.2 Applications

Multivariate normal log-likelihood. For $y \sim N_n(0,\Sigma)$,

$$\log f(y) = -\frac{n}{2}\log(2\pi) - \frac{1}{2}\log\det\Sigma - \frac{1}{2}y^\top\Sigma^{-1}y.$$

- Method 1: invert Σ ($2n^3$), form the quadratic form ($2n^2$), compute the determinant by LU ($\frac{2}{3}n^3$).
- Method 2: factor $\Sigma = LL^\top$ ($\frac{1}{3}n^3$), solve $Lx = y$ by forward substitution (n^2), then $y^\top\Sigma^{-1}y = x^\top x$ ($2n$) and $\log\det\Sigma = 2\sum_i \log\ell_{ii}$ (n).

Method 2 is about 8 times cheaper, more accurate, and its log-determinant cannot overflow. For many evaluation points, factor once and pay n^2 per point.

Linear regression. For $X \in \mathbb{R}^{n\times p}$, solve the normal equations $X^\top X\beta = X^\top y$ by Cholesky:

form $X^\top X$	np^2 (symmetric)
form $X^\top y$	$2np$
Cholesky of $X^\top X$	$\frac{1}{3}p^3$
two triangular solves	$2p^2$
$(X^\top X)^{-1}$ for standard errors	$\frac{4}{3}p^3$

In total $np^2 + \frac{1}{3}p^3$ flops ($np^2 + \frac{5}{3}p^3$ with standard errors): fastest when $n \gg p$, but forming $X^\top X$ squares the condition number (Lecture 4).

3.3 The sweep operator

For a symmetric $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ with $a_{kk} \neq 0$, **sweeping** on the kth diagonal entry maps A to $\hat{A} = (\hat{a}_{ij})$ with

$$\hat{a}_{kk} = -\frac{1}{a_{kk}}, \qquad \hat{a}_{ik} = \frac{a_{ik}}{a_{kk}}, \qquad \hat{a}_{kj} = \frac{a_{kj}}{a_{kk}}, \qquad \hat{a}_{ij} = a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}} \quad (i, j \neq k).$$

The last formula is one Gaussian elimination step (Schur complement update), so a sweep costs about n^2 flops ($n^2/2$ by symmetry), in place. The **inverse sweep** on entry k is

$$\check{a}_{kk} = -\frac{1}{a_{kk}}, \qquad \check{a}_{ik} = -\frac{a_{ik}}{a_{kk}}, \qquad \check{a}_{kj} = -\frac{a_{kj}}{a_{kk}}, \qquad \check{a}_{ij} = a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}};$$

which restores A (e.g. $\hat{a}_{ij} - \hat{a}_{ik}\hat{a}_{kj}/\hat{a}_{kk} = a_{ij} - a_{ik}a_{kj}/a_{kk} + a_{ik}a_{kj}/a_{kk} = a_{ij}$).

Lemma 3.2 (Exchange property)

Let $U = [u_1, \ldots, u_n]$ and $V = [v_1, \ldots, v_n]$ be $p \times n$ with $V = UA$. After sweeping on entry k,

$$[v_1, \ldots, v_{k-1}, -u_k, v_{k+1}, \ldots, v_n] = [u_1, \ldots, u_{k-1}, v_k, u_{k+1}, \ldots, u_n]\,\hat{A}.$$

Sweeping exchanges u_k and v_k (up to sign).

Proof. Column k of the right side is $-v_k/a_{kk} + \sum_{i\neq k} u_i a_{ik}/a_{kk} = -(v_k - \sum_{i\neq k} u_i a_{ik})/a_{kk} = -u_k a_{kk}/a_{kk} = -u_k$, using $v_k = \sum_i u_i a_{ik}$. For $j \neq k$, column j is

$$v_k\frac{a_{kj}}{a_{kk}} + \sum_{i\neq k} u_i\Big(a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}}\Big) = \sum_{i\neq k} u_i a_{ij} + \frac{a_{kj}}{a_{kk}}\Big(v_k - \sum_{i\neq k} u_i a_{ik}\Big) = \sum_{i\neq k} u_i a_{ij} + a_{kj}u_k = v_j.$$

■

Proposition 3.3

Partition $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$ with $A_{11} \in \mathbb{R}^{k\times k}$ invertible. Sweeping on the diagonal entries $1, \ldots, k$ (in any order, as long as each pivot is nonzero) gives

$$\hat{A} = \begin{pmatrix} -A_{11}^{-1} & A_{11}^{-1}A_{12} \\ A_{21}A_{11}^{-1} & A_{22} - A_{21}A_{11}^{-1}A_{12} \end{pmatrix}.$$

Proof. Apply the lemma to $A = IA$ ($U = I$, $V = A$) for each sweep. After sweeping $1, \ldots, k$,

$$\begin{pmatrix} -I_k & A_{12} \\ 0 & A_{22} \end{pmatrix} = \begin{pmatrix} A_{11} & 0 \\ A_{21} & I \end{pmatrix}\begin{pmatrix} \hat{A}_{11} & \hat{A}_{12} \\ \hat{A}_{21} & \hat{A}_{22} \end{pmatrix}.$$

The blocks give $A_{11}\hat{A}_{11} = -I$, $A_{11}\hat{A}_{12} = A_{12}$, $A_{21}\hat{A}_{11} + \hat{A}_{21} = 0$, $A_{21}\hat{A}_{12} + \hat{A}_{22} = A_{22}$. The result depends only on the swept set, so sweeps commute. ■

Regression by sweeping. Sweeping $\left(\begin{smallmatrix} X^\top X & X^\top y \\ y^\top X & y^\top y \end{smallmatrix}\right)$ on its first p diagonal entries gives

$$\begin{pmatrix} -(X^\top X)^{-1} & (X^\top X)^{-1}X^\top y \\ y^\top X(X^\top X)^{-1} & y^\top y - y^\top X(X^\top X)^{-1}X^\top y \end{pmatrix} = \begin{pmatrix} -\sigma^{-2}\operatorname{Cov}(\hat{\beta}) & \hat{\beta} \\ \hat{\beta}^\top & \|y - \hat{y}\|^2 \end{pmatrix}:$$

coefficients, unscaled covariance and RSS at once. Hence **stepwise regression**: adding variable j is one sweep on entry j, removing it one inverse sweep, each $O(p^2)$; the swept matrix holds the current $\hat{\beta}$, standard errors and RSS, from which the F (or $t^2 = \hat{\beta}_j^2/\widehat{\operatorname{Var}}\hat{\beta}_j$) statistics are read off. A (numerically) zero pivot flags a variable collinear with those in the model. The price is the conditioning of the normal equations.

3.4 Selected exercises

Motivation. Cholesky versus LU with a determinant test, the entry bound behind symmetric pivoting, and one stepwise-regression sweep.

Exercise 3.1 ($LDL^\top$ and a determinant test)

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with nonsingular leading principal submatrices $A_k = A[1{:}k, 1{:}k]$, $k = 1, \dots, n$, and put $\det A_0 = 1$.
(a) Show that $A = LDL^\top$ for a unique unit lower triangular L and diagonal D, and that $d_k = \det A_k / \det A_{k-1}$.
(b) Show that A is positive definite if and only if $\det A_k > 0$ for all k, and that then $LD^{1/2}$ is the Cholesky factor of A.

Solution. [Supplement]

(a) *Existence.* By Theorem 2.2, $A = LU$ uniquely with L unit lower triangular, and $A_k = L_k U_k$ gives $\det A_k = \prod_{i\le k} u_{ii}$, so $u_{kk} = \det A_k / \det A_{k-1} \neq 0$. With $D = \operatorname{diag}(u_{11}, \dots, u_{nn})$ and $M = D^{-1}U$ unit upper triangular, $A = LDM$ and $A = A^\top = M^\top(DL^\top)$ is another LU decomposition; by uniqueness $M^\top = L$, i.e. $A = LDL^\top$.

Uniqueness. $A = L'D'L'^\top$ gives the LU decomposition $L'(D'L'^\top)$, so $L' = L$, $D'L^\top = DL^\top$ and $D' = D$.

(b) If A is positive definite, so is each A_k (x supported on the first k coordinates), and $\det A_k$, a product of positive eigenvalues, is positive. Conversely, if all $\det A_k > 0$ then all $d_k > 0$, and for $x \neq 0$, $y = L^\top x \neq 0$,

$$x^\top A x = y^\top D y = \sum_k d_k y_k^2 > 0.$$

So $A = (LD^{1/2})(LD^{1/2})^\top$ with positive diagonal $\sqrt{d_k}$, the Cholesky factor by uniqueness in Theorem 3.1. In Example 3.1 the pivots $4, \frac{1}{4}, \frac{9}{4}, \frac{1}{4}$ are the d_k, and $\ell_{kk} = \sqrt{d_k}$. ■

Exercise 3.2 (Entries of a positive semidefinite matrix)

Let A be symmetric positive semidefinite. Show:
(a) $|a_{ij}| \le \sqrt{a_{ii}a_{jj}}$ for all i, j;
(b) if $a_{ii} = 0$, then the whole i-th row and column of A vanish;
(c) $\max_{i,j}|a_{ij}| = \max_i a_{ii}$.

Solution. [Supplement] For $s, t \in \mathbb{R}$ and $i \neq j$,

$$0 \le (se_i + te_j)^\top A(se_i + te_j) = s^2 a_{ii} + 2st\, a_{ij} + t^2 a_{jj}. \qquad (\dagger)$$

(b) If $a_{ii} = 0$, $t = 1$ gives $2sa_{ij} + a_{jj} \ge 0$ for all s, so $a_{ij} = 0$ for $j \neq i$; by symmetry the column vanishes too.

(a) If $a_{ii} = 0$ or $a_{jj} = 0$, both sides are 0 by (b). If $a_{ii} > 0$, $s = -a_{ij}$, $t = a_{ii}$ in (†) give $0 \le a_{ii}(a_{ii}a_{jj} - a_{ij}^2)$. The case $i = j$ is trivial.

(c) By (a), $|a_{ij}| \le \sqrt{a_{ii}a_{jj}} \le \max\{a_{ii}, a_{jj}\}$, and the a_{ii} are among the $|a_{ij}|$.

Hence the largest remaining diagonal entry dominates the remaining block, justifying symmetric pivoting. ■

Exercise 3.3 (What one sweep does in stepwise regression)

Let $X_1 \in \mathbb{R}^{n\times q}$ and $x \in \mathbb{R}^n$ be such that $[X_1, x]$ has full column rank, $y \in \mathbb{R}^n$, $H_1 = X_1(X_1^\top X_1)^{-1}X_1^\top$, $r_x = (I - H_1)x$ and $r_y = (I - H_1)y$. Order the cross-product matrix of (X_1, x, y) as

$$T = \begin{pmatrix} X_1^\top X_1 & X_1^\top x & X_1^\top y \\ x^\top X_1 & x^\top x & x^\top y \\ y^\top X_1 & y^\top x & y^\top y \end{pmatrix}.$$

(a) Show that after sweeping on the first q diagonal entries, the lower-right 2×2 block is $\begin{pmatrix} \|r_x\|^2 & r_x^\top r_y \\ r_y^\top r_x & \|r_y\|^2 \end{pmatrix}$. In particular the next pivot is nonzero.

(b) Sweeping next on the entry of x, show that the residual sum of squares drops from $\|r_y\|^2$ to $\|r_y\|^2 - (r_x^\top r_y)^2/\|r_x\|^2$, and that the coefficient of x in the regression of y on $[X_1, x]$ is $r_x^\top r_y/\|r_x\|^2$.

Solution. [Supplement]

(a) Proposition 3.3 with $A_{11} = X_1^\top X_1$ (invertible), $A_{21} = [x, y]^\top X_1$, $A_{22} = [x, y]^\top [x, y]$ gives the swept block

$$A_{22} - A_{21}A_{11}^{-1}A_{12} = [x, y]^\top (I - H_1)[x, y].$$

As $I - H_1$ is symmetric idempotent, $u^\top (I - H_1)v = \{(I - H_1)u\}^\top (I - H_1)v$, giving the entries. $r_x \neq 0$ since $x \notin \mathcal{R}(X_1)$, so the pivot $\|r_x\|^2 > 0$.

(b) Sweeping on the x-entry with $a_{xx} = \|r_x\|^2$, $a_{xy} = r_x^\top r_y$, $a_{yy} = \|r_y\|^2$ gives

$$\hat{a}_{yy} = a_{yy} - \frac{a_{yx}a_{xy}}{a_{xx}} = \|r_y\|^2 - \frac{(r_x^\top r_y)^2}{\|r_x\|^2}, \qquad \hat{a}_{xy} = \frac{a_{xy}}{a_{xx}} = \frac{r_x^\top r_y}{\|r_x\|^2}.$$

By Proposition 3.3 this equals T swept on all entries of $X = [X_1, x]$, so by regression by sweeping the (y, y) entry is $\|y - \hat{y}\|^2$ and the (x, y) entry is the coefficient of x in $\hat{\beta} = (X^\top X)^{-1}X^\top y$. The drop is ≥ 0, zero iff $r_x \perp r_y$; the coefficient is the slope of r_y on r_x (Frisch–Waugh). ■

Lecture 4. Least Squares: QR Decomposition

4.1 QR decomposition and least squares

Besides Cholesky on the normal equations, least squares can use the QR decomposition (R's `lm`). Let $X \in \mathbb{R}^{n\times p}$ have full column rank ($n \geq p$).

Definition (QR decomposition)

- **Full QR**: $X = QR$, $Q \in \mathbb{R}^{n\times n}$ orthogonal, $R \in \mathbb{R}^{n\times p}$ upper triangular (lower $(n-p)\times p$ block 0). The first p columns of Q are an orthonormal basis of $\mathcal{R}(X)$, the last $n-p$ of $\mathcal{N}(X^\top) = \mathcal{R}(X)^\perp$.
- **Reduced QR**: $X = Q_1 R_1$ with $Q_1 \in \mathbb{R}^{n\times p}$, $Q_1^\top Q_1 = I_p$ (but $Q_1 Q_1^\top \neq I_n$), and $R_1 \in \mathbb{R}^{p\times p}$ upper triangular with positive diagonal.

As $X^\top X = R^\top Q^\top Q R = R_1^\top R_1$, QR of X gives the Cholesky factor of the Gram matrix for free.

Least squares via QR. The normal equations become $R_1^\top R_1 \beta = R_1^\top Q_1^\top y$, i.e. ($R_1$ invertible)

$$R_1 \beta = Q_1^\top y,$$

one back substitution. (Equivalently $\|y - X\beta\|^2 = \|Q_1^\top y - R_1\beta\|^2 + \|Q_2^\top y\|^2$, so RSS $= \|Q_2^\top y\|^2$.) This beats the normal equations because

$$\kappa(X^\top X) = \kappa(X)^2,$$

(singular values of $X^\top X$ are squares of those of X). Standard errors use $(X^\top X)^{-1} = R_1^{-1} R_1^{-\top}$, again by triangular solves.

4.2 Gram–Schmidt

Gram–Schmidt orthonormalization of the columns $x_1, \ldots, x_p$ gives the reduced QR of any full-column-rank X:

(1) $q_1 = x_1 / \|x_1\|_2$.
(2) For $k = 2, \ldots, p$: $v_k = x_k - \sum_{j<k} \langle q_j, x_k \rangle q_j$ and $q_k = v_k / \|v_k\|_2$.

The q_k are orthonormal with $\operatorname{span}\{x_1, \ldots, x_k\} = \operatorname{span}\{q_1, \ldots, q_k\}$. With $r_{jk} = \langle q_j, x_k \rangle$ ($j < k$), $r_{kk} = \|v_k\|_2$, $r_{jk} = 0$ ($j > k$), step (2) reads $x_k = \sum_{j\leq k} r_{jk} q_j$, i.e. $X = Q_1 R_1$.

Classical vs modified. Classical Gram–Schmidt (CGS) projects the original x_k; for nearly collinear columns the computed q's are far from orthogonal. **Modified** Gram–Schmidt (MGS) projects the *current* partially orthogonalized vector, one direction at a time:

$$v_k^{(1)} = x_k, \qquad v_k^{(j+1)} = v_k^{(j)} - \langle q_j, v_k^{(j)} \rangle q_j, \quad j = 1, \ldots, k-1, \qquad q_k = v_k^{(k)} / \|v_k^{(k)}\|.$$

They agree in exact arithmetic; in floating point MGS loses much less orthogonality, though not none. Both cost $\sum_k 4n(k-1) \approx 2np^2$ flops and give only the reduced QR.

Example 4.1 (Where CGS breaks)

Let $x_1 = (1, \epsilon, 0, 0)^\top$, $x_2 = (1, 0, \epsilon, 0)^\top$, $x_3 = (1, 0, 0, \epsilon)^\top$ with $[1 + \epsilon^2] = 1$. Then $r_{11} = [\sqrt{1+\epsilon^2}] = 1$ and $q_1 = x_1$.

- $k = 2$: $r_{12} = q_1^\top x_2 = 1$, $v_2 = (0, -\epsilon, \epsilon, 0)^\top$, $q_2 = (0, -1, 1, 0)^\top/\sqrt{2}$.
- $k = 3$, CGS: with the *original* x_3, $r_{13} = 1$ and $r_{23} = q_2^\top x_3 = 0$, so $v_3 = (0, -\epsilon, 0, \epsilon)^\top$, $q_3 = (0, -1, 0, 1)^\top/\sqrt{2}$, and $q_2^\top q_3 = \frac{1}{2}$: orthogonality is lost.
- $k = 3$, MGS: $v_3^{(2)} = (0, -\epsilon, 0, \epsilon)^\top$ and $q_2^\top v_3^{(2)} = \epsilon/\sqrt{2} \neq 0$, so $v_3 = (0, -\frac{\epsilon}{2}, -\frac{\epsilon}{2}, \epsilon)^\top$, $q_3 = (0, -1, -1, 2)^\top/\sqrt{6} \perp q_2$.

CGS rounds the tiny true $q_2^\top x_3$ away; MGS recomputes it from a vector where it is not negligible. MGS fails in turn for almost parallel columns, e.g. $x_1 = (0.7, 0.7 + \epsilon_{\text{mach}})^\top$, $x_2 = (1/\sqrt{2}, 1/\sqrt{2})^\top$: v_2 is a difference of nearly equal vectors whose direction is rounding noise (with two columns CGS does the same operations). Householder QR, which subtracts no projections, gives an orthogonal Q to machine precision in both cases.

4.3 Householder reflections

Motivation. Orthogonal reflections preserve lengths, so triangularizing X by them solves least squares without squaring the condition number.

Proposition 4.1 (Householder reflector)

For $x, w \in \mathbb{R}^n$ with $\|x\|_2 = \|w\|_2$ and $x \neq w$, let $u = (x - w)/\|x - w\|_2$ and $H = I - 2uu^\top$. Then H is symmetric and orthogonal and $Hx = w$.

Proof. $H^\top = H$ and $H^2 = I - 4uu^\top + 4u(u^\top u)u^\top = I$. With $x^\top x = w^\top w$, $\|x - w\|^2 = 2x^\top x - 2w^\top x$, so

$$Hx = x - 2\frac{(x-w)^\top x}{\|x-w\|^2}(x-w) = x - \frac{x^\top x - w^\top x}{x^\top x - w^\top x}(x-w) = w.$$

■

H reflects across the hyperplane $u^\perp$. Householder QR takes H_1 with $w = \pm\|x_1\|e_1$ to zero column 1 below the diagonal, then $H_2 = \text{diag}(1, \tilde{H}_2)$ on rows 2:n, and so on:

$$H_p \cdots H_2 H_1 X = \begin{pmatrix} R_1 \\ 0 \end{pmatrix}, \qquad Q = H_1 H_2 \cdots H_p.$$

This is the *full* QR. Take $w = -\text{sign}(x_1)\|x\|e_1$, so the first entry of $x - w$ is $x_1 + \text{sign}(x_1)\|x\|$, a sum of like-signed numbers; the other sign gives $x_1 - \text{sign}(x_1)\|x\|$, catastrophically small when $x \approx \|x\|e_1$, and u would be noise.

Implementation and cost. H_j is never formed: $HX = X - 2u(u^\top X)$ costs $4np$. At stage j, $\tilde{u}_j$ costs $3(n - j + 1)$ and applying $\tilde{H}_j$ to the $(n - j + 1) \times (p - j + 1)$ block $4(n - j + 1)(p - j + 1)$; in total

$$\sum_{j=1}^{p} 4(n - j + 1)(p - j + 1) \approx 2np^2 - \tfrac{2}{3}p^3 \text{ flops.}$$

In place: R_1 overwrites the upper triangle and the Householder vectors (first entry 1) are stored below. $Q^\top y$ or Qv apply the reflections in turn; forming Q costs another $2np^2 - \frac{2}{3}p^3$. Householder QR is backward stable; LAPACK `geqrf` (`geqp3` with column pivoting) and R's `qr` use it.

Least squares without forming Q. Apply each reflection to y when computed ($4(n-j+1)$ at stage j); y is overwritten by $Q^\top y = \binom{Q_1^\top y}{Q_2^\top y}$ with no $n \times n$ matrix stored. Then

- $\hat\beta$ solves $R_1\beta = Q_1^\top y$ (the first p entries), by back substitution;
- $\mathrm{RSS} = \|Q_2^\top y\|^2$ (last $n-p$ entries), $\hat\sigma^2 = \mathrm{RSS}/(n-p)$, or RSS/n for the MLE;
- $\widehat{\mathrm{Cov}}(\hat\beta) = \hat\sigma^2 R_1^{-1} R_1^{-\top}$ by triangular solves with the stored R_1.

Total: $2np^2 - \frac{2}{3}p^3 + O(np)$ flops, $O(np)$ memory.

4.4 Givens rotations

A Givens rotation $G(i,k,\theta)$ is the identity except for entries $c, s, -s, c$ in rows and columns i, k ($c = \cos\theta$, $s = \sin\theta$); it is orthogonal. For $y = G^\top x$,

$$y_i = cx_i - sx_k, \quad y_k = sx_i + cx_k, \quad y_j = x_j \ (j \neq i,k),$$

and $c = x_i/\sqrt{x_i^2 + x_k^2}$, $s = -x_k/\sqrt{x_i^2 + x_k^2}$ give $y_k = 0$. $G^\top$ changes only two rows ($6m$ flops). Givens QR zeros one entry at a time, $G_t^\top \cdots G_1^\top X = \binom{R_1}{0}$, in about $3np^2 - p^3$ flops: more than Householder, but suited to sparse or structured matrices (e.g. adding a row to a QR). Each rotation is stored as one number in the zeroed position.

4.5 Conditioning of least squares

For nonsquare X, $\kappa(X) = \sigma_{\max}(X)/\sigma_{\min}(X) = \sigma_{\max}(X)\sigma_{\max}(X^\dagger)$ ($X^\dagger$ the Moore–Penrose pseudoinverse). The condition number of the least squares *problem* is

$$\kappa(X) + \frac{\kappa(X)^2 \tan\theta}{\eta}, \qquad \cos\theta = \frac{\|X\beta\|}{\|y\|}, \quad \eta = \frac{\|X\|\|\beta\|}{\|X\beta\|}.$$

For nearly collinear X and non-small residuals the $\kappa(X)^2$ term dominates, so stable algorithms matter.

Example 4.2

$X = \begin{pmatrix} 1 & 1 \\ 10^{-3} & 0 \\ 0 & 10^{-3} \end{pmatrix}$ has full rank, but with six significant digits $X^\top X = \begin{pmatrix} 1+10^{-6} & 1 \\ 1 & 1+10^{-6} \end{pmatrix}$ rounds to the singular $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$; QR on X itself loses nothing.

method	flops	stability	notes
Cholesky on $X^\top X$	$np^2 + \frac{1}{3}p^3$	least stable	fastest; $X^\top X$ can be updated with streaming data
Householder QR	$2np^2 - \frac{2}{3}p^3$	stable	default in R, MATLAB, Julia
MGS	$2np^2$	stable-ish	gives Q_1 explicitly
SVD	$4n^2p + 8np^2 + 9p^3$	most stable	also handles rank deficiency

No method dominates: Cholesky is twice as fast for $n \gg p$ and needs only $X^\top X$ (which fits in memory when X may not); QR is more accurate.

4.6 Underdetermined least squares

If rank $X < p$ (e.g. $n < p$), the normal equations have infinitely many solutions and need an extra criterion.

- **Minimum ℓ_2 norm**: the solution of $X^\top X\beta = X^\top y$ with smallest $\|\beta\|_2$ is $\hat\beta = X^\dagger y$, equal to $X^\top(XX^\top)^{-1}y$ for full row rank, and the ridge limit $\lim_{\lambda\downarrow 0}(X^\top X + \lambda I)^{-1}X^\top y$.
- **Minimum ℓ_0 "norm"** $\|\beta\|_0 = \#\{j : \beta_j \neq 0\}$: the sparsest solution, i.e. best subset selection (combinatorial).
- **Minimum ℓ_1 norm**: a convex relaxation (basis pursuit, related to the lasso), a linear program that under suitable conditions recovers the sparsest solution.

4.7 Selected exercises

Motivation. Uniqueness of reduced QR, the minimum-norm solution as ridgeless limit, and leverages from Q_1.

Exercise 4.1 (Uniqueness of the reduced QR decomposition)

Let $X \in \mathbb{R}^{n\times p}$ have full column rank.
(a) Show that there is exactly one reduced QR decomposition $X = Q_1R_1$ with $Q_1^\top Q_1 = I_p$ and R_1 upper triangular with positive diagonal, and that Gram–Schmidt produces it.
(b) Householder QR may return an R_1 with negative diagonal entries. How is it converted to the form in (a)?

Solution. [Supplement]

(a) $X^\top X$ is positive definite ($v^\top X^\top Xv = \|Xv\|^2 > 0$ for $v \neq 0$). If $X = Q_1R_1$ as stated, $X^\top X = R_1^\top R_1$ with $R_1^\top$ lower triangular with positive diagonal, so by Theorem 3.1 $R_1^\top$ is the Cholesky factor, determined by X, and $Q_1 = XR_1^{-1}$ is determined too. Existence: Gram–Schmidt (Section 4.2) gives $X = Q_1R_1$ with $r_{kk} = \|v_k\|_2 > 0$ ($x_k \notin \operatorname{span}\{x_1, \dots, x_{k-1}\}$).

(b) Let $S = \operatorname{diag}(\operatorname{sign} r_{11}, \dots, \operatorname{sign} r_{pp})$, nonsingular with $S^2 = I$. Then $X = (Q_1S)(SR_1)$ with orthonormal columns and diagonal $|r_{kk}| > 0$, the unique decomposition of (a): the methods agree up to signs of columns of Q_1 and rows of R_1. ■

Exercise 4.2 (The minimum-norm solution)

Let $X \in \mathbb{R}^{n\times p}$ with rank $X = n < p$ and $y \in \mathbb{R}^n$, and put $\beta^\dagger = X^\top(XX^\top)^{-1}y$.
(a) Show that $\beta^\dagger$ solves the normal equations and is the unique solution of smallest Euclidean norm.
(b) Show that $(X^\top X + \lambda I_p)^{-1}X^\top y = X^\top(XX^\top + \lambda I_n)^{-1}y$ for $\lambda > 0$, and deduce that ridge regression converges to $\beta^\dagger$ as $\lambda \downarrow 0$.

Solution. [Supplement]

(a) $z^\top XX^\top z = \|X^\top z\|^2 > 0$ for $z \neq 0$, so $XX^\top$ is invertible. $X^\top$ is injective, so $X^\top X\beta = X^\top y \iff X\beta = y$. $X\beta^\dagger = y$, and every solution is $\beta = \beta^\dagger + d$ with $Xd = 0$. With $\beta^\dagger = X^\top z$, $z = (XX^\top)^{-1}y$, $\beta^{\dagger\top}d = z^\top Xd = 0$, so

$$\|\beta\|^2 = \|\beta^\dagger\|^2 + \|d\|^2 \geq \|\beta^\dagger\|^2,$$

with equality iff $d = 0$.

(b) For $\lambda > 0$ both $X^\top X + \lambda I_p$ and $XX^\top + \lambda I_n$ are positive definite. Multiplying $X^\top (XX^\top + \lambda I_n) = (X^\top X + \lambda I_p) X^\top$ by $(X^\top X + \lambda I_p)^{-1}$ on the left and $(XX^\top + \lambda I_n)^{-1}$ on the right,

$$X^\top (XX^\top + \lambda I_n)^{-1} = (X^\top X + \lambda I_p)^{-1} X^\top.$$

Apply to y. As inversion is continuous at $XX^\top$, the ridge solution tends to $\beta^\dagger$. ■

Exercise 4.3 (Leverages from the QR decomposition)

Let $X \in \mathbb{R}^{n \times p}$ have full column rank with reduced QR decomposition $X = Q_1 R_1$, and let $H = X(X^\top X)^{-1} X^\top$ be the hat matrix.
(a) Show that $H = Q_1 Q_1^\top$.
(b) Show that the leverage h_{ii} is the squared norm of the i-th row of Q_1, that $0 \le h_{ii} \le 1$, and that $\sum_i h_{ii} = p$.
(c) After a Householder QR, how can all leverages be computed without forming an $n \times n$ matrix?

Solution. [Supplement]

(a) $X^\top X = R_1^\top R_1$ with R_1 invertible, so $(X^\top X)^{-1} = R_1^{-1} R_1^{-\top}$ and

$$H = Q_1 R_1 R_1^{-1} R_1^{-\top} R_1^\top Q_1^\top = Q_1 Q_1^\top.$$

(b) $h_{ii} = e_i^\top Q_1 Q_1^\top e_i = \|Q_1^\top e_i\|^2 \ge 0$, the squared norm of row i of Q_1. With $Q = [Q_1, Q_2]$ orthogonal, $1 = \|Q^\top e_i\|^2 = \|Q_1^\top e_i\|^2 + \|Q_2^\top e_i\|^2 \ge h_{ii}$. And $\sum_i h_{ii} = \operatorname{tr}(Q_1 Q_1^\top) = \operatorname{tr}(Q_1^\top Q_1) = p$.

(c) $Q_1 = H_1 H_2 \cdots H_p \binom{I_p}{0}$: apply the stored reflections, last first, to $\binom{I_p}{0}$, at most $4np$ flops each, so $O(np^2)$ flops and $O(np)$ memory, plus $2np$ for the row norms; H is never formed. ■

Lecture 5. Iterative Methods for Linear Systems

5.1 Direct versus iterative methods

Direct methods (LU, Cholesky, QR, SVD) have a fixed cost and are accurate; they suit dense matrices of moderate size. **Iterative methods** (Jacobi, Gauss–Seidel, SOR, conjugate gradient, GMRES) produce approximations whose accuracy grows with the iterations; they suit very large, sparse or structured systems, parallel computing and warm starts.

Example 5.1 (PageRank)

Let $a_{ij} = 1$ if page i links to page j, and $r_i = \sum_j a_{ij}$. A random surfer at a page with $r_i > 0$ follows a random link with probability p and otherwise jumps to a uniform page; from a page without links it jumps uniformly. The transition probabilities are

$$p_{ij} = \frac{1-p}{n} + p\frac{a_{ij}}{r_i} \qquad (a_{ij}/r_i := 1/n \text{ if } r_i = 0),$$

and the scores are the stationary distribution: $x^\top P = x^\top$, i.e. $(I - P^\top)x = 0$. For $n \approx 10^9$, LU takes $\frac{2}{3}(10^9)^3/10^{12}$ seconds, about 20 million years at a teraflop; iterative methods need only products with the sparse link matrix. $I - P^\top$ is never formed: row i of $(I - P^\top)x$ is $x_i - \frac{1-p}{n}\sum_{j:r_j>0} x_j - \frac{1}{n}\sum_{j:r_j=0} x_j - p\sum_{j\to i} x_j/r_j$, the last sum over pages linking to i. With the two global sums kept as running totals, a Gauss–Seidel or SOR sweep costs $O(n+\mathrm{nnz}(A))$.

5.2 Splitting and convergence

Split $A = M - K$ with M invertible and easy to invert. Then $Ax = b$ iff $x = M^{-1}Kx + M^{-1}b$: a solution is a fixed point of

$$x^{(t+1)} = Rx^{(t)} + c, \qquad R = M^{-1}K, \quad c = M^{-1}b.$$

Norms. A vector norm satisfies $\|x\| \geq 0$, $\|x\| = 0 \iff x = 0$, $\|cx\| = |c|\|x\|$ and the triangle inequality (e.g. $\ell_1, \ell_2, \ell_\infty$); a matrix norm is also submultiplicative, $\|AB\| \leq \|A\|\|B\|$. Examples: Frobenius $\|A\|_F = \sqrt{\mathrm{tr}(AA^\top)}$ and the **operator norm** $\|A\| = \sup_{y\neq 0}\|Ay\|/\|y\|$, with

$$\|A\|_1 = \max_j \sum_i |a_{ij}|, \qquad \|A\|_\infty = \max_i \sum_j |a_{ij}|, \qquad \|A\|_2 = \sqrt{\rho(A^\top A)},$$

where $\rho(B) = \max_i|\lambda_i(B)|$ is the **spectral radius**.

If $Ax^\star = b$, then $x^{(t+1)} - x^\star = R(x^{(t)} - x^\star)$, so $\|R\| < 1$ for some operator norm implies convergence.

Theorem 5.1

For every operator norm, $\rho(A) \leq \|A\|$. Conversely, for every $\epsilon > 0$ there is an operator norm with $\|A\| \leq \rho(A) + \epsilon$. Hence $\rho(A) = \inf\|A\|$ over operator norms, and $\rho(A) < 1$ iff $\|A\| < 1$ for some operator norm.

Proof. If $Av = \lambda v$, $v \neq 0$ (complex in general), then $|\lambda|\|v\| = \|Av\| \leq \|A\|\|v\|$.

Conversely, let $A = STS^{-1}$ (Schur) with T upper triangular, eigenvalues on the diagonal, and $D_\delta = \mathrm{diag}(1, \delta, \ldots, \delta^{n-1})$. Then $T_\delta = D_\delta^{-1}TD_\delta = (SD_\delta)^{-1}A(SD_\delta)$ has entries $\delta^{j-i}t_{ij}$ $(j \geq i)$, with

off-diagonal part shrinking as $\delta \to 0$. $\|x\|_\delta = \|(SD_\delta)^{-1}x\|_\infty$ is a vector norm ($M = (SD_\delta)^{-1}$ is invertible, so $\|Mx\|_\infty = 0 \iff x = 0$ and the other axioms transfer). With $y = (SD_\delta)^{-1}x$, its induced norm is

$$\|A\|_\delta = \sup_{y \neq 0} \frac{\|(SD_\delta)^{-1}A(SD_\delta)y\|_\infty}{\|y\|_\infty} = \|T_\delta\|_\infty = \max_i \sum_{j \geq i} |t_{ij}| \delta^{j-i},$$

which is at most $\rho(A) + \epsilon$ for small δ. ■

Corollary 5.2

If A is invertible, the iteration $x^{(t+1)} = Rx^{(t)} + c$ converges to $A^{-1}b$ for every $x^{(0)}$ if and only if $\rho(R) < 1$.

Proof. If $\rho(R) < 1$, take $\|R\| < 1$ by Theorem 5.1: $\|x^{(t)} - x^\star\| \leq \|R\|^t \|x^{(0)} - x^\star\| \to 0$ (this direction does not need A invertible). Conversely, if $Rv = \lambda v$ with $|\lambda| \geq 1$, start at $x^{(0)} = x^\star + v$: $\|x^{(t)} - x^\star\| = |\lambda|^t \|v\| \geq \|v\| \neq 0$. ■

5.3 Classical iterations

Write $A = L + D + U$ (strictly lower, diagonal, strictly upper parts).

Jacobi. $M = D$, $K = -(L + U)$:

$$x_i^{(t+1)} = \frac{1}{a_{ii}} \Big(b_i - \sum_{j \neq i} a_{ij} x_j^{(t)} \Big), \qquad x^{(t+1)} = x^{(t)} + D^{-1}(b - Ax^{(t)}).$$

Gauss–Seidel. $M = D + L$ (forward substitution), $K = -U$: new components are used immediately,

$$x_i^{(t+1)} = \frac{1}{a_{ii}} \Big(b_i - \sum_{j<i} a_{ij} x_j^{(t+1)} - \sum_{j>i} a_{ij} x_j^{(t)} \Big).$$

This is coordinate descent for linear systems. In code: writing each new x_i back into x before the next component is Gauss–Seidel; replacing x only after the sweep is Jacobi. Gauss–Seidel needs no second vector but is sequential; Jacobi vectorizes and parallelizes.

Successive over-relaxation (SOR). Weight the Gauss–Seidel update against the old value:

$$x_i^{(t+1)} = (1 - \omega) x_i^{(t)} + \frac{\omega}{a_{ii}} \Big(b_i - \sum_{j<i} a_{ij} x_j^{(t+1)} - \sum_{j>i} a_{ij} x_j^{(t)} \Big),$$

i.e. $M = \frac{1}{\omega} D + L$, $K = \frac{1-\omega}{\omega} D - U$. $\omega = 1$ is Gauss–Seidel; over-relaxation $\omega > 1$ can accelerate convergence substantially.

Proposition 5.3

If A is strictly diagonally dominant by rows, $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$, then the Jacobi and Gauss–Seidel iterations converge for every starting point.

Proof. *Jacobi.* $R = -D^{-1}(L+U)$ has $\|R\|_\infty = \max_i \sum_{j\neq i}|a_{ij}|/|a_{ii}| < 1$. *Gauss–Seidel.* Let $Rv = \lambda v$, $R = -(D+L)^{-1}U$, so $-Uv = \lambda(D+L)v$, and take $|v_i| = \|v\|_\infty = 1$. Row i gives $\lambda a_{ii}v_i = -\sum_{j>i} a_{ij}v_j - \lambda\sum_{j<i} a_{ij}v_j$, so

$$|\lambda|\Big(|a_{ii}| - \sum_{j<i}|a_{ij}|\Big) \le \sum_{j>i}|a_{ij}|,$$

and the right side is less than the bracket by dominance. So $|\lambda| < 1$. ■

Jacobi also converges for strict *column* dominance $|a_{jj}| > \sum_{i\neq j}|a_{ij}|$: $R = -D^{-1}(L+U)$ is similar to $DRD^{-1} = -(L+U)D^{-1}$, whose $\|\cdot\|_1$ is $\max_j \sum_{i\neq j}|a_{ij}|/|a_{jj}| < 1$.

Example 5.2 (Jacobi for the second-difference matrix)

Let $A = \text{tridiag}(-1, 2, -1) \in \mathbb{R}^{n\times n}$, with eigenvalues $\lambda_i = 2 - 2\cos\frac{i\pi}{n+1}$ and orthonormal eigenvectors v_i. Jacobi has $D = 2I$, $R = I - \frac{1}{2}A$, with the same eigenvectors and eigenvalues

$$\mu_i = 1 - \tfrac{1}{2}\lambda_i = \cos\frac{i\pi}{n+1}, \qquad \rho(R) = \cos\frac{\pi}{n+1} < 1.$$

It converges slowly: $\rho(R) \approx 1 - \frac{\pi^2}{2(n+1)^2}$, so a constant-factor error reduction takes $O(n^2)$ sweeps. With $\varepsilon^{(k)} = \sum_i a_i^{(k)} v_i$, $a_i^{(k)} = \mu_i^k a_i^{(0)}$: each mode decays at rate $|\mu_i|$. Modes near $i = (n+1)/2$ ($\mu_i \approx 0$) die at once; the smoothest ($i = 1$) and most oscillatory ($i = n$, $\mu_n = -\mu_1$) persist. This motivates multigrid, which treats smooth error on coarser grids.

Cost. A dense sweep costs $2n^2$ flops, so iteration beats LU only with $o(n)$ sweeps. The gain is for sparse or structured A, where Av costs $O(n)$ or $O(n\log n)$ and A need not be stored (e.g. $(Av)_i = v_i - v_{i+1}$).

5.4 Conjugate gradient

For symmetric positive definite A, $Ax = b$ is equivalent to minimizing $\frac{1}{2}x^\top Ax - b^\top x$ (gradient $Ax - b$). **Conjugate gradient** minimizes along mutually A-conjugate directions; in exact arithmetic it terminates in at most n steps, and with preconditioning it converges far sooner. It is the method of choice for huge structured positive definite systems (in a classical plasma physics example: Gauss–Seidel 208,000 iterations, block SOR 765, incomplete-Cholesky-preconditioned CG 25). Details belong to an advanced course.

On the Poisson matrix (sparse, symmetric positive definite, block tridiagonal) Jacobi needs thousands of iterations, Gauss–Seidel and SOR fewer, CG far fewer, all much cheaper than dense Cholesky: *exploit structure.*

5.5 Selected exercises

Motivation. The eigenvalues of Example 5.2, the SOR range $0 < \omega < 2$, Gauss–Seidel for positive definite systems, and finite termination of conjugate directions.

Exercise 5.1 (Eigenpairs of the second-difference matrix)

Let $A = \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{n\times n}$. For $i = 1, \dots, n$ let $\theta_i = i\pi/(n+1)$ and $v_i \in \mathbb{R}^n$ with entries $(v_i)_j = \sin(j\theta_i)$. Show that $Av_i = (2 - 2\cos\theta_i)v_i$, that these are all the eigenvalues of A, that the v_i are mutually orthogonal, and that A is positive definite.

Solution. [Supplement] Fix i, $\theta = \theta_i$, $s_j = \sin(j\theta)$, $j = 0, \dots, n+1$. As $s_0 = s_{n+1} = 0$, $(Av_i)_j = -s_{j-1} + 2s_j - s_{j+1}$ for all $j = 1, \dots, n$, and $\sin((j-1)\theta) + \sin((j+1)\theta) = 2\sin(j\theta)\cos\theta$ gives

$$(Av_i)_j = 2s_j - 2s_j\cos\theta = (2 - 2\cos\theta)s_j.$$

$v_i \neq 0$ as $s_1 = \sin\theta \neq 0$. cos is strictly decreasing on $(0, \pi)$, so the n values λ_i are distinct, hence all the eigenvalues, and the v_i are orthogonal ($\lambda_i v_k^\top v_i = v_k^\top A v_i = (Av_k)^\top v_i = \lambda_k v_k^\top v_i$). Finally $\lambda_i > 0$ as $\cos\theta_i < 1$. ■

Exercise 5.2 (SOR requires $0 < \omega < 2$)

Let A have nonzero diagonal entries and let $R_\omega = M^{-1}K$ with $M = \frac{1}{\omega}D + L$, $K = \frac{1-\omega}{\omega}D - U$ be the SOR iteration matrix ($\omega \neq 0$). Show that $\rho(R_\omega) \geq |\omega - 1|$, and conclude that SOR converges for every starting point only if $0 < \omega < 2$.

Solution. [Supplement] M is lower triangular with diagonal D/ω and K upper triangular with diagonal $(1-\omega)D/\omega$, so

$$\det R_\omega = \frac{\det K}{\det M} = \frac{\prod_i (1-\omega)a_{ii}/\omega}{\prod_i a_{ii}/\omega} = (1-\omega)^n.$$

Each of the n eigenvalues has modulus at most $\rho(R_\omega)$, so $|1-\omega|^n \leq \rho(R_\omega)^n$. For $\omega \leq 0$ or $\omega \geq 2$, $\rho(R_\omega) \geq 1$, and Corollary 5.2 gives non-convergence for some start. ■

Exercise 5.3 (Gauss–Seidel converges for positive definite systems)

Let A be symmetric positive definite. Show that the Gauss–Seidel iteration converges for every starting point.

Solution. [Supplement] Here $U = L^\top$, $R = -(D+L)^{-1}L^\top$; by Corollary 5.2 show $|\lambda| < 1$.

Step 1. Let $Rv = \lambda v$, $v \in \mathbb{C}^n \setminus \{0\}$, so $-L^\top v = \lambda(D+L)v$. Multiply by v^*, with $\delta = v^*Dv$, $\sigma = v^*Lv$; L real gives $v^*L^\top v = \bar{\sigma}$, so

$$-\bar{\sigma} = \lambda(\delta + \sigma).$$

Step 2. $\delta > 0$ as $a_{ii} > 0$. With $v = x + iy$, $v^*Av = x^\top Ax + y^\top Ay > 0$, i.e. $a := \delta + 2\operatorname{Re}\sigma > 0$. And $\delta + \sigma \neq 0$, else $\bar{\sigma} = 0$, so $\sigma = \delta = 0$.

Step 3. With $\sigma = s + it$,

$$|\lambda|^2 = \frac{|\sigma|^2}{|\delta + \sigma|^2} = \frac{s^2 + t^2}{(\delta + s)^2 + t^2},$$

which is < 1 iff $s^2 < (\delta + s)^2$ iff $\delta(\delta + 2s) > 0$, true as $\delta > 0$ and $\delta + 2s = a > 0$. ■

Exercise 5.4 (Conjugate directions terminate in n steps)

Let A be symmetric positive definite, $\varphi(x) = \frac{1}{2}x^\top Ax - b^\top x$, and let $d_0, \dots, d_{n-1}$ be nonzero and A-conjugate: $d_i^\top A d_j = 0$ for $i \neq j$. From any x_0 define

$$x_{k+1} = x_k + \alpha_k d_k, \qquad \alpha_k = \frac{d_k^\top r_k}{d_k^\top A d_k}, \qquad r_k = b - Ax_k.$$

(a) Show that $d_0, \dots, d_{n-1}$ are linearly independent.
(b) Show that α_k minimizes $\alpha \mapsto \varphi(x_k + \alpha d_k)$.
(c) Show that $d_j^\top r_k = 0$ for all $j < k$, and conclude that $x_n = A^{-1}b$.

Solution. [Supplement]

(a) $\sum_i c_i d_i = 0$ times $d_j^\top A$ gives $c_j d_j^\top A d_j = 0$, so $c_j = 0$.

(b) $\nabla\varphi = -r$, so $\varphi(x_k + \alpha d_k) = \varphi(x_k) - \alpha d_k^\top r_k + \frac{\alpha^2}{2} d_k^\top A d_k$, strictly convex in α with derivative zero at $\alpha = \alpha_k$.

(c) Induction on k ($k = 0$ is vacuous). As $r_{k+1} = r_k - \alpha_k A d_k$,

$$d_k^\top r_{k+1} = d_k^\top r_k - \alpha_k d_k^\top A d_k = 0, \qquad d_j^\top r_{k+1} = d_j^\top r_k - \alpha_k d_j^\top A d_k = d_j^\top r_k = 0 \quad (j < k),$$

by the choice of α_k, conjugacy and induction. For $k = n$, r_n is orthogonal to n independent vectors by (a), so $r_n = 0$.

Conjugate gradient generates the d_k on the fly by A-conjugating successive residuals, with one product Ad_k per step. ■

Lecture 6. Nonlinear Equations

We seek x with $f(x) = 0$ for nonlinear f; there is no direct method, so we iterate. A critical point of g is a root of $f = g'$ (or ∇g), so root finding covers maximum likelihood, nonlinear least squares, Bayes and MAP estimation. Examples: a root of $f(x) = x^4 - \sin x$; the maximizer of $g(x) = \log x/(1+x)$, a root of $g'(x) = \{1 + x^{-1} - \log x\}/(1+x)^2$.

6.1 Bisection

If f is continuous on $[l, r]$ with $f(l)f(r) \le 0$, there is a root in $[l, r]$. At $m = (l+r)/2$: stop if $f(m) = 0$; else replace r by m if $f(l)f(m) < 0$, and l by m otherwise. After n steps the bracket has length $2^{-n}(r^{(0)} - l^{(0)})$. Bisection gains one bit per step but always works. Typical use: quantiles, $F(x) - u = 0$ for a continuous cdf F; if $F = \int_{-\infty}^{x} f$ needs numerical integration (Lecture 8), this gives the inverse cdf method of Lecture 9 when F^{-1} has no closed form.

6.2 Fixed-point iteration

x is a **fixed point** of F if $F(x) = x$. Choose F with $f(x) = 0 \iff F(x) = x$ and iterate $x^{(k+1)} = F(x^{(k)})$ (Lecture 5 is the case $F(x) = Rx + c$).

Theorem 6.1 (Contraction mapping)

Let $I \subset \mathbb{R}$ be a closed interval and $F : I \to I$ with $|F(x) - F(y)| \le L|x - y|$ for all $x, y \in I$, where $0 \le L < 1$. Then F has a unique fixed point $x^\star \in I$, the iteration converges to it from every $x^{(0)} \in I$, and

$$|x^{(n)} - x^\star| \le \frac{L^n}{1-L}|x^{(1)} - x^{(0)}|.$$

Proof. $|x^{(k+1)} - x^{(k)}| = |F(x^{(k)}) - F(x^{(k-1)})| \le L|x^{(k)} - x^{(k-1)}| \le \cdots \le L^k|x^{(1)} - x^{(0)}|$. For $m > n$, by the triangle inequality,

$$|x^{(m)} - x^{(n)}| \le \sum_{k=n}^{m-1} L^k|x^{(1)} - x^{(0)}| \le \frac{L^n}{1-L}|x^{(1)} - x^{(0)}|. \qquad (*)$$

So the sequence is Cauchy, with limit $x^\star \in I$ (I closed), and continuity gives $x^\star = \lim F(x^{(k)}) = F(x^\star)$. Another fixed point $y^\star$ would satisfy $|x^\star - y^\star| \le L|x^\star - y^\star|$, so $x^\star = y^\star$. Let $m \to \infty$ in $(*)$. ■

The geometric rate L^n is **linear convergence**.

Proposition 6.2

If F is differentiable on $I = [a, b]$, then $|F(x) - F(y)| \le L|x - y|$ for all $x, y \in I$ if and only if $|F'(x)| \le L$ for all $x \in I$. Hence the best Lipschitz constant is $\sup_I |F'|$.

Proof. ($\Rightarrow$) Difference quotients at x are bounded by L, hence so is their limit $F'(x)$.

($\Leftarrow$) Mean value theorem: $|F(x) - F(y)| = |F'(c)||x - y| \le L|x - y|$. ■

Example 6.1 (Relaxed fixed-point iteration)

To solve $f(x) = 0$, iterate $F(x) = x + \alpha f(x)$, $\alpha \neq 0$. By Proposition 6.2 and Theorem 6.1 it converges if F maps an interval I around the root into itself and $\sup_I |1 + \alpha f'| < 1$. This needs f' of constant sign; if $0 < m \leq f' \leq M$ on I, every $\alpha \in (-2/M, 0)$ works and $\alpha = -2/(m + M)$ gives the best $L = (M - m)/(M + m)$. Newton's method is the adaptive choice $\alpha = -1/f'(x^{(t)})$, making $F'(x^\star) = 0$. For $f = g'$ and $\alpha = -\gamma$ this is gradient descent $x^{(t+1)} = x^{(t)} - \gamma g'(x^{(t)})$ (Lecture 7), and the condition is the step-size bound $\gamma < 2/\sup g''$.

Example 6.2

For $f(x) = x^4 - \sin x$ on $(0, 1]$, both $F_1(x) = (\sin x)^{1/4}$ and $F_2(x) = \sin x / x^3$ fix the root. F_1 maps $I = [0.2, 1]$ into itself and $F_1'(x) = \cos x / \{4(\sin x)^{3/4}\}$ is decreasing with $0 < F_1'(1) < F_1'(0.2) < 1$: a contraction, so the iteration converges. F_2 maps $[0.9, 1]$ onto $[0.841, 1.075]$ with $F_2'(1) < -1$: it expands and diverges. The choice of F matters.

6.3 Newton's method

Motivation. Newton's method replaces f by its tangent, doubling the correct digits near a simple root, at the price of derivatives and possible divergence.

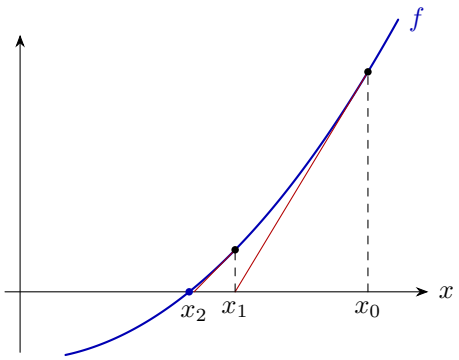


Figure. *Newton's method: $x_{k+1} = x_k - f(x_k)/f'(x_k)$ is the zero of the tangent at x_k.*

Linearizing, $0 = f(x^\star) \approx f(x^{(k)}) + f'(x^{(k)})(x^\star - x^{(k)})$, suggests

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}.$$

i.e. fixed-point iteration with $F(x) = x - f(x)/f'(x)$, for which

$$F'(x) = \frac{f(x) f''(x)}{f'(x)^2}.$$

Theorem 6.3 (Local quadratic convergence)

Let $f(x^\star) = 0$, $f'(x^\star) \neq 0$, and f be three times continuously differentiable near $x^\star$. Then for $x^{(0)}$ close enough to $x^\star$, Newton's iterates converge to $x^\star$ and

$$\lim_{k \to \infty} \frac{|x^{(k+1)} - x^\star|}{|x^{(k)} - x^\star|^2} = \frac{1}{2}|F''(x^\star)| = \frac{|f''(x^\star)|}{2|f'(x^\star)|}.$$

Proof. $F'(x^\star) = 0$, so $|F'| \le \frac{1}{2}$ on a closed neighbourhood I of $x^\star$, which F maps into itself; Theorem 6.1 gives convergence. By Taylor,

$$x^{(k+1)} - x^\star = F(x^{(k)}) - F(x^\star) = F'(x^\star)(x^{(k)} - x^\star) + \tfrac{1}{2}F''(\tilde{x})(x^{(k)} - x^\star)^2 = \tfrac{1}{2}F''(\tilde{x})(x^{(k)} - x^\star)^2,$$

$\tilde{x}$ between $x^{(k)}$ and $x^\star$. Divide and let $k \to \infty$; differentiating F' gives $F''(x^\star) = f''(x^\star)/f'(x^\star)$. ■

Quadratic convergence roughly doubles the correct digits per step, but only *locally*: from a poor start Newton's method can wander, reach an unintended root, or diverge (e.g. roots of $\cos x$). In practice it is safeguarded by bisection or line search.

6.4 Systems of equations

For $f : \mathbb{R}^n \to \mathbb{R}^n$ with Jacobian $Jf(x) = (\partial f_i/\partial x_j)$, Newton's method is

$$x^{(k+1)} = x^{(k)} - [Jf(x^{(k)})]^{-1} f(x^{(k)}),$$

solving $Jf(x^{(k)})\Delta x = -f(x^{(k)})$ (e.g. by LU), never inverting. Cheaper variants replace the Jacobian:

- modified Newton: the fixed $J = Jf(x^{(0)})$ (factor once);
- Jacobi: $\operatorname{diag} Jf(x^{(k)})$;
- Gauss–Seidel: the lower triangle of $Jf(x^{(k)})$;
- SOR: $\frac{1}{\omega}\operatorname{diag} Jf(x^{(k)})$ plus the strict lower triangle.

They converge more slowly (typically linearly) with much cheaper steps.

6.5 Selected exercises

Motivation. Global convergence for convex increasing f, the multiple-root case, and an application with exact error squaring.

Exercise 6.1 (Global convergence for convex increasing functions)

Let $f : \mathbb{R} \to \mathbb{R}$ be continuously differentiable and convex with $f' > 0$, and let $f(x^\star) = 0$.
(a) Show that from every $x^{(0)}$ the Newton iterates satisfy $x^{(k)} \ge x^\star$ for $k \ge 1$, decrease monotonically from $k = 1$ on, and converge to $x^\star$.
(b) Apply this to compute $\log a$ $(a > 0)$ without logarithms.

Solution. [Supplement]

(a)

Step 1. (Overshoot.) Convexity gives $f(y) \ge f(x) + f'(x)(y - x)$; with $x = x^{(k)}$, $y = x^{(k+1)}$ the right side is 0, so $f(x^{(k+1)}) \ge 0 = f(x^\star)$, and $x^{(k+1)} \ge x^\star$ for $k \ge 0$ as f is strictly increasing.

Step 2. (Monotonicity.) For $k \ge 1$, $f(x^{(k)}) \ge 0$ and $f'(x^{(k)}) > 0$, so $x^{(k+1)} \le x^{(k)}$.

Step 3. (Limit.) $(x^{(k)})_{k\ge1}$ is nonincreasing and bounded below by $x^\star$, so $x^{(k)} \to \ell \ge x^\star$. By continuity of f, f' and $f'(\ell) > 0$, $\ell = \ell - f(\ell)/f'(\ell)$, so $f(\ell) = 0$ and $\ell = x^\star$ (unique root).

(b) $f(x) = e^x - a$ is convex, $f' = e^x > 0$, root $\log a$:

$$x^{(k+1)} = x^{(k)} - \frac{e^{x^{(k)}} - a}{e^{x^{(k)}}} = x^{(k)} - 1 + a\,e^{-x^{(k)}},$$

converging to $\log a$ from every start by (a), quadratically by Theorem 6.3 ($f'(\log a) = a \ne 0$). ■

Exercise 6.2 (Newton's method at a multiple root)

Let $f(x) = (x - x^\star)^m g(x)$ with an integer $m \geq 2$, g continuously differentiable near $x^\star$ and $g(x^\star) \neq 0$.
(a) Show that Newton's method converges to $x^\star$ from every point close enough to $x^\star$, but only linearly: $\lim_k (x^{(k+1)} - x^\star)/(x^{(k)} - x^\star) = (m-1)/m$.
(b) Show that the modified iteration $x^{(k+1)} = x^{(k)} - m f(x^{(k)})/f'(x^{(k)})$ satisfies $|x^{(k+1)} - x^\star| \leq C|x^{(k)} - x^\star|^2$ near $x^\star$.

Solution. [Supplement] Write $e = x - x^\star$. Then $f'(x) = e^{m-1}\{mg(x) + eg'(x)\}$ with bracket $\to mg(x^\star) \neq 0$, so nonzero on some $|e| < \delta_0$, where $f'(x) \neq 0$ for $x \neq x^\star$ and

$$\frac{f(x)}{f'(x)} = \frac{e\, g(x)}{mg(x) + eg'(x)}.$$

(An iterate equal to $x^\star$ stops at the root.)

(a) The Newton map satisfies

$$F(x) - x^\star = e\Big(1 - \frac{g(x)}{mg(x) + eg'(x)}\Big) = e\,\varphi(x), \qquad \varphi(x) = \frac{(m-1)g(x) + eg'(x)}{mg(x) + eg'(x)}.$$

φ is continuous with $\varphi(x^\star) = (m-1)/m < 1$, so $|\varphi| \leq q < 1$ on some $|e| < \delta \leq \delta_0$. There $|e_{k+1}| \leq q|e_k|$, the iterates stay and $e_k \to 0$, and $e_{k+1}/e_k = \varphi(x^{(k)}) \to (m-1)/m$.

(b) For $G(x) = x - mf(x)/f'(x)$,

$$G(x) - x^\star = e\Big(1 - \frac{mg(x)}{mg(x) + eg'(x)}\Big) = e^2\,\psi(x), \qquad \psi(x) = \frac{g'(x)}{mg(x) + eg'(x)},$$

with ψ continuous, hence bounded by some C on a closed interval around $x^\star$. So $|e_{k+1}| \leq Ce_k^2$: quadratic convergence for $C|e_0| < 1$ (e_0 in that interval). This needs m; otherwise apply Newton to f/f', which has a simple root at $x^\star$. ■

Exercise 6.3 (Division by multiplication)

Let $a > 0$ and apply Newton's method to $f(x) = 1/x - a$.
(a) Show that the iteration is $x^{(k+1)} = x^{(k)}(2 - ax^{(k)})$, which uses no division.
(b) With $r_k = 1 - ax^{(k)}$, show that $r_{k+1} = r_k^2$.
(c) Conclude that $x^{(k)} \to 1/a$ if and only if $0 < x^{(0)} < 2/a$, that the convergence is then quadratic, and describe what happens otherwise.

Solution. [Supplement]

(a) $f'(x) = -1/x^2$, so $x - f(x)/f'(x) = x + x^2(1/x - a) = 2x - ax^2$.

(b) $1 - ax(2 - ax) = 1 - 2ax + a^2x^2 = (1 - ax)^2$.

(c) By (b), $r_k = r_0^{2^k}$ and $x^{(k)} = (1 - r_k)/a$. If $|r_0| < 1$, i.e. $0 < x^{(0)} < 2/a$, then $x^{(k)} \to 1/a$ with

$$\Big|x^{(k+1)} - \frac{1}{a}\Big| = \frac{r_k^2}{a} = a\Big|x^{(k)} - \frac{1}{a}\Big|^2,$$

so the correct bits double exactly. If $|r_0| = 1$ ($x^{(0)} \in \{0, 2/a\}$), $x^{(k)} = 0$ for $k \geq 1$; if $|r_0| > 1$, $x^{(k)} \to -\infty$. Division routines use this iteration after a table lookup gives a start with small $|r_0|$. ■

Lecture 7. Optimization

We consider $\min_{x\in\mathbb{R}^d} f(x)$; maximization is minimization of $-f$.

7.1 Golden section search

For $d = 1$, let f be continuous and **strictly unimodal** on $[a, b]$: $f(\lambda x+(1-\lambda)y) < \max\{f(x), f(y)\}$ for $x \neq y$, $\lambda \in (0, 1)$. Keep $a < x_1 < b$ with $f(x_1) < \min\{f(a), f(b)\}$, so the minimum is in (a, b), and place x_0 in the longer of (a, x_1), (x_1, b); say $a < x_0 < x_1$.

- If $f(x_0) > f(x_1)$, the minimum is in (x_0, b): set $(a, x_1, b) \leftarrow (x_0, x_1, b)$.
- Otherwise it is in (a, x_1): set $(a, x_1, b) \leftarrow (a, x_0, x_1)$.

For equally good outcomes and self-similarity, require $b - x_0 = x_1 - a$ and that the interior point divide the new interval in the same proportion. With $\alpha = (x_0 - a)/(b - a)$, $y = (x_1 - x_0)/(b - a)$: $\alpha + y + \alpha = 1$ and $1 : \alpha = (y + \alpha) : y$, so

$$\alpha = \frac{3-\sqrt{5}}{2}, \qquad 1-\alpha = \frac{\sqrt{5}-1}{2} \approx 0.618,$$

the golden section. Each step shrinks the interval by 0.618 with one evaluation: linear, but guaranteed to find the global minimum of a unimodal function.

7.2 Newton's method

At a minimizer $\nabla f(x^\star) = 0$. Newton–Raphson on ∇f minimizes the local quadratic model

$$f(x) \approx f(x^{(t)}) + \nabla f(x^{(t)})^\top (x - x^{(t)}) + \tfrac{1}{2}(x - x^{(t)})^\top \nabla^2 f(x^{(t)})(x - x^{(t)}),$$

giving the **naive Newton** iteration

$$x^{(t+1)} = x^{(t)} - [\nabla^2 f(x^{(t)})]^{-1} \nabla f(x^{(t)}).$$

It converges quadratically near a minimizer with positive definite Hessian, but it is not a descent method: it can converge to a maximum or saddle, or diverge (try $f = \sin$).

7.3 Practical Newton: positive definite approximations and line search

Motivation. A positive definite Hessian substitute and a line search keep Newton's fast local convergence while guaranteeing descent.

Use a positive definite $H^{(t)}$ and a step size s:

$$x^{(t+1)} = x^{(t)} + s\,\Delta x^{(t)}, \qquad \Delta x^{(t)} = -[H^{(t)}]^{-1} \nabla f(x^{(t)}).$$

Proposition 7.1 (Descent direction)

If $H^{(t)}$ is positive definite and $\nabla f(x^{(t)}) \neq 0$, then $f(x^{(t)} + s\Delta x^{(t)}) < f(x^{(t)})$ for all sufficiently small $s > 0$.

Proof. By Taylor, $f(x^{(t)} + s\Delta x^{(t)}) - f(x^{(t)}) = -s\,\nabla f^\top [H^{(t)}]^{-1} \nabla f + o(s)$ with a positive quadratic form. ∎

($\{\nabla f^\top H^{-1} \nabla f\}^{1/2}$ is the **Newton decrement**.) For strictly convex f the true Hessian works (**damped Newton**), still with a line search for global convergence; $H^{(t)} = I$ is gradient descent.

Line search. Choose s exactly (minimize $f(x+s\Delta x)$, e.g. golden section) or approximately by step halving or **backtracking** (Armijo): with $\alpha \in (0, \frac{1}{2})$, $\beta \in (0,1)$, start at $s=1$ and set $s \leftarrow \beta s$ while

$$f(x+s\Delta x) > f(x) + \alpha s \nabla f(x)^\top \Delta x.$$

This accepts s once f is below a line with slope a fraction α of the initial slope, which holds for small s by Proposition 7.1; it costs only function evaluations. A common slip: with the trial point $x_s = x + s\Delta x$ formed first, the bound is $f(x) + \alpha \nabla f(x)^\top (x_s - x)$, which already contains s; writing $\alpha s \nabla f(x)^\top (x_s - x)$ counts s twice and is too lenient. (For IRLS, the step-s trial point is the weighted least squares fit to $z = \eta + sW^{-1}(y-\mu)$, which is linear in z and returns $\beta + s\Delta\beta$.)

7.4 Fisher scoring

For maximum likelihood, $f = -\ell(\theta)$. **Fisher scoring** replaces $-\nabla^2 \ell$ by the expected information

$$I(\theta) = E[-\nabla^2 \ell(\theta)] = E[\nabla \ell(\theta) \nabla \ell(\theta)^\top] \succeq 0,$$

(equality when differentiation and expectation interchange), giving

$$\theta^{(t+1)} = \theta^{(t)} + s[I(\theta^{(t)})]^{-1} \nabla \ell(\theta^{(t)}).$$

Example 7.1 (Logistic regression)

$y_i \sim \mathrm{Ber}(p_i)$ with $\operatorname{logit} p_i = \eta_i = x_i^\top \beta$. Then

$$\ell(\beta) = \sum_i \{y_i x_i^\top \beta - \log(1 + e^{x_i^\top \beta})\}, \qquad \nabla \ell = X^\top (y - p),$$

$$-\nabla^2 \ell = X^\top W X, \qquad W = \mathrm{diag}\{p_i(1-p_i)\}.$$

The Hessian is free of y, so $I(\beta) = -\nabla^2 \ell$ and Newton = Fisher scoring. Rearranged,

$$\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} z^{(t)}, \qquad z^{(t)} = X\beta^{(t)} + s(W^{(t)})^{-1}(y - p^{(t)}),$$

a weighted least squares fit to the **working responses** $z^{(t)}$: **iteratively reweighted least squares** (IRLS). Any weighted least squares solver fits logistic regression.

Example 7.2 (Poisson regression)

$y_i \sim \mathrm{Poi}(\lambda_i)$, $\log \lambda_i = x_i^\top \beta$: $\ell(\beta) = \sum_i \{y_i x_i^\top \beta - e^{x_i^\top \beta} - \log y_i!\}$, $\nabla \ell = X^\top (y - \lambda)$, $-\nabla^2 \ell = X^\top W X$, $W = \mathrm{diag}(\lambda_i)$; again Newton = Fisher scoring = IRLS with $z = X\beta + W^{-1}(y - \lambda)$.

7.5 Generalized linear models

An **exponential dispersion family** has density

$$p(y \mid \eta, \phi) = \exp\left\{\frac{y\eta - b(\eta)}{a(\phi)} + c(y, \phi)\right\},$$

with natural parameter η, mean $\mu = b'(\eta)$, variance $b''(\eta)a(\phi)$. A **GLM** sets $g(\mu) = x^\top \beta$; the **canonical link** $g = (b')^{-1}$ makes $\eta = x^\top \beta$.

family	canonical link	variance function
normal	$\eta = \mu$	1
Poisson	$\eta = \log \mu$	μ
binomial	$\eta = \log \frac{\mu}{1-\mu}$	$\mu(1-\mu)$
gamma	$\eta = \mu^{-1}$	μ^2
inverse Gaussian	$\eta = \mu^{-2}$	μ^3

Proposition 7.2

With $a(\phi) \equiv 1$, $\mu_i = g^{-1}(x_i^\top \beta)$ and $\sigma_i^2 = b''(\eta_i)$:

$$\nabla \ell(\beta) = \sum_i \frac{(y_i - \mu_i)}{\sigma_i^2 g'(\mu_i)} x_i, \qquad I(\beta) = \sum_i \frac{1}{\sigma_i^2 g'(\mu_i)^2} x_i x_i^\top = X^\top W X.$$

Fisher scoring is IRLS with weights $w_i = 1/\{\sigma_i^2 g'(\mu_i)^2\}$. For the canonical link, $-\nabla^2 \ell = I(\beta)$, so Newton = Fisher scoring = IRLS and $-\ell$ is convex.

Proof. $\ell_i = y_i \eta_i - b(\eta_i)$, so $\partial \ell_i / \partial \eta_i = y_i - \mu_i$. As $d\mu = b''(\eta) d\eta = \sigma^2 d\eta$ and $d\mu = x^\top d\beta / g'(\mu)$, $\partial \eta_i / \partial \beta = x_i / \{\sigma_i^2 g'(\mu_i)\}$: the score. Differentiating again gives $-x_i x_i^\top / \{\sigma_i^2 g'(\mu_i)^2\}$ plus a term proportional to $y_i - \mu_i$ (mean 0): $I(\beta)$. For the canonical link $\partial \eta_i / \partial \beta = x_i$ and $\nabla^2 \ell = -\sum_i b''(\eta_i) x_i x_i^\top = -I(\beta)$, free of y_i ($\sigma_i^2 g'(\mu_i) = 1$). ■

For non-canonical links (e.g. probit) Fisher scoring and Newton differ and the likelihood need not be concave (probit happens to be log-concave). R's `glm` uses Fisher scoring.

Example 7.3 (Probit regression)

$y_i \sim \text{Ber}(p_i)$, $p_i = \Phi(\eta_i)$, $\eta_i = \beta_0 + \beta_1 x_i$; write $\Phi_i = \Phi(\eta_i)$, $\phi_i = \phi(\eta_i)$, $\mathbf{x}_i = (1, x_i)^\top$. Then

$$\ell(\beta) = \sum_i \{y_i \log \Phi_i + (1 - y_i) \log(1 - \Phi_i)\}, \qquad \nabla \ell = \sum_i \frac{(y_i - \Phi_i)\phi_i}{\Phi_i(1 - \Phi_i)} \mathbf{x}_i.$$

With $\phi'(z) = -z\phi(z)$,

$$\nabla^2 \ell = \sum_i f_i \, \mathbf{x}_i \mathbf{x}_i^\top, \qquad f_i = -y_i \frac{\phi_i(\eta_i \Phi_i + \phi_i)}{\Phi_i^2} - (1 - y_i) \frac{\phi_i \{\phi_i - \eta_i(1 - \Phi_i)\}}{(1 - \Phi_i)^2},$$

which depends on y_i (non-canonical link). With $Ey_i = \Phi_i$ the Fisher information is $X^\top W X$, $w_i = \phi_i^2 / \{\Phi_i(1 - \Phi_i)\}$, as in Proposition 7.2 ($\sigma_i^2 = \Phi_i(1 - \Phi_i)$, $g'(\mu_i) = 1/\phi_i$). The IRLS step is

$$\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} z^{(t)}, \qquad z_i^{(t)} = \eta_i^{(t)} + \frac{y_i - \Phi_i^{(t)}}{\phi_i^{(t)}}.$$

7.6 Nonlinear least squares: Gauss–Newton

(Gauss located Ceres in 1801 by least squares.) Minimize

$$f(\beta) = \tfrac{1}{2} \sum_{i=1}^{n} \{y_i - \mu_i(\beta)\}^2,$$

e.g. a logistic growth curve $\mu(x; \beta) = \beta_3 / (1 + e^{-\beta_1 - \beta_2 x})$. With $J(\beta)$ the $n \times p$ Jacobian with rows $\nabla \mu_i(\beta)^\top$,

$$\nabla f = -J^\top (y - \mu), \qquad \nabla^2 f = J^\top J - \sum_i (y_i - \mu_i) \nabla^2 \mu_i.$$

Gauss–Newton keeps only the positive semidefinite $J^\top J$:

$$\beta^{(t+1)} = \beta^{(t)} + s(J^\top J)^{-1} J^\top (y - \mu(\beta^{(t)})),$$

i.e. each step regresses the residuals on J. It suits small residuals or nearly linear μ, and is Fisher scoring for $y_i \sim N(\mu_i(\beta), \sigma^2)$, as $E[-\nabla^2 \ell] \propto J^\top J$.

7.7 Gradient descent

$$x^{(t+1)} = x^{(t)} - \gamma_t \nabla f(x^{(t)}).$$

$\Delta x = -\nabla f / \|\nabla f\|$ minimizes the linear model $\nabla f^\top \Delta x$ over $\|\Delta x\|_2 \le 1$ (steepest descent); the model is unbounded, so a step size is always needed. Steps are cheap and Hessian-free, attractive at large scale, but the method zigzags, converges slowly, and needs smooth f.

Theorem 7.3

Let f be convex and differentiable with L-Lipschitz gradient, $\|\nabla f(x) - \nabla f(y)\|_2 \le L\|x - y\|_2$, and let f attain its minimum $p^\star$ at $x^\star$. With constant step $\gamma \in (0, 1/L]$,

$$f(x^{(t)}) - p^\star \le \frac{\|x^{(0)} - x^\star\|_2^2}{2\gamma t}.$$

Proof. The Lipschitz gradient gives $f(y) \le f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2$; with $x^+ = x - \gamma \nabla f(x)$, $\gamma L \le 1$: $f(x^+) \le f(x) - \frac{\gamma}{2}\|\nabla f(x)\|^2$. Convexity gives $f(x) \le p^\star + \nabla f(x)^\top (x - x^\star)$, so

$$f(x^+) - p^\star \le \nabla f(x)^\top (x - x^\star) - \frac{\gamma}{2}\|\nabla f(x)\|^2 = \frac{1}{2\gamma}(\|x - x^\star\|^2 - \|x^+ - x^\star\|^2).$$

Summing telescopes: $\sum_{k=1}^t \{f(x^{(k)}) - p^\star\} \le \frac{1}{2\gamma}\|x^{(0)} - x^\star\|^2$, and the left side is $\ge t\{f(x^{(t)}) - p^\star\}$ as $f(x^{(k)})$ is nonincreasing. ∎

The $O(1/t)$ rate is *sublinear*; strong convexity gives linear convergence, never quadratic.

Example 7.4

- Least squares $f(\beta) = \frac{1}{2}\|X\beta - y\|^2$: $L = \sigma_{\max}(X)^2$, so $\gamma \le 1/\sigma_{\max}(X)^2$.
- Logistic regression: $L = \frac{1}{4}\|X^\top X\|_2$ as $p(1 - p) \le \frac{1}{4}$. The minimum may not exist (separable data); a ridge penalty $\frac{\rho}{2}\|\beta\|^2$ fixes this.
- Poisson regression: $-X^\top (y - e^{X\beta})$ is not globally Lipschitz, so a constant step can fail and line search is needed. An L-Lipschitz gradient forces $v^\top \nabla^2 f(\beta) v \le L\|v\|^2$, but if $x_j^\top v > 0$ for some j,

$$v^\top \nabla^2 f(tv) v = \sum_i e^{t x_i^\top v} (x_i^\top v)^2 \ge e^{t x_j^\top v} (x_j^\top v)^2 \to \infty \quad (t \to \infty).$$

Example 7.5 (Multinomial logistic regression)

With K classes, dummy-coded $y_i \in \{0, 1\}^K$ and $\log(p_{ik}/p_{iK}) = x_i^\top \beta_k$ ($k < K$, $\beta_K = 0$), $p_{ik} = e^{x_i^\top \beta_k}/(1 + \sum_{l<K} e^{x_i^\top \beta_l})$. With $B = [\beta_1, \ldots, \beta_{K-1}]$, Y_{-K} the first $K - 1$ columns of Y

and $P = (p_{ik})_{n\times(K-1)}$,

$$\ell(B) = \sum_i \sum_k y_{ik} \log p_{ik} = \operatorname{tr}(Y_{-K}^\top X B) - \sum_i \log\Big(1 + \sum_{l<K} e^{x_i^\top \beta_l}\Big),$$

$$\nabla_{\beta_k} \ell = \sum_i x_i (y_{ik} - p_{ik}), \qquad \text{so} \qquad B^{(t+1)} = B^{(t)} + \gamma X^\top (Y_{-K} - P^{(t)})$$

is gradient ascent ($K = 2$: logistic regression), $O(npK)$ per step with no $p(K-1)$-dimensional Hessian; the Hessian is bounded by $\frac{1}{2}\|X\|_2^2$, so a constant step works.

Example 7.6 (Quantiles by optimization)

The α-quantile of X (cdf F, density $f > 0$) solves $F(x) = \alpha$; Newton gives $x^{(t+1)} = x^{(t)} - \{F(x^{(t)}) - \alpha\}/f(x^{(t)})$, i.e. Newton minimization of ψ with $\psi' = F - \alpha$, $\psi'' = f > 0$. As $F(x) = E[\mathbf{1}\{X \le x\}]$,

$$\psi(x) = E\big[(x - X)\mathbf{1}\{X \le x\}\big] - \alpha x = E\big[\rho_\alpha(X - x)\big] + \text{const}, \qquad \rho_\alpha(u) = u\{\alpha - \mathbf{1}(u < 0)\},$$

the expected **check loss** of quantile regression, convex. Gradient descent $x^{(t+1)} = x^{(t)} - \gamma_t\{F(x^{(t)}) - \alpha\}$ needs no density; if only sampling from F is easy, use the unbiased one-sample gradient,

$$x^{(t+1)} = x^{(t)} - \gamma_t(\mathbf{1}\{X_t \le x^{(t)}\} - \alpha), \qquad X_t \overset{\text{iid}}{\sim} F,$$

with $\sum_t \gamma_t = \infty$, $\sum_t \gamma_t^2 < \infty$ (e.g. $\gamma_t = c/t$): this *stochastic gradient* (Robbins–Monro) algorithm converges to the quantile a.s. Alternatively, the sample α-quantile of N draws has standard error $\sqrt{\alpha(1-\alpha)/N}/f(x_\alpha)$.

7.8 Coordinate descent

Minimize over one coordinate (or block) at a time, cycling:

$$x_j^{(t+1)} = \arg\min_{x_j} f(x_1^{(t+1)}, \ldots, x_{j-1}^{(t+1)}, x_j, x_{j+1}^{(t)}, \ldots, x_d^{(t)}).$$

Each one-dimensional problem is often easy (sometimes closed form), and the objective decreases monotonically: the optimization analogue of Gauss–Seidel. For nondifferentiable convex f it can stick at a non-optimal "corner" where no single coordinate move helps.

7.9 Selected exercises

Motivation. Linear convergence under strong convexity, termination of backtracking, and coordinate descent as Gauss–Seidel.

Exercise 7.1 (Linear convergence under strong convexity)

Let f be differentiable with L-Lipschitz gradient and μ-*strongly convex*, $\mu > 0$:

$$f(y) \ge f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2 \quad \text{for all } x, y,$$

and let $x^\star$ minimize f, $p^\star = f(x^\star)$.
(a) Show that $\|\nabla f(x)\|^2 \ge 2\mu\{f(x) - p^\star\}$ for all x.

(b) Show that gradient descent with $\gamma = 1/L$ satisfies $f(x^{(t)}) - p^\star \le (1-\mu/L)^t\{f(x^{(0)}) - p^\star\}$.
(c) For least squares $f(\beta) = \frac{1}{2}\|X\beta - y\|^2$ with X of full column rank, identify the rate in terms of $\kappa(X)$.

Solution. [Supplement]

(a) The right side of the strong convexity inequality is minimized over y at $y - x = -\nabla f(x)/\mu$, giving $f(y) \ge f(x) - \|\nabla f(x)\|^2/(2\mu)$ for all y; take $y = x^\star$.

(b) As for Theorem 7.3, with $x^+ = x - \frac{1}{L}\nabla f(x)$,

$$f(x^+) \le f(x) - \frac{1}{2L}\|\nabla f(x)\|^2 \le f(x) - \frac{\mu}{L}\{f(x) - p^\star\}$$

by (a); subtract $p^\star$ and induct. Here $\mu \le L$ (compare the lower bound with the quadratic upper bound at $y \ne x$), so $0 \le 1 - \mu/L < 1$.

(c) $\nabla^2 f = X^\top X$ has eigenvalues in $[\sigma_{\min}(X)^2, \sigma_{\max}(X)^2]$; the exact quadratic Taylor expansion gives $\mu = \sigma_{\min}(X)^2$, $L = \sigma_{\max}(X)^2$. The rate is $1 - 1/\kappa(X)^2$: about $\kappa(X)^2$ iterations per constant-factor reduction. ■

Exercise 7.2 (Backtracking terminates)

Let ∇f be L-Lipschitz, $\nabla f(x) \ne 0$, $\Delta x = -\nabla f(x)$, $\alpha \in (0, \frac{1}{2}]$ and $\beta \in (0, 1)$.
(a) Show that the Armijo condition $f(x + s\Delta x) \le f(x) + \alpha s \nabla f(x)^\top \Delta x$ holds for every $s \in (0, 1/L]$.
(b) Deduce that backtracking from $s = 1$ stops after finitely many reductions, at some $s \ge s_{\min} = \min\{1, \beta/L\}$, and that f then decreases by at least $\alpha s_{\min}\|\nabla f(x)\|^2$.
(c) If f is bounded below, show that gradient descent with backtracking satisfies $\nabla f(x^{(t)}) \to 0$, without any convexity assumption.

Solution. [Supplement] Write $g = \nabla f(x)$.

(a) $f(x - sg) \le f(x) - s(1 - \frac{Ls}{2})\|g\|^2$, and for $s \le 1/L$, $1 - \frac{Ls}{2} \ge \frac{1}{2} \ge \alpha$, so $f(x - sg) \le f(x) + \alpha s\, g^\top \Delta x$.

(b) Trial steps $1, \beta, \beta^2, \dots$ reach $\beta^j \le 1/L$, so by (a) one is accepted. Either $s = 1$, or $s = \beta^j$ ($j \ge 1$) follows a rejected $\beta^{j-1} > 1/L$, so $s > \beta/L$. Either way $s \ge s_{\min}$ and $f(x^+) \le f(x) - \alpha s_{\min}\|g\|^2$.

(c) Summing (b), $\alpha s_{\min} \sum_{t<T} \|\nabla f(x^{(t)})\|^2 \le f(x^{(0)}) - \inf f < \infty$, so the terms tend to 0. ■

Exercise 7.3 (Coordinate descent for least squares is Gauss–Seidel)

Let $f(\beta) = \frac{1}{2}\|y - X\beta\|^2$ with columns $x_1, \dots, x_p$ of X all nonzero, and $G = X^\top X$.
(a) Show that one cycle of coordinate descent is exactly one Gauss–Seidel sweep for $G\beta = X^\top y$.
(b) Deduce that coordinate descent converges to the least squares solution when X has full column rank.
(c) Show that a cycle can be carried out in $O(np)$ flops without forming G.

Solution. [Supplement]

(a) In β_j alone, f is quadratic with leading coefficient $\frac{1}{2}\|x_j\|^2 > 0$ and $\partial f/\partial \beta_j = (G\beta)_j - (X^\top y)_j$, minimized by

$$\beta_j \leftarrow \frac{1}{g_{jj}}\Big((X^\top y)_j - \sum_{l \ne j} g_{jl}\beta_l\Big),$$

with β_l, $l < j$, already updated and $l > j$ old: the Gauss–Seidel update for $G\beta = X^\top y$.

(b) G is positive definite, so Exercise 5.3 gives convergence to $G^{-1}X^\top y = \hat{\beta}$.

(c) Keep $r = y - X\beta$; as $x_j^\top r = (X^\top y)_j - \sum_l g_{jl}\beta_l$, the update is

$$\beta_j \leftarrow \beta_j + \frac{x_j^\top r}{\|x_j\|^2}, \qquad r \leftarrow r - x_j\,\Delta\beta_j,$$

$O(n)$ per coordinate ($\|x_j\|^2$ precomputed), $O(np)$ per cycle, versus $O(np^2)$ to form G. ■

Lecture 8. Numerical Integration

We want $\int_a^b f(x)\,dx$. Statistical integrals rarely have closed forms:

- expectations, $E\sin X = \int \sin(x) f_X(x)\,dx$;
- Bayesian normalizing constants $\int f(x \mid \theta)\pi(\theta)\,d\theta$;
- likelihoods of mixed models, e.g. a Poisson GLM with random intercept, $\prod_i \int \prod_j f(y_{ij} \mid \gamma_i)\phi(\gamma_i)\,d\gamma_i$.

Approaches: quadrature (now) and Monte Carlo (Lecture 10).

8.1 Newton–Cotes rules

With $h = (b-a)/n$, $x_i = a + ih$, approximate f on each $[x_i, x_{i+1}]$ by an interpolating polynomial of degree m, integrate it exactly, and add up.

- $m = 0$, **Riemann**: $\int_{x_i}^{x_{i+1}} f \approx h f(x_i)$.
- $m = 1$, **trapezoidal**: $\int_{x_i}^{x_{i+1}} f \approx \frac{h}{2}\{f(x_i) + f(x_{i+1})\}$, so $T(n) = h\{\frac{1}{2}f(a) + f(x_1) + \cdots + f(x_{n-1}) + \frac{1}{2}f(b)\}$.
- $m = 2$, **Simpson**: $\int_{x_i}^{x_{i+1}} f \approx \frac{h}{6}\{f(x_i) + 4f(\frac{x_i + x_{i+1}}{2}) + f(x_{i+1})\}$.

Proposition 8.1 (Accuracy)

For smooth f the total errors are $O(n^{-1})$ (Riemann), $O(n^{-2})$ (trapezoidal), $O(n^{-4})$ (Simpson); the local errors are

$$\frac{h^2}{2}f'(x_i) + O(h^3), \qquad -\frac{h^3}{12}f''(x_i) + O(h^4), \qquad -\frac{h^5}{2880}f^{(4)}(x_i) + O(h^6).$$

Proof. Integrate the Taylor expansion of f at x_i over $[x_i, x_i + h]$:

$$\int_{x_i}^{x_{i+1}} f = h f(x_i) + \frac{h^2}{2}f'(x_i) + \frac{h^3}{6}f''(x_i) + \frac{h^4}{24}f^{(3)}(x_i) + \frac{h^5}{120}f^{(4)}(x_i) + O(h^6).$$

Riemann: error $\frac{h^2}{2}f'(x_i) + O(h^3)$. *Trapezoid*: $\frac{h}{2}\{f(x_i) + f(x_{i+1})\} = h f(x_i) + \frac{h^2}{2}f' + \frac{h^3}{4}f'' + O(h^4)$, error $(\frac{1}{6} - \frac{1}{4})h^3 f'' + O(h^4) = -\frac{h^3}{12}f''(x_i) + O(h^4)$. *Simpson*: expanding at midpoint and endpoint, the rule matches through h^4 and differs by $(\frac{1}{120} - \frac{5}{576})h^5 f^{(4)} = -\frac{h^5}{2880}f^{(4)}$. Summing $n = (b-a)/h$ local errors gives global orders h, h^2, h^4. ■

Simpson needs about twice the evaluations per interval, but wins by orders of magnitude at equal cost. For $\int_0^2 e^{-x}dx = 1 - e^{-2} \approx 0.864665$ and $n = 10$ the errors are 8.9×10^{-2}, 2.9×10^{-3}, 4.8×10^{-7}; log–log slopes against n are -1, -2, -4.

Example 8.1 (An infinite range)

Newton–Cotes rules need a finite interval. For $\int_1^\infty e^{-x}x^{-1/2}dx$ substitute $t = 1/x$:

$$\int_1^\infty e^{-x}x^{-1/2}dx = \int_0^1 e^{-1/t}t^{-3/2}\,dt.$$

At $t = 0$ the integrand is $0 \cdot \infty =$ `NaN` in floating point, so trapezoid and Simpson break there.

But $e^{-1/t}t^{-3/2} \to 0$ with all derivatives as $t \downarrow 0$; defining it as 0 at $t = 0$ gives a smooth integrand, and the rules converge at their usual rates to $\sqrt{\pi}\,\mathrm{erfc}(1) = \Gamma(\frac{1}{2}, 1) \approx 0.278806$. (Or use an open rule: midpoint, Gaussian quadrature.)

General m: interpolation. For distinct nodes $x_0, \ldots, x_m$, the **Lagrange polynomials** $L_j(x) = \prod_{k \neq j} \frac{x - x_k}{x_j - x_k}$ satisfy $L_j(x_k) = \delta_{jk}$, so $p = \sum_j y_j L_j$ interpolates (x_j, y_j).

Theorem 8.2 (Polynomial interpolation)

For $m + 1$ distinct points there is a unique polynomial of degree at most m through $(x_0, y_0), \ldots, (x_m, y_m)$.

Proof. Existence: Lagrange form. Another q gives $p - q$ of degree $\leq m$ with $m + 1$ roots, so $p = q$. ■

Corollary 8.3

If f is a polynomial of degree at most m, then $\int_a^b f = \sum_j w_j f(x_j)$ with $w_j = \int_a^b L_j(x)\,dx$.

Proof. By uniqueness, $f = \sum_j f(x_j) L_j$; integrate. ■

8.2 Gaussian quadrature

Newton–Cotes fixes equally spaced nodes and chooses weights to integrate degree m exactly; choosing the nodes too gains much more.

$\mathcal{P}^m$, the polynomials of degree $\leq m$ on (a, b), has inner product

$$\langle p, q \rangle = \int_a^b p(x) q(x) w(x)\,dx$$

for a positive weight w. Gram–Schmidt on $1, x, x^2, \ldots$ gives orthonormal $q_0, q_1, \ldots, q_k$ of exact degree k and orthogonal to $\mathcal{P}^{k-1}$; for $w \equiv 1$ on $(-1, 1)$, the **Legendre polynomials**.

Lemma 8.4

The roots of q_k are real, simple and lie in (a, b).

Proof. Let $r_1 < \cdots < r_l$ be the sign changes of q_k in (a, b). If $l < k$, $q_k \prod_{i \leq l}(x - r_i)$ has constant sign and is not 0, so its integral is nonzero, yet $\prod_{i \leq l}(x - r_i) \in \mathcal{P}^{k-1}$ is orthogonal to q_k. So $l = k$: k distinct real roots in (a, b). ■

Theorem 8.5 (Gaussian quadrature)

Let $x_0, \ldots, x_m$ be the roots of q_{m+1} and $w_j = \int_a^b L_j(x) w(x)\,dx$. Then

$$\int_a^b f(x) w(x)\,dx = \sum_{j=0}^{m} w_j f(x_j)$$

for every polynomial f of degree at most $2m + 1$, and all $w_j > 0$. Degree $2m + 1$ cannot be improved.

Proof. Divide: $f = gq_{m+1} + r$, $g, r \in \mathcal{P}^m$. Then $\int gq_{m+1}w = 0$, so $\int fw = \int rw$, and $f(x_j) = r(x_j)$; by Corollary 8.3 (weight w) $\int rw = \sum w_j r(x_j) = \sum w_j f(x_j)$. Positivity: L_k^2 has degree $2m$, so $0 < \int L_k^2 w = \sum_j w_j L_k(x_j)^2 = w_k$. Optimality: $f = \prod_j (x - x_j)^2$ has degree $2m + 2$ and $\int fw > 0$, while the rule gives 0. ■

$m + 1$ well-chosen nodes are exact to degree $2m + 1$, against m for equal spacing.

Weight functions. Other weights give other orthogonal families, for integrals $\int g(x)w(x)\,dx$:

polynomials	weight $w(x)$	domain	related distribution
Legendre	1	$(-1, 1)$	uniform
Laguerre	$x^\alpha e^{-x}$	$(0, \infty)$	gamma
Hermite	e^{-x^2}	$(-\infty, \infty)$	normal
Jacobi	$(1-x)^\alpha(1+x)^\beta$	$(-1, 1)$	beta (after rescaling)

Each satisfies a three-term recursion $q_m(x) = (\alpha_m + \beta_m x)q_{m-1}(x) - \gamma_m q_{m-2}(x)$ and has *interlacing* roots (those of q_{m+1} separate those of q_m); the nodes are found by Newton's method using these, or as eigenvalues of a tridiagonal matrix.

Gauss–Hermite quadrature. $\int f(x)\,dx = \int \{f(x)e^{x^2}\}e^{-x^2}dx \approx \sum_j w_j f(x_j)e^{x_j^2}$, good only if $f(x)e^{x^2}$ is nearly polynomial. For $f(x) = e^{-(x-\mu)^2/(2\sigma^2)}$ with μ far from 0 or σ^2 far from $\frac{1}{2}$, the nodes miss the mass. Remedy: *centre and scale*, $z = (x - \mu)/(\sqrt{2}\sigma)$ gives $\sqrt{2}\sigma \int e^{-z^2}dz$, exact with one node.

Example 8.2 (Binomial–logit posterior)

$x \sim B(n, p)$, $p = e^\theta/(1 + e^\theta)$, prior $\theta \sim N(0, \sigma^2)$. The normalizing constant is $\int \bar{g}$, $\bar{g}(\theta) = e^{\theta x}(1 + e^\theta)^{-n}e^{-\theta^2/(2\sigma^2)}$. Gauss–Hermite rescaled only by σ does badly for $x/n = 0.9$: the integrand peaks near $\theta = \log 9 = 2.2$, far from the nodes. Centring at the maximizer θ^*, $\int \bar{g} = \int \{\bar{g}(z + \theta^*)e^{z^2}\}e^{-z^2}dz$, is accurate with few nodes.

Example 8.3 (Poisson posterior: centre *and* scale)

$y_1, \ldots, y_{10}$ Poisson with mean e^α, prior $\alpha \sim N(0, 100)$, data $11, 19, 27, 12, 14, 11, 10, 13, 15, 10$ ($\sum y_i = 142$). The log unnormalized posterior $h(\alpha) = 142\alpha - 10e^\alpha - \alpha^2/200$ has mode $\hat{\alpha} = 2.65306$ and scale $\hat{s} = \{-h''(\hat{\alpha})\}^{-1/2} = 0.08392$. With $\alpha = \hat{\alpha} + \sqrt{2}\hat{s}z$,

$$E[g(\alpha) \mid y] = \frac{\int g(\hat{\alpha} + \sqrt{2}\hat{s}z)e^{h(\hat{\alpha}+\sqrt{2}\hat{s}z)-h(\hat{\alpha})+z^2}e^{-z^2}dz}{\int e^{h(\hat{\alpha}+\sqrt{2}\hat{s}z)-h(\hat{\alpha})+z^2}e^{-z^2}dz},$$

both by Gauss–Hermite ($\sqrt{2}\hat{s}$ cancels; subtracting $h(\hat{\alpha}) \approx 235$ prevents overflow). The posterior is nearly normal, so the bracketed functions are nearly polynomial:

nodes	posterior mean	posterior variance
2	2.651881	0.0070417
5	2.649534	0.0070677
10	2.6495297	0.00706793
15	2.649529692	0.007067929

Fifteen nodes match adaptive integration to 10 digits. Uncentred, the nodes sit near 0, where the posterior (mean 2.65, sd 0.084) is numerically zero, and the rule fails completely.

Example 8.4 (Variance of the sample median)

For $X_1, \ldots, X_n$ i.i.d. $N(0,1)$, $n = 2m+1$, the median $X_{(m+1)}$ has density $n\binom{n-1}{m}\Phi(x)^m\{1-\Phi(x)\}^m\phi(x)$ (one observation in $[x, x+dx)$, exactly m of the other $n-1$ below). By symmetry $EX_{(m+1)} = 0$ and

$$\begin{aligned}\operatorname{Var} X_{(m+1)} &= \int_{-\infty}^{\infty} x^2 n \binom{n-1}{m} \Phi(x)^m\{1-\Phi(x)\}^m\phi(x)\,dx \\ &= \frac{2n}{\sqrt{\pi}}\binom{n-1}{m}\int u^2\Phi(\sqrt{2}u)^m\{1-\Phi(\sqrt{2}u)\}^m e^{-u^2}du,\end{aligned}$$

with $x = \sqrt{2}u$. Gauss–Hermite with k nodes:

k	2	4	8	16	32	64	exact	$\frac{\pi}{2n}$
$n = 11$	0.117	0.238	0.234	0.161	0.1379	0.13716	0.13716	0.1428
$n = 21$	0.007	0.057	0.127	0.123	0.0840	0.07349	0.07326	0.0748

Convergence is slow and erratic: $\Phi^m(1-\Phi)^m$ has width of order $n^{-1/2}$, while the nodes suit unit variance; rescaling by the asymptotic sd fixes this. The asymptotic variance $1/\{4n\phi(0)^2\} = \pi/(2n)$ is 4.1% ($n = 11$) and 2.1% ($n = 21$) too large.

Example 8.5 (Generalized Gauss–Laguerre)

With $x = u^2$, $\int_0^\infty \sqrt{x}e^{-x}dx = 2\int_0^\infty u^2e^{-u^2}du = \frac{\sqrt{\pi}}{2}$ $(= \Gamma(\frac{3}{2}))$. For $\int_0^\infty f(x)\sqrt{x}e^{-x}dx$ use the weight $x^{1/2}e^{-x}$ (generalized Laguerre, $\alpha = \frac{1}{2}$): for $f \equiv 1$ one node $(w_0 = \Gamma(\frac{3}{2}))$ is exact. For $f = \log x$, exact value $\Gamma'(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}(2 - \gamma - 2\log 2) \approx 0.032338$, two errors remain.

(i) *Method error*: $\log x$ is not polynomial and is singular at 0, so the error decays only algebraically in m.

(ii) *Floating-point error*: nodes and weights are computed approximately, the smallest node is near 0 where $\log x$ is large, the largest weights may underflow, and the small result (0.03) arises by cancellation, magnifying rounding errors.

8.3 Laplace approximation

For $\int f(x)e^{-ag(x)}dx$ with large a and g uniquely minimized at 0, the integrand concentrates near 0, and $g(x) \approx g(0) + \frac{1}{2}g''(0)x^2$ gives

$$\int f(x)e^{-ag(x)}dx \approx f(0)e^{-ag(0)}\sqrt{\frac{2\pi}{ag''(0)}}:$$

integration becomes optimization.

Theorem 8.6 (Laplace)

Suppose (1) for every $\delta > 0$ there is $\rho > 0$ with $g(x) - g(0) \geq \rho$ for $|x| \geq \delta$;
(2) g is twice continuously differentiable near 0 with $g'(0) = 0$, $g''(0) > 0$;
(3) f is continuous near 0 with $f(0) > 0$;

(4) $\int |f|e^{-ag} < \infty$ for $a \geq a_0$. Then

$$\lim_{a\to\infty} \frac{\int f(x)e^{-ag(x)}dx}{f(0)e^{-ag(0)}\sqrt{2\pi/(ag''(0))}} = 1.$$

Proof. Take $g(0) = 0$. Given $\epsilon \in (0, \frac{1}{2}g''(0))$, choose δ with, for $|x| < \delta$,

$$(1-\epsilon)f(0) \leq f(x) \leq (1+\epsilon)f(0), \qquad \tfrac{1}{2}(g''(0)-2\epsilon)x^2 \leq g(x) \leq \tfrac{1}{2}(g''(0)+2\epsilon)x^2.$$

Tails. For $|x| \geq \delta$, $g \geq \rho$, so for $a \geq a_0$,

$$\int_{|x|\geq\delta} |f|e^{-ag} = \int_{|x|\geq\delta} |f|e^{-a_0 g}e^{-(a-a_0)g} \leq e^{-(a-a_0)\rho}\int |f|e^{-a_0 g} = O(e^{-\rho a}).$$

Centre, upper bound. $\int_{|x|<\delta} fe^{-ag} \leq (1+\epsilon)f(0)\int e^{-\frac{a}{2}(g''(0)-2\epsilon)x^2}dx = (1+\epsilon)f(0)\sqrt{\frac{2\pi}{a(g''(0)-2\epsilon)}}$. *Centre, lower bound.* $\int_{|x|<\delta} fe^{-ag} \geq (1-\epsilon)f(0)\int_{|x|<\delta} e^{-\frac{a}{2}(g''(0)+2\epsilon)x^2}dx$, the Gaussian tail over $|x| \geq \delta$ being $O(e^{-\lambda a})$, $\lambda = \frac{1}{2}(g''(0)+2\epsilon)\delta^2$. Multiply by $\sqrt{a}$ and let $a \to \infty$:

$$(1-\epsilon)f(0)\sqrt{\frac{2\pi}{g''(0)+2\epsilon}} \leq \liminf \sqrt{a}\int fe^{-ag} \leq \limsup \sqrt{a}\int fe^{-ag} \leq (1+\epsilon)f(0)\sqrt{\frac{2\pi}{g''(0)-2\epsilon}}.$$

Let $\epsilon \to 0$. ■

Example 8.6 (Stirling's formula)

With $x = tz$, $\Gamma(t+1) = t^{t+1}\int_0^\infty e^{-t(z-\log z)}dz$; $g(z) = z - \log z$ has minimum $g(1) = 1$ with $g''(1) = 1$, so $\Gamma(t+1) \sim \sqrt{2\pi}\, t^{t+1/2}e^{-t}$.

Posterior expectations. With log-likelihood $\ell_n(\theta)$, log-prior $\pi(\theta)$ and $\tilde{h} = \log h > 0$,

$$E[h(\Theta) \mid x] = \frac{\int e^{\tilde{h}+\ell_n+\pi}d\theta}{\int e^{\ell_n+\pi}d\theta}.$$

Apply Laplace separately at the maximizers $\tilde{\theta}$ of $\tilde{h}+\ell_n+\pi$ and $\hat{\theta}$ of $\ell_n+\pi$ (posterior mode):

$$E[h(\Theta) \mid x] \approx \exp\{(\tilde{h}+\ell_n+\pi)(\tilde{\theta}) - (\ell_n+\pi)(\hat{\theta})\}\sqrt{\frac{(\ell_n+\pi)''(\hat{\theta})}{(\tilde{h}+\ell_n+\pi)''(\tilde{\theta})}}.$$

For large n the relative error is $O(n^{-2})$: the leading errors of numerator and denominator cancel.

Example 8.7 (Beta moments)

For $\Theta \sim \text{Beta}(\alpha, \beta)$, exactly $E\Theta^k = \frac{\alpha(\alpha+1)\cdots(\alpha+k-1)}{(\alpha+\beta)\cdots(\alpha+\beta+k-1)}$. With $g(\theta) = -(\alpha-1)\log\theta - (\beta-1)\log(1-\theta)$, minimized at $\hat{\theta} = \frac{\alpha-1}{\alpha+\beta-2}$,

$$\int_0^1 \theta^{\alpha-1}(1-\theta)^{\beta-1}d\theta \approx \Big(\frac{\alpha-1}{\alpha+\beta-2}\Big)^{\alpha-1}\Big(\frac{\beta-1}{\alpha+\beta-2}\Big)^{\beta-1}\sqrt{\frac{2\pi(\alpha-1)(\beta-1)}{(\alpha+\beta-2)^3}},$$

and likewise with $\alpha+k$ for the numerator. As $\alpha, \beta \to \infty$ with $\alpha/(\alpha+\beta) \to p$, both the moment and its approximation tend to p^k.

Unlike quadrature, whose node count grows exponentially with dimension, Laplace extends directly to $\mathbb{R}^d$: $\int fe^{-ag} \approx f(\hat{x})e^{-ag(\hat{x})}(2\pi/a)^{d/2}\det\{\nabla^2 g(\hat{x})\}^{-1/2}$.

8.4 Selected exercises

Motivation. Simpson as extrapolated trapezoid, the three-term recursion, and the two-point Gauss rule by hand.

Exercise 8.1 (Simpson's rule as an extrapolated trapezoid)

Let $T(n)$ be the composite trapezoidal rule and $S(n)$ the composite Simpson rule on n intervals of length $h = (b-a)/n$ (Simpson using the midpoints $m_i = x_i + h/2$).
(a) Show that $S(n) = \frac{1}{3}\{4T(2n) - T(n)\}$.
(b) Suppose $T(n) = I + ch^2 + O(h^4)$, where $I = \int_a^b f$ and c does not depend on h. Show that $\frac{1}{3}\{4T(2n) - T(n)\} = I + O(h^4)$, in agreement with Proposition 8.1.

Solution. [Supplement]

(a) $T(2n)$ uses the x_i and the midpoints m_i, spacing $h/2$:

$$T(n) = h\Big\{\tfrac{1}{2}f(a) + \sum_{i=1}^{n-1} f(x_i) + \tfrac{1}{2}f(b)\Big\},$$
$$T(2n) = \frac{h}{2}\Big\{\tfrac{1}{2}f(a) + \sum_{i=1}^{n-1} f(x_i) + \sum_{i=0}^{n-1} f(m_i) + \tfrac{1}{2}f(b)\Big\}.$$

Hence

$$4T(2n) - T(n) = h\Big\{\tfrac{1}{2}f(a) + \sum_{i=1}^{n-1} f(x_i) + \tfrac{1}{2}f(b)\Big\} + 2h\sum_{i=0}^{n-1} f(m_i),$$

and dividing by 3,

$$\frac{4T(2n) - T(n)}{3} = \frac{h}{6}\Big\{f(a) + 2\sum_{i=1}^{n-1} f(x_i) + f(b) + 4\sum_{i=0}^{n-1} f(m_i)\Big\} = \sum_{i=0}^{n-1} \frac{h}{6}\{f(x_i) + 4f(m_i) + f(x_{i+1})\},$$

which is $S(n)$ (each interior node lies in two intervals).

(b) $T(2n) = I + \frac{c}{4}h^2 + O(h^4)$, so $4T(2n) - T(n) = 3I + O(h^4)$. This is *Richardson extrapolation* (cancelling the leading error); repeating it gives Romberg integration. ■

Exercise 8.2 (The three-term recursion)

Let $\langle p, q\rangle = \int_a^b pqw$ with a positive weight w having finite moments, and let $p_0 = 1, p_1, p_2, \ldots$ be the *monic* orthogonal polynomials (p_k of degree k with leading coefficient 1, orthogonal to $\mathcal{P}^{k-1}$); set $p_{-1} = 0$.
(a) Show that for $k \geq 0$

$$p_{k+1}(x) = (x - a_k)p_k(x) - b_k p_{k-1}(x), \qquad a_k = \frac{\langle xp_k, p_k\rangle}{\langle p_k, p_k\rangle}, \quad b_k = \frac{\langle p_k, p_k\rangle}{\langle p_{k-1}, p_{k-1}\rangle} > 0$$

(b_0 is irrelevant since $p_{-1} = 0$).
(b) For the Legendre weight $w \equiv 1$ on $(-1, 1)$, compute p_1 and p_2.

Solution. [Supplement]

(a)

Step 1. $q = p_{k+1} - xp_k$ has degree $\leq k$ (both monic of degree $k+1$), so $q = \sum_{j \leq k} c_j p_j$, $c_j = \langle q, p_j\rangle / \langle p_j, p_j\rangle$.

Step 2. $\langle xp, r\rangle = \langle p, xr\rangle$, so for $j \le k$, as $\langle p_{k+1}, p_j\rangle = 0$,

$$\langle q, p_j\rangle = -\langle xp_k, p_j\rangle = -\langle p_k, xp_j\rangle.$$

Step 3. For $j \le k-2$, $xp_j \in \mathcal{P}^{k-1}$, so $c_j = 0$. For $j = k$, $c_k = -a_k$. For $j = k-1$, $xp_{k-1} = p_k + (\text{degree} \le k-1)$, so $\langle p_k, xp_{k-1}\rangle = \langle p_k, p_k\rangle$ and $c_{k-1} = -b_k$.

Step 4. So $p_{k+1} = xp_k - a_k p_k - b_k p_{k-1}$, with $b_k > 0$ a ratio of squared norms. Normalizing $q_k = p_k/\|p_k\|$ gives the recursion quoted in the lecture.

(b) By symmetry $a_0 = \int_{-1}^{1} x\,dx/2 = 0$, so $p_1 = x$; $a_1 = \int_{-1}^{1} x^3 dx / \int_{-1}^{1} x^2 dx = 0$ and $b_1 = \frac{2/3}{2} = \frac{1}{3}$, so $p_2(x) = x^2 - \frac{1}{3}$. ■

Exercise 8.3 (Two-point Gauss–Legendre rule)

(a) Using Exercise 8.2(b), derive the Gaussian quadrature rule with two nodes for $\int_{-1}^{1} f(x)\,dx$. Verify directly that it integrates $1, x, x^2, x^3$ exactly but not x^4.
(b) Transfer the rule to an interval $[c-r, c+r]$.

Solution. [Supplement]

(a) By Theorem 8.5 with $m = 1$, the nodes are the roots $x_{0,1} = \mp 1/\sqrt{3}$ of p_2, and

$$w_0 = \int_{-1}^{1} \frac{x - 1/\sqrt{3}}{-2/\sqrt{3}}\,dx = \int_{-1}^{1} \Big(\frac{1}{2} - \frac{\sqrt{3}}{2}x\Big)dx = 1,$$

and $w_1 = 1$ by symmetry: $\int_{-1}^{1} f \approx f(-1/\sqrt{3}) + f(1/\sqrt{3})$. Check: 1 gives 2; x, x^3 give 0 (odd); x^2 gives $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$. For x^4 the rule gives $\frac{2}{9} \ne \frac{2}{5}$: exactness stops at $2m + 1 = 3$.

(b) With $x = c + rt$, $\int_{c-r}^{c+r} f\,dx = r\int_{-1}^{1} f(c+rt)\,dt \approx r\{f(c - r/\sqrt{3}) + f(c + r/\sqrt{3})\}$, exact for cubics: two evaluations, where Simpson needs three. ■

Lecture 9. Random Number Generation

9.1 Uniform random numbers

All random number generation starts from i.i.d. Unif$(0,1)$ variates; computers produce deterministic **pseudo-random** sequences that only look random.

Multiplicative congruential generators. For a large prime m and $a \in \{2, \dots, m-1\}$, iterate $x_{k+1} = ax_k \bmod m$ and output $u_k = x_k/m$.

Proposition 9.1

The map $f(x) = ax \bmod m$ is a bijection of $\{1, \dots, m-1\}$. The sequence is periodic with period $n = \min\{k \geq 1 : a^k \equiv 1 \bmod m\}$, which divides $m-1$.

Proof. m is prime, so $\gcd(a, m) = 1$ and $ax \not\equiv 0$ for $1 \leq x \leq m-1$; $ax \equiv ay$ gives $m \mid x - y$, so $x = y$. Thus f is injective, hence bijective. As $x_k = a^k x_0 \bmod m$, the sequence returns to x_0 exactly when $a^k \equiv 1$, and Fermat's little theorem $a^{m-1} \equiv 1$ makes the period divide $m-1$. ■

x_0 is the **seed**. For a **primitive root** a (period $m-1$) the sequence runs through $\{1, \dots, m-1\}$; a classical choice is $m = 2^{31} - 1$, $a = 7^5 = 16807$, period $2^{31} - 2$ (early MATLAB).

Mixed congruential generators. $x_{i+1} = (ax_i + c) \bmod m$, $c \neq 0$, allows $m = 2^M$ (free reduction in binary). By Hull–Dobell the period is the full m iff $\gcd(c, m) = 1$, every prime factor of m divides $a - 1$, and $4 \mid a - 1$ when $4 \mid m$; both $m = 2048$, $a = 1229$, $c = 1$ and the old C-library $m = 2^{32}$, $a = 1103515245$, $c = 12345$ qualify. The first gives only 2048 values (multiples of $1/2048$) and repeats after 2048 draws; the second has poor low-order bits (period 2^k modulo 2^k; the last bit alternates). Plots of (u_i, u_{i+1}) show the lattice structure of both.

A good generator has a long period and passes tests of uniformity and independence. Congruential p-tuples $(u_k, \dots, u_{k+p-1})$ lie on few parallel hyperplanes in $[0,1]^p$ (Marsaglia; RANDU puts all triples on 15 planes); shuffling helps. R's default **Mersenne Twister** (Matsumoto–Nishimura, 1998) has period $2^{19937} - 1$ and is equidistributed in up to 623 dimensions.

9.2 The inverse CDF method

Motivation. If U is uniform, $F^{-1}(U) \sim F$: the simplest sampler when the quantile function is explicit.

For a cdf F define the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, $0 < u < 1$.

Theorem 9.2

Let $X \sim F$ and $U \sim$ Unif$(0,1)$.

(a) $F^{-1}(U)$ has cdf F.

(b) If F is continuous, $F(X) \sim$ Unif$(0,1)$.

(c) In general, $P\{F(X) \leq y\} \leq y$ for $y \in [0,1]$.

Proof. (a) We show $\{F^{-1}(u) \le t\} = \{u \le F(t)\}$. By monotonicity and right continuity, $\{x : F(x) \ge u\} = [F^{-1}(u), \infty)$, so $F(F^{-1}(u)) \ge u$. If $F^{-1}(u) \le t$ then $u \le F(F^{-1}(u)) \le F(t)$; if $u \le F(t)$ then t is in the set, so $F^{-1}(u) \le t$. Hence $P(F^{-1}(U) \le t) = P(U \le F(t)) = F(t)$.

(b) $P\{F(X) \le F(t)\} = F(t)$: the event is $\{X \le t\} \cup \{X > t, F(X) = F(t)\}$, and the second set, where F is flat, has probability 0. For $u \in (0,1)$ continuity gives $F(t) = u$, so $P\{F(X) \le u\} = u$.

(c) By (a) take $X = F^{-1}(U)$; $F(F^{-1}(U)) \ge U$ gives $\{F(X) \le y\} \subset \{U \le y\}$. ■

Example 9.1

- Exponential, $F(x) = 1 - e^{-\lambda x}$: $F^{-1}(u) = -\frac{1}{\lambda}\log(1-u)$, or $-\frac{1}{\lambda}\log U$ as $1 - U \overset{d}{=} U$.
- Cauchy(μ, β): $F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{x-\mu}{\beta}$, $F^{-1}(u) = \mu + \beta\tan(\pi u - \frac{\pi}{2})$.
- Discrete uniform on $\{1, \dots, k\}$: $F^{-1}(u) = \lceil ku \rceil$.
- Geometric(p) on $\{1, 2, \dots\}$: $\lceil Y \rceil$ for $Y \sim \text{Exp}(\lambda)$, $1 - e^{-\lambda} = p$, so $X = \lceil \log U / \log(1-p) \rceil$.

If F^{-1} has no closed form but F (or only f) is computable, solve $F(x) = u$ numerically per draw: bisection, or Newton $x \leftarrow x - \{F(x) - u\}/f(x)$ (Lecture 6), with $F = \int_{-\infty}^{x} f$ by quadrature if needed. This is exact to tolerance but costly, and heavy tails (e.g. t_4) need a wide bracket.

9.3 Normal random numbers

Φ^{-1} has no closed form (accurate rational approximations exist).

Box–Muller. For independent $X, Y \sim N(0,1)$ in polar coordinates (R, Θ) the joint density is $\frac{1}{2\pi}e^{-r^2/2}r$, so $\Theta \sim \text{Unif}(0, 2\pi)$ independently of R, with $P(R^2 > s) = e^{-s/2}$ ($R^2 \sim$ Exp, mean 2). So for independent $U, V \sim \text{Unif}(0,1)$,

$$R = \sqrt{-2\log U}, \quad \Theta = 2\pi V, \qquad (X, Y) = (R\cos\Theta, R\sin\Theta)$$

are independent standard normals.

Marsaglia's polar method. If $(U, V) = (T\cos\Phi, T\sin\Phi)$ is uniform on the unit disk, $P(T^2 \le t, \Phi \le \phi) = t\frac{\phi}{2\pi}$, so $T^2 \sim \text{Unif}(0,1)$ and $\Phi \sim \text{Unif}(0, 2\pi)$ independently; $-2\log T^2$ replaces R^2 and $(U/T, V/T)$ replaces $(\cos\Theta, \sin\Theta)$:

(1) draw U, V i.i.d. Unif$(-1, 1)$;
(2) if $T^2 = U^2 + V^2 \ge 1$, return to (1);
(3) output $X = U\sqrt{-2\log T^2/T^2}$ and $Y = V\sqrt{-2\log T^2/T^2}$.

It avoids trigonometric functions at the price of $4/\pi$ times as many pairs (a rejection step, Section 9.5).

9.4 Generating from definitions

- Bernoulli(p): $\mathbf{1}\{U \le p\}$. Binomial(n, p): sum of n Bernoullis. Negative binomial(r, p): sum of r geometrics.
- Poisson(λ): the N with $\prod_{i=1}^{N} U_i \ge e^{-\lambda} > \prod_{i=1}^{N+1} U_i$, since $T_i = -\log U_i/\lambda$ are Exp(λ) interarrival times and N counts arrivals in $[0,1]$: $T_1 + \cdots + T_N \le 1 < T_1 + \cdots + T_{N+1}$.

- $\chi^2(\nu)$: $\sum_{i\le\nu} Z_i^2$; for even ν, also $-2\log\prod_{i\le\nu/2} U_i$, since $\chi^2(\nu) = \text{Gamma}(\nu/2, 2)$.
- $t(\nu) = Z/\sqrt{X/\nu}$ and $F(\nu_1, \nu_2) = (X_1/\nu_1)/(X_2/\nu_2)$ from independent normal and chi-square variates.

9.5 Acceptance–rejection sampling

Motivation. With a known density up to a constant, sample an easier envelope and accept with probability proportional to the ratio; accepted draws are exact.

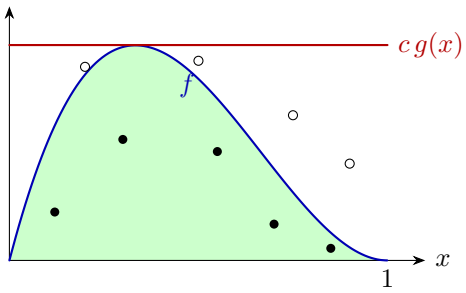


Figure. *Acceptance–rejection for a Beta(2, 3) target f under the uniform envelope $c\,g$: filled points are accepted, open ones rejected; the acceptance rate is $1/c$.*

To sample a density f without tractable cdf (von Neumann):

(1) find a density g that is easy to sample and a constant c with $f \le cg =: h$ (an *envelope*);
(2) draw $X \sim g$ and an independent $U \sim \text{Unif}(0, 1)$;
(3) accept X if $U \le f(X)/h(X)$; otherwise return to (2).

The **body** of a nonnegative function f is $B_f = \{(x, y) : 0 \le y \le f(x)\}$, with area $|B_f| = \int f$.

Theorem 9.3

If $X \sim g$ and $U \sim \text{Unif}(0, 1)$ are independent, then $(X, cg(X)U)$ is uniform on B_{cg}.

Proof. Given $X = x$, $Y = cg(x)U$ has density $\frac{1}{cg(x)}\mathbf{1}\{0 \le y \le cg(x)\}$; times $g(x)$, the joint density is $\frac{1}{c}\mathbf{1}_{B_{cg}}$, and $|B_{cg}| = c$. ■

Corollary 9.4

An accepted X has density f, and each proposal is accepted with probability $1/c$.

Proof. Acceptance is $(X, Y) \in B_f \subset B_{cg}$, so the accepted (X, Y) is uniform on B_f, with x-marginal $\int_0^{f(x)} dy/|B_f| = f(x)$. The acceptance probability is $|B_f|/|B_{cg}| = 1/c$. ■

So the envelope should be tight ($c \approx 1$); the number of proposals per draw is geometric with mean c. f need only be known up to a constant shared with the envelope.

Example 9.2 (Uniform on the disk)

In Marsaglia's method $f = \frac{1}{\pi}\mathbf{1}\{u^2 + v^2 < 1\}$, $g = \frac{1}{4}\mathbf{1}_{[-1,1]^2}$, $c = 4/\pi$, and $f/(cg)$ is the disk indicator: accept iff $U^2 + V^2 < 1$.

Example 9.3 (Beta variates from uniforms)

For Beta(a,b), $a,b>1$, the density is bounded by its mode value $d_{\max}$ at $(a-1)/(a+b-2)$. With $g=\text{Unif}(0,1)$, $c=d_{\max}$: keep $Y\sim\text{Unif}(0,1)$ iff $U\le f(Y)/d_{\max}$, at acceptance rate $1/d_{\max}$ (1000 proposals give about $1000/d_{\max}$ draws). Simple, but wasteful for peaked densities (large a,b).

Example 9.4 (Gamma variates)

As $c\,\text{Gamma}(\alpha,1)=\text{Gamma}(\alpha,c)$ and $\text{Gamma}(\alpha,1)\overset{d}{=}\text{Gamma}(\lfloor\alpha\rfloor,1)\oplus\text{Gamma}(\alpha-\lfloor\alpha\rfloor,1)$ (the first a sum of exponentials), take $0<\alpha<1$. Then $x^{\alpha-1}e^{-x}\le x^{\alpha-1}$ on $[0,1]$ and $\le e^{-x}$ on $(1,\infty)$, giving $h(x)=\{x^{\alpha-1}\mathbf{1}_{[0,1]}+e^{-x}\mathbf{1}_{(1,\infty)}\}/\Gamma(\alpha)$, $c=(\frac{1}{\alpha}+\frac{1}{e})/\Gamma(\alpha)$. $g=h/c$ has cdf $G(x)=x^\alpha/(1+\alpha/e)$ on $[0,1]$ and $1-\alpha e^{-x}/(1+\alpha/e)$ for $x>1$, with closed-form inverse

$$G^{-1}(u)=\begin{cases}\{(1+\alpha/e)u\}^{1/\alpha}, & u\le 1/(1+\alpha/e),\\ -\log\{(1/\alpha+1/e)(1-u)\}, & \text{otherwise.}\end{cases}$$

Example 9.5 (Gamma with $\alpha>1$: a Cauchy envelope)

Let $\tilde f(x)=x^{\alpha-1}e^{-x}$ $(x>0)$, $\alpha>1$, $\mu=\alpha-1$, $\sigma^2=2\alpha-1$, and

$$g(x)=\frac{\mu^\mu e^{-\mu}}{1+(x-\mu)^2/\sigma^2},$$

a multiple of the Cauchy(μ,σ) density. *Domination*: $k(x)=x^{\alpha-1}e^{-x}\{\sigma^2+(x-\mu)^2\}$ has $k'(x)=-x^{\alpha-2}e^{-x}(x-\alpha)^2(x-\mu)$, ≥ 0 for $x<\mu$ and ≤ 0 for $x>\mu$, so $k\le k(\mu)=\sigma^2\mu^\mu e^{-\mu}$, i.e. $\tilde f\le g$ on $(0,\infty)$. *Algorithm*: $Y=\mu+\sigma\tan\{\pi(V-\frac{1}{2})\}$, $V\sim\text{Unif}(0,1)$; reject if $Y\le 0$; else accept if $U\le\tilde f(Y)/g(Y)$. *Efficiency*: $\int g=\pi\sigma\mu^\mu e^{-\mu}$ and $\int\tilde f=\Gamma(\alpha)$, so the acceptance probability is

$$\frac{\Gamma(\alpha)}{\pi\sqrt{2\alpha-1}\,(\alpha-1)^{\alpha-1}e^{-(\alpha-1)}}\approx 0.47,\ 0.50,\ 0.54,\ 0.56\quad(\alpha=1.5,2,5,100),$$

increasing to $1/\sqrt{\pi}\approx 0.564$ by Stirling, $\Gamma(\mu+1)\sim\sqrt{2\pi\mu}\,\mu^\mu e^{-\mu}$: about two proposals per draw for every α.

9.6 Multivariate distributions

Batch methods.

- Multinomial$(n;p_1,\dots,p_k)$: draw n categories (j when $U\in[\sum_{l<j}p_l,\sum_{l\le j}p_l)$) and count. (Independent Poissons λp_j conditioned on sum n also work, inefficiently.)
- $N(\mu,\Sigma)$: $\mu+LZ$, $Z\sim N(0,I_p)$, $LL^\top=\Sigma$: Cholesky (cheapest) or $Q\Lambda^{1/2}Q^\top$ (handles singular Σ). For large p the factorization is the bottleneck.
- Multivariate $t_\nu(\mu,\Sigma)$: $\mu+X/\sqrt{W/\nu}$ with $X\sim N(0,\Sigma)$ independent of $W\sim\chi^2(\nu)$.

Sequential (conditional) sampling. Draw components one at a time from their conditionals:

- Multinomial: $X_1\sim B(n,p_1)$, then $X_2\mid X_1=x_1\sim B(n-x_1,p_2/(1-p_1))$, and so on.

- Multivariate normal: $X_1 \sim N(\mu_1, \sigma_{11})$, then $(X_2, \ldots, X_p) \mid X_1 = x_1 \sim N(\mu_2 + \Sigma_{21}(x_1 - \mu_1)/\sigma_{11},\ \Sigma_{22} - \Sigma_{21}\Sigma_{12}/\sigma_{11})$, and recurse.

9.7 Selected exercises

Motivation. Weak low-order bits of power-of-two congruential generators, and an optimal normal sampler by rejection.

Exercise 9.1 (Low-order bits of a mixed congruential generator)

Let $x_{i+1} = (ax_i + c) \bmod 2^M$ and, for $1 \le k \le M$, $y_i = x_i \bmod 2^k$ (the k lowest bits of x_i).
(a) Show that $y_{i+1} = (ay_i + c) \bmod 2^k$, and conclude that (y_i) is eventually periodic with period at most 2^k.
(b) Show that if a and c are odd, the last bit of x_i alternates between 0 and 1.

Solution. [Supplement]

(a) $x_{i+1} = ax_i + c - q2^M$ and $2^k \mid 2^M$, so $x_{i+1} \equiv ax_i + c \equiv ay_i + c \pmod{2^k}$: $y_{i+1} = (ay_i + c) \bmod 2^k$, a fixed function of y_i with at most 2^k values. Two of $y_0, \ldots, y_{2^k}$ coincide, $y_s = y_{s+d}$ with $1 \le d \le 2^k$, and then $y_{i+d} = y_i$ for $i \ge s$.

(b) $k = 1$: a, c odd give $y_{i+1} = (y_i + 1) \bmod 2$.

So for $m = 2^{32}$, $a = 1103515245$, $c = 12345$ the last k bits repeat with period at most 2^k despite full period 2^{32}: use the high-order bits ($u_i = x_i/2^M$), never $x_i \bmod 2$ as a coin. ■

Exercise 9.2 (Normal variates from a Laplace envelope)

Let ϕ be the $N(0,1)$ density and $g_\lambda(x) = \frac{\lambda}{2} e^{-\lambda|x|}$ the Laplace density, $\lambda > 0$.
(a) Find the smallest $c(\lambda)$ with $\phi \le c(\lambda) g_\lambda$.
(b) Find the λ that maximizes the acceptance probability, and that probability.
(c) Show that for the optimal λ the sampler can be run with exponential variates only.

Solution. [Supplement]

(a) $\phi(x)/g_\lambda(x) = \frac{2}{\lambda\sqrt{2\pi}} \exp(-\frac{x^2}{2} + \lambda|x|)$, and $-\frac{t^2}{2} + \lambda t$ is maximized over $t \ge 0$ at $t = \lambda$, value $\lambda^2/2$:

$$c(\lambda) = \sup_x \frac{\phi(x)}{g_\lambda(x)} = \sqrt{\frac{2}{\pi}}\, \frac{e^{\lambda^2/2}}{\lambda}.$$

(b) Maximize $1/c(\lambda)$ (Corollary 9.4): $\log c(\lambda) = \text{const} + \frac{\lambda^2}{2} - \log\lambda$ has derivative $\lambda - 1/\lambda$ and second derivative $1 + \lambda^{-2} > 0$, so $\lambda = 1$, $c(1) = \sqrt{2e/\pi} \approx 1.3155$, acceptance ≈ 0.760.

(c) $\phi(x)/\{c(1)g_1(x)\} = \exp\{-\frac{1}{2}(|x| - 1)^2\}$. With $X = SE_1$ ($E_1 \sim \text{Exp}(1)$, independent random sign S), $U \le e^{-(E_1-1)^2/2}$ iff $E_2 \ge (E_1 - 1)^2/2$ for $E_2 = -\log U \sim \text{Exp}(1)$. So draw E_1, E_2 until $E_2 \ge (E_1 - 1)^2/2$ and return $\pm E_1$; on average 1.3155 pairs per variate. ■

Lecture 10. Monte Carlo Integration

10.1 Naive Monte Carlo

To compute $\mu = \mathbb{E}f(X) = \int f(x)p(x)\,dx$, draw $X_1, \dots, X_n$ i.i.d. from p and use

$$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} f(X_i).$$

It is unbiased, $\hat{\mu}_n \to \mu$ a.s. when $\mathbb{E}|f(X)| < \infty$, and $\sqrt{n}(\hat{\mu}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$ when $\sigma^2 = \operatorname{Var} f(X) < \infty$; with the sample variance $\hat{\sigma}^2$, $\hat{\mu}_n \pm 1.96\,\hat{\sigma}/\sqrt{n}$ is an approximate 95% interval.

The error is $O_p(n^{-1/2})$ *in every dimension*, while grid quadrature suffers the curse of dimensionality: quadrature (Lecture 8) wins in one or two dimensions, Monte Carlo in high dimensions. The rate cannot be improved much, but σ^2 can: **variance reduction**.

Example 10.1

- $\int_0^\infty \sqrt{x}e^{-x}dx = \Gamma(3/2) = \sqrt{\pi}/2 = \mathbb{E}\sqrt{X}$, $X \sim \text{Exp}(1)$, with finite variance.
- $\int_0^1 x^{-1/2}dx = 2 = \mathbb{E}U^{-1/2}$, but $\mathbb{E}U^{-1} = \infty$: consistent with infinite variance, so the CLT interval is meaningless and running averages jump. Substituting $x = y^2$ gives $\int_0^1 2\,dy$, estimated exactly: flattening the integrand is the cheapest variance reduction.
- Buffon's needle: a needle of length L on lines spaced $D \geq L$ crosses one with probability $2L/(\pi D)$. With centre distance $Y \sim \text{Unif}(0, D/2)$ and angle $\Theta \sim \text{Unif}(0, \pi/2)$ independent, it crosses iff $Y \leq \frac{L}{2}\sin\Theta$, with probability $\frac{2}{D} \cdot \frac{2}{\pi}\int_0^{\pi/2} \frac{L}{2}\sin\theta\,d\theta = \frac{2L}{\pi D}$. With k crossings in n throws, $\hat{p} = k/n$ is unbiased for p, and $\hat{\pi} = 2Ln/(Dk)$; it is biased ($E[1/\hat{p}] > 1/p$ by Jensen), indeed $E\hat{\pi} = \infty$ as $P(k = 0) > 0$. The delta method with $g(p) = 2L/(Dp)$, $g'(p)^2 = \pi^2/p^2$, gives

  $$\operatorname{Var}\hat{\pi} \approx g'(p)^2 \frac{p(1-p)}{n} = \frac{\pi^2(1-p)}{np},$$

  decreasing in p; $L = D$ gives $\operatorname{Var}\hat{\pi} \approx 5.63/n$, so three digits need tens of millions of throws.

10.2 Importance sampling

For a density $h > 0$ wherever $fp \neq 0$,

$$\mu = \int f(x)\frac{p(x)}{h(x)}h(x)\,dx = \mathbb{E}_h\Big[f(Y)\frac{p(Y)}{h(Y)}\Big], \qquad \hat{\mu}_h = \frac{1}{n}\sum_{i=1}^{n} f(Y_i)w(Y_i), \quad Y_i \overset{\text{iid}}{\sim} h,$$

with **importance weights** $w = p/h$: unbiased, variance $\frac{1}{n}\{\int f^2p^2/h - \mu^2\}$. Useful when p is hard to sample or rarely visits where f is large.

Proposition 10.1

The variance of $\hat{\mu}_h$ is minimized by $h^\star \propto |f|\,p$. If $f \geq 0$, then $h^\star = fp/\mu$ and the variance is zero.

Proof. By Cauchy–Schwarz, for admissible h,

$$\Big(\int |f|p\Big)^2 = \Big(\int \frac{|f|p}{\sqrt{h}}\sqrt{h}\Big)^2 \le \int \frac{f^2p^2}{h}\int h = \int \frac{f^2p^2}{h},$$

with equality iff $h \propto |f|p$. If $f \ge 0$, $fw \equiv \mu$ under $h^\star$. ■

$h^\star$ needs μ, but guides the choice: h roughly proportional to $|f|p$, with tails at least as heavy as fp, or the weights explode and the variance may be infinite. Diagnose with the spread of the weights.

Example 10.2 (Normal tail probability)

Estimate $\pi(c) = P(Z > c) = \Phi(-c)$ for large c. The naive estimator has relative error $\sqrt{(1-\pi(c))/(n\pi(c))}$, exploding in c. Sample $Y_i \sim N(c,1)$ instead; the weight is $\phi(y)/\phi(y-c) = e^{-cy+c^2/2}$, so

$$\hat\pi_{\text{imp}}(c) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{Y_i > c\} e^{-cY_i + c^2/2}.$$

Its second moment per draw is

$$e^{c^2}\int_c^\infty e^{-2cy}\phi(y-c)\,dy = e^{c^2}\int_c^\infty \phi(y+c)\,dy = e^{c^2}\,\Phi(-2c),$$

as $-2cy - \frac{1}{2}(y-c)^2 = -\frac{1}{2}(y+c)^2$; so $\operatorname{Var}\hat\pi_{\text{imp}}(c) = \frac{1}{n}\{e^{c^2}\Phi(-2c) - \pi(c)^2\}$. By Mills' ratio $\Phi(-t) \approx \phi(t)/t$, the squared relative error is about $c\sqrt{\pi/2}/n$ against $\sqrt{2\pi}\,ce^{c^2/2}/n$ naively: the relative error grows like $c^{1/2}$ instead of $c^{1/2}e^{c^2/4}$.

10.3 Antithetic variates

If $X \overset{d}{=} X'$ are negatively correlated,

$$\operatorname{Var}\Big[\frac{f(X)+f(X')}{2}\Big] = \frac{\operatorname{Var} f(X)}{2}(1+\rho), \qquad \rho = \operatorname{Corr}(f(X), f(X')),$$

beating two independent draws when $\rho < 0$. Natural pairs: $(U, 1-U)$ for uniforms, $(X, 2m-X)$ for distributions symmetric about m.

Proposition 10.2

If f and g are both nondecreasing (or both nonincreasing), then $\operatorname{Cov}(f(X), g(X)) \ge 0$ whenever it exists. Hence $\operatorname{Cov}(h(U), h(1-U)) \le 0$ for monotone h.

Proof. With Y an independent copy of X, $\{f(X)-f(Y)\}\{g(X)-g(Y)\} \ge 0$, with expectation

$$2\mathbb{E}f(X)g(X) - 2\mathbb{E}f(X)\mathbb{E}g(X) = 2\operatorname{Cov}(f(X), g(X)) \ge 0.$$

For the consequence take $f = h$, $g(u) = -h(1-u)$, monotone in the same direction. ■

Example 10.3

For $\mu = \mathbb{E}e^U = e - 1$: $\operatorname{Var} e^U = \frac{e^2-1}{2} - (e-1)^2 \approx 0.2420$ and $\operatorname{Cov}(e^U, e^{1-U}) = e - (e-1)^2 \approx -0.2342$, so $\frac{1}{2}(e^U + e^{1-U})$ has variance ≈ 0.0039 against 0.1210 for two independent draws: about 97% less.

10.4 Control variates

If $Y = g(X)$ has known mean μ_Y, $f(X) - c(Y - \mu_Y)$ is unbiased for every c, with variance

$$\operatorname{Var} f(X) + c^2 \operatorname{Var} Y - 2c \operatorname{Cov}(f(X), Y),$$

minimized at $c^\star = \operatorname{Cov}(f(X), Y)/\operatorname{Var} Y$, giving $\operatorname{Var} f(X)\,(1 - \rho^2)$, $\rho = \operatorname{Corr}(f(X), Y)$. In practice $c^\star$ is the least-squares slope of $f(X_i)$ on Y_i (negligible bias for large n).

Example 10.4

For $\mathbb{E}e^U$ with $Y = U$ ($\mu_Y = \frac{1}{2}$, $\operatorname{Var} U = \frac{1}{12}$): $\operatorname{Cov}(e^U, U) = 1 - \frac{e-1}{2} \approx 0.1409$, $c^\star \approx 1.6903$, and the variance per draw drops from 0.2420 to $0.2420 - 12(0.1409)^2 \approx 0.0039$.

10.5 A comparison: the size of a test

Example 10.5

For $X_1, \ldots, X_{25}$ i.i.d. Poisson(λ), the CLT test of $H_0 : \lambda = 2$ vs $\lambda > 2$ rejects when $(\bar{X} - 2)/\sqrt{2/25} > 1.645$, i.e. $S = \sum X_i > c = \lfloor 25(2 + 1.645\sqrt{2/25}) \rfloor = 61$. As $S \sim \text{Poi}(25\lambda)$, the size is $p = 1 - F_{50}(61) = 0.05568$. Estimate p by N simulated S:

- *Standard*: $\hat{p} = \frac{1}{N} \sum \mathbf{1}\{S_j > c\}$ has variance $p(1-p)/N = 0.0526/N$.
- *Antithetic*: with $S_j = F_{50}^{-1}(U_j)$, $S_j' = F_{50}^{-1}(1 - U_j)$, both indicators equal 1 only if $F_{50}(c) < U_j < 1 - F_{50}(c)$, impossible as $F_{50}(c) > \frac{1}{2}$; so the covariance is $-p^2$ and N pairs have variance $p(1-2p)/(2N) = 0.0247/N$ against $0.0263/N$ for $2N$ independent draws. Only 6%: a rare-event indicator is monotone but nearly constant.
- *Importance sampling*: $S_j \sim h = \text{Poi}(25\lambda')$ with $\lambda' = 2.4653$ (the threshold, so about half the draws reject), weight $w(s) = e^{-25(\lambda - \lambda')}(\lambda/\lambda')^s$; the variance per draw is $\sum_{s>c} f_{50}(s)^2/h(s) - p^2 = 0.0063$ (f_{50} the Poi(50) pmf): 8.3 times smaller.

Each gives a CLT interval $\hat{p} \pm z_{1-\alpha/2}\sqrt{\widehat{\text{Var}}}$. For the power at $\lambda \in [2.2, 4]$, replace F_{50} by $F_{25\lambda}$. Once rejection is not rare ($\lambda \geq 2.5$, power ≥ 0.5) the fixed envelope $\lambda' = 2.4653$ loses its advantage, and for large λ the weights $(\lambda/\lambda')^s$ make importance sampling *worse* than standard Monte Carlo: per draw 0.299 vs 0.248 at $\lambda = 2.5$, and 17.3 vs 0.053 at $\lambda = 3$. An envelope tuned to one λ does not suit all.

10.6 Selected exercises

Motivation. How too-light tails of the sampling density ruin importance sampling, exactly, in a Gaussian case.

Exercise 10.1 (Importance densities with light tails)

Let $p = N(0, 1)$ and $h = N(0, s)$ with variance $s > 0$, and $w = p/h$.
(a) Show that $\mathbb{E}_h[w(Y)^2] < \infty$ if and only if $s > \frac{1}{2}$, and compute it.
(b) Using $f \equiv 1$ (so that the estimator is the average weight, estimating 1), find the variance per draw and the best s.

Solution. [Supplement]

(a) $\mathbb{E}_h[w^2] = \int p^2/h$, and

$$\frac{p(x)^2}{h(x)} = \frac{e^{-x^2}/(2\pi)}{e^{-x^2/(2s)}/\sqrt{2\pi s}} = \sqrt{\frac{s}{2\pi}}\exp\Big\{-x^2\Big(1-\frac{1}{2s}\Big)\Big\}.$$

For $s \le \frac{1}{2}$ the exponent is ≥ 0 and the integral diverges. For $s > \frac{1}{2}$, with $\int e^{-cx^2}dx = \sqrt{\pi/c}$, $c = 1 - \frac{1}{2s} > 0$,

$$\mathbb{E}_h[w^2] = \sqrt{\frac{s}{2\pi}}\sqrt{\frac{\pi}{1-1/(2s)}} = \frac{s}{\sqrt{2s-1}}.$$

(b) The variance per draw is $s/\sqrt{2s-1} - 1$, with derivative $(s-1)/(2s-1)^{3/2}$: decreasing on $(\frac{1}{2}, 1)$, increasing on $(1, \infty)$, zero at $s = 1$ ($h = p$). It blows up as $s \downarrow \frac{1}{2}$ and is infinite for $s \le \frac{1}{2}$, though the estimator stays unbiased and consistent: lighter tails than $|f|p$ give rare, enormous weights. Heavier tails cost moderately, e.g. $\frac{2}{\sqrt{3}} - 1 \approx 0.155$ at $s = 2$. ■

Lecture 11. Markov Chain Monte Carlo

Markov chain Monte Carlo (MCMC) simulates π by running a Markov chain with limiting distribution π and averaging along the path; an ergodic theorem replaces the law of large numbers.

11.1 Markov chains

A discrete-time Markov chain on a countable state space S is a sequence $\{X_n\}$ with

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \ldots, X_0) = P(X_{n+1} = j \mid X_n = i) = P_{ij}.$$

We assume time homogeneity. The rows of the **transition matrix** $P = (P_{ij})$ sum to one. Examples: simple random walk on $\mathbb{Z}$ ($P_{i,i+1} = p$, $P_{i,i-1} = 1 - p$); gambler's ruin on $\{0, \ldots, N\}$, absorbing at 0 and N.

Chapman–Kolmogorov. $P_{ij}^{(n)} = P(X_{m+n} = j \mid X_m = i)$ satisfies $P^{(m+n)} = P^{(m)}P^{(n)}$ (condition on X_m), so $P^{(n)} = P^n$, and $X_0 \sim \pi^{(0)}$ (row vector) gives $X_n \sim \pi^{(0)}P^n$.

Classification.

- $i \to j$ (j *reachable*) if $P_{ij}^{(n)} > 0$ for some $n \geq 0$; i, j *communicate* if $i \to j$ and $j \to i$. The chain is **irreducible** if there is one communication class. The random walk is irreducible; gambler's ruin has classes $\{0\}$, $\{1, \ldots, N-1\}$, $\{N\}$.
- The **period** of i is $d(i) = \gcd\{n \geq 1 : P_{ii}^{(n)} > 0\}$; the chain is **aperiodic** if $d(i) = 1$ for all i.
- With $\tau_{ii} = \min\{n \geq 1 : X_n = i\}$ given $X_0 = i$, i is *recurrent* if $P(\tau_{ii} < \infty) = 1$, else *transient*; recurrent i is **positive recurrent** if $\mathbb{E}\tau_{ii} < \infty$, else null recurrent. These are class properties.
- **Ergodic** means aperiodic and positive recurrent. On a finite space an irreducible chain is positive recurrent, so irreducible and aperiodic means ergodic.

Stationary distributions. π is **stationary** if $\pi P = \pi$, $\pi \geq 0$, $\pi\mathbf{1} = 1$. If $P_{ij}^{(n)} \to \pi_j$ for all i, π is stationary: $\pi_k = \lim_n (P^{n+1})_{ik} = \sum_j \lim_n (P^n)_{ij} P_{jk} = (\pi P)_k$ (finite S; in general by dominated convergence).

Theorem 11.1 (Limit theorems; stated without proof)

(a) An irreducible chain has a unique stationary distribution if and only if it is positive recurrent.

(b) If the chain is irreducible and ergodic, then $\lim_n P_{ij}^{(n)} = \pi_j$ exists independently of i, π is the unique stationary distribution, and $\pi_j = 1/\mathbb{E}\tau_{jj}$ is also the long-run fraction of time spent at j.

(c) If the chain is irreducible and aperiodic and a stationary distribution π exists, then $P_{ij}^{(n)} \to \pi_j$, and the chain is positive recurrent, hence ergodic.

Theorem 11.2 (Ergodic theorem)

If $\{X_n\}$ is irreducible and ergodic with stationary distribution π, and $\mathbb{E}_\pi|g(X)| < \infty$, then

$$\frac{1}{n}\sum_{k=1}^{n} g(X_k) \to \mathbb{E}_\pi g(X) \quad \text{almost surely, from any starting state.}$$

This extends the strong law to Markov chains and underlies MCMC: build an irreducible, aperiodic, positive recurrent chain with stationary distribution f, run it, and average.

Detailed balance. Stationarity is usually guaranteed by **detailed balance**: $\pi_i P_{ij} = \pi_j P_{ji}$ for all i, j.

Proposition 11.3

If π satisfies detailed balance, it is stationary. Moreover the stationary chain is *reversible*: $(X_0, \dots, X_n)$ and $(X_n, \dots, X_0)$ have the same distribution.

Proof. $(\pi P)_j = \sum_i \pi_i P_{ij} = \pi_j \sum_i P_{ji} = \pi_j$. Reversibility: apply detailed balance repeatedly,

$$\pi_{i_0} P_{i_0 i_1} P_{i_1 i_2} \cdots P_{i_{n-1} i_n} = P_{i_1 i_0} \pi_{i_1} P_{i_1 i_2} \cdots = \cdots = \pi_{i_n} P_{i_n i_{n-1}} \cdots P_{i_1 i_0}.$$

■

General state spaces. P_{ij} becomes a **transition kernel** $K(x, y)$, the conditional density of X_{n+1} at y given $X_n = x$, $\int K(x, y)\, dy = 1$ (point masses allowed). Stationarity reads $\pi(y) = \int \pi(x) K(x, y)\, dx$, implied by $\pi(x)K(x, y) = \pi(y)K(y, x)$; the limit theorems carry over.

11.2 The Metropolis–Hastings algorithm

Motivation. Metropolis–Hastings builds a chain with the target as stationary distribution from density ratios alone, so normalizing constants are never needed.

Let f be the target (possibly unnormalized) and $g(\cdot \mid x)$ an easy **proposal**. Start at $X^{(0)}$ with $f(X^{(0)}) > 0$; for $t = 0, 1, \dots$:

(1) draw a candidate $\tilde{X} \sim g(\cdot \mid X^{(t)})$;

(2) compute the **Metropolis–Hastings ratio**

$$R(X^{(t)}, \tilde{X}) = \frac{f(\tilde{X})\, g(X^{(t)} \mid \tilde{X})}{f(X^{(t)})\, g(\tilde{X} \mid X^{(t)})};$$

(3) set $X^{(t+1)} = \tilde{X}$ with probability $\alpha(X^{(t)}, \tilde{X}) = \min\{R, 1\}$, and $X^{(t+1)} = X^{(t)}$ otherwise.

Metropolis et al. (1953) used a symmetric proposal $g(y \mid x) = g(x \mid y)$ (e.g. random walk $y = x + \epsilon$), giving $R = f(\tilde{X})/f(X^{(t)})$: uphill moves always accepted, downhill sometimes. Hastings (1970) allowed asymmetric g.

Theorem 11.4

The Metropolis–Hastings chain satisfies detailed balance with respect to f; hence f is a stationary distribution.

Proof. The transition kernel is

$$K(x,y) = \alpha(x,y)g(y \mid x) + \{1 - r(x)\}\delta_x(y), \qquad r(x) = \int \alpha(x,y)g(y \mid x)\,dy,$$

with $r(x)$ the acceptance probability from x. For the continuous part,

$$f(x)\alpha(x,y)g(y \mid x) = \min\{f(x)g(y \mid x),\, f(y)g(x \mid y)\} = f(y)\alpha(y,x)g(x \mid y),$$

the middle being symmetric in (x,y). The point mass $f(x)\{1-r(x)\}\delta_x(y)$ lives at $y = x$, where it equals $f(y)\{1-r(y)\}\delta_y(x)$. So $f(x)K(x,y) = f(y)K(y,x)$; apply Proposition 11.3. ■

The other conditions of Theorem 11.2 depend on g:

- *Irreducibility*: if $f > 0$ on S and the chain with kernel g is irreducible.
- *Aperiodicity* usually comes from rejections: e.g. in a discrete chain with $g(i \mid i) > 0$ and a proposable j with $f(j) < f(i)$, $\alpha(i,j) < 1$, so $r(i) < 1$, $K(i,i) > 0$ and $d(i) = 1$.

Example 11.1 (Posterior sampling)

For a posterior $p(\theta \mid y) = c\,p(y \mid \theta)p(\theta)$ with unknown c,

$$R(\theta, \tilde{\theta}) = \frac{p(y \mid \tilde{\theta})p(\tilde{\theta})g(\theta \mid \tilde{\theta})}{p(y \mid \theta)p(\theta)g(\tilde{\theta} \mid \theta)},$$

free of c. The **independence chain** $g(\cdot \mid \theta) = p(\cdot)$ (prior proposals) gives the likelihood ratio $R = p(y \mid \tilde{\theta})/p(y \mid \theta)$.
E.g. for $X \sim \text{Bin}(n,p)$, $p = e^\theta/(1+e^\theta)$, $\theta \sim N(0,\sigma^2)$ (Example 8.2), the posterior $\propto e^{\theta x}(1+e^\theta)^{-n}e^{-\theta^2/(2\sigma^2)}$ has no closed-form normalizer; propose $\tilde{\theta} \sim N(0,\sigma^2)$ and accept with probability $\min\{1, e^{(\tilde{\theta}-\theta)x}\{(1+e^\theta)/(1+e^{\tilde{\theta}})\}^n\}$.

11.3 The Gibbs sampler

For $X = (X_1, \ldots, X_p)$, with X_{-i} all coordinates but i, the Gibbs sampler updates coordinates in turn from their **full conditionals** (often standard):

$$X_i^{(t+1)} \sim f_{X_i \mid X_{-i}}(\cdot \mid X_1^{(t+1)}, \ldots, X_{i-1}^{(t+1)}, X_{i+1}^{(t)}, \ldots, X_p^{(t)}), \qquad i = 1, \ldots, p,$$

always using the latest values (as in Gauss–Seidel, Lecture 5).

Example 11.2 (Bivariate normal)

For $(X_1, X_2) \sim N(0, \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right))$, $X_1 \mid X_2 = x_2 \sim N(\rho x_2, 1-\rho^2)$ and symmetrically: $X_1^{(t+1)} \sim N(\rho X_2^{(t)}, 1-\rho^2)$, then $X_2^{(t+1)} \sim N(\rho X_1^{(t+1)}, 1-\rho^2)$. $X_2^{(t)}$ is autoregressive with coefficient ρ^2, so mixing is slow for $|\rho| \approx 1$.

Proposition 11.5

Each coordinate update of the Gibbs sampler is a Metropolis–Hastings step whose proposal is always accepted. Hence f is stationary for the Gibbs sampler.

Proof. Fix i and x. The proposal is $g_i(\tilde{x} \mid x) = f_{X_i|X_{-i}}(\tilde{x}_i \mid x_{-i})\,\delta(\tilde{x}_{-i} - x_{-i})$. On $\{\tilde{x}_{-i} = x_{-i}\}$ (probability one), factor $f(x) = f_{X_{-i}}(x_{-i}) f_{X_i|X_{-i}}(x_i \mid x_{-i})$:

$$R(x, \tilde{x}) = \frac{f_{X_{-i}}(x_{-i}) f_{X_i|X_{-i}}(\tilde{x}_i \mid x_{-i}) \cdot f_{X_i|X_{-i}}(x_i \mid x_{-i})}{f_{X_{-i}}(x_{-i}) f_{X_i|X_{-i}}(x_i \mid x_{-i}) \cdot f_{X_i|X_{-i}}(\tilde{x}_i \mid x_{-i})} = 1.$$

By Theorem 11.4 each update leaves f invariant, hence so does their composition. ■

So Gibbs is Metropolis–Hastings with a coordinate-dependent proposal (the composed chain need not be reversible, but f is stationary).

Example 11.3 (Ising model)

Place p spins $x_i \in \{-1, +1\}$ around a circle ($x_{p+1} = x_1$) with

$$p(x) = \frac{1}{Z} \exp\Big(\beta \sum_{j=1}^{p} x_j x_{j+1}\Big), \qquad \beta > 0,$$

favouring aligned neighbours. Z has 2^p terms, but the full conditional of x_i involves only its neighbours and Z cancels. With $s_i = x_{i-1} + x_{i+1}$,

$$P(x_i = +1 \mid x_{-i}) = \frac{e^{\beta s_i}}{e^{\beta s_i} + e^{-\beta s_i}} = \frac{1}{1 + e^{-2\beta s_i}},$$

logistic in the neighbours; equivalently the spin flips with probability $1/\{1 + \exp(2\beta x_i s_i)\}$.

11.4 Practical issues

- *Burn-in.* Discard the initial, non-stationary part of the run, judged by trace plots and convergence diagnostics.
- *Mixing and stopping.* Strong autocorrelation (Example 11.2 with $|\rho| \approx 1$, or a badly scaled random walk) makes the effective sample size much smaller than the number of iterations; Monte Carlo standard errors must account for it (e.g. batch means).
- *Dependence.* For nearly independent draws, run N parallel chains from dispersed starts and keep the final values, or *thin* one chain to $Y_m = X_{mk}$. Thinning saves storage, not error: for $\mathbb{E}_\pi g$ the full chain is never worse.

11.5 Selected exercises

Motivation. The two-state chain in closed form, slow mixing of the bivariate normal Gibbs sampler, and why thinning never helps.

Exercise 11.1 (The two-state chain)

Let $P = \left(\begin{smallmatrix} 1-a & a \\ b & 1-b \end{smallmatrix}\right)$ with $a, b \in (0, 1]$, and put $\lambda = 1 - a - b$.

(a) Show that $\pi = \frac{1}{a+b}(b, a)$ satisfies detailed balance, hence is stationary.
(b) Show that $P^n = \Pi + \lambda^n E$ for $n \geq 1$, where $\Pi = \mathbf{1}\pi$ (both rows equal to π) and $E = \frac{1}{a+b}\begin{pmatrix} a & -a \\ -b & b \end{pmatrix}$.
(c) Deduce that $P_{ij}^{(n)} \to \pi_j$ geometrically fast unless $a = b = 1$, and explain the exception.

Solution. [Supplement]

(a) $\pi_1 P_{12} = \frac{ba}{a+b} = \pi_2 P_{21}$; apply Proposition 11.3.

(b)

Step 1. ($n = 1$.) $\Pi + \lambda E$ has $(1,1)$ entry $\frac{b+(1-a-b)a}{a+b} = 1 - a$, $(1,2)$ entry $\frac{a-(1-a-b)a}{a+b} = a$, and likewise b, $1 - b$ in row 2: $P = \Pi + \lambda E$.

Step 2. (Products.) $\Pi^2 = \mathbf{1}(\pi\mathbf{1})\pi = \Pi$; $E\mathbf{1} = 0$ gives $E\Pi = 0$; $\pi E = \frac{1}{(a+b)^2}(ba - ab, -ba + ab) = 0$ gives $\Pi E = 0$; and $E^2 = \frac{1}{(a+b)^2}\begin{pmatrix} a^2+ab & -a^2-ab \\ -ab-b^2 & ab+b^2 \end{pmatrix} = E$.

Step 3. (Induction.) If $P^n = \Pi + \lambda^n E$, then $P^{n+1} = (\Pi + \lambda^n E)(\Pi + \lambda E) = \Pi + \lambda^{n+1} E$ by Step 2.

(c) By (b), $|P_{ij}^{(n)} - \pi_j| \leq |\lambda|^n \max\{a, b\}/(a + b)$, with $\lambda \in [-1, 1)$ and $\lambda = -1$ only for $a = b = 1$; otherwise convergence is geometric at rate $|\lambda|$. For $a = b = 1$ the chain alternates deterministically (period 2), P^n alternates between I and P, and there is no limit, though $\pi = (\frac{1}{2}, \frac{1}{2})$ is stationary. ■

Exercise 11.2 (Mixing of the bivariate normal Gibbs sampler)

In Example 11.2, with $|\rho| < 1$:
(a) Show that $X_2^{(t+1)} = \rho^2 X_2^{(t)} + \varepsilon_t$ with $\varepsilon_t \sim N(0, 1 - \rho^4)$ independent of $X_2^{(0)}, \ldots, X_2^{(t)}$.
(b) Show that $N(0, 1)$ is stationary for $(X_2^{(t)})$, and that started from it, $\operatorname{Corr}(X_2^{(t)}, X_2^{(t+k)}) = \rho^{2k}$.
(c) Show that for the stationary chain, $n \operatorname{Var}(\frac{1}{n}\sum_{t=1}^n X_2^{(t)}) \to \frac{1+\rho^2}{1-\rho^2}$, and interpret.

Solution. [Supplement]

(a) $X_1^{(t+1)} = \rho X_2^{(t)} + \sqrt{1-\rho^2} Z_1$, $X_2^{(t+1)} = \rho X_1^{(t+1)} + \sqrt{1-\rho^2} Z_2$, with $Z_1, Z_2 \sim N(0, 1)$ independent of each other and the past, so

$$X_2^{(t+1)} = \rho^2 X_2^{(t)} + \varepsilon_t, \qquad \varepsilon_t = \rho\sqrt{1-\rho^2} Z_1 + \sqrt{1-\rho^2} Z_2,$$

and $\operatorname{Var} \varepsilon_t = (1 - \rho^2)(\rho^2 + 1) = 1 - \rho^4$.

(b) $X_2^{(t)} \sim N(0, 1)$ gives $X_2^{(t+1)} \sim N(0, \rho^4 + 1 - \rho^4)$. Iterating (a), $X_2^{(t+k)} = \rho^{2k} X_2^{(t)}$ plus a combination of $\varepsilon_t, \ldots, \varepsilon_{t+k-1}$, independent of $X_2^{(t)}$, so the correlation is ρ^{2k}.

(c) With $r = \rho^2 \in [0, 1)$ and unit variances,

$$n \operatorname{Var}\Big(\frac{1}{n}\sum_{t=1}^n X_2^{(t)}\Big) = \frac{1}{n}\sum_{s,t=1}^n r^{|s-t|} = 1 + 2\sum_{k=1}^{n-1}\Big(1 - \frac{k}{n}\Big) r^k.$$

By monotone convergence this tends to $1 + 2r/(1 - r) = (1 + r)/(1 - r)$: n Gibbs draws are worth about $n(1 - \rho^2)/(1 + \rho^2)$ independent ones (effective sample size), about $n/100$ for $\rho = 0.99$. ■

Exercise 11.3 (Thinning never helps)

Let (X_t) be a stationary sequence (the joint law of $(X_{t+s})_{t\ge 1}$ does not depend on s) and $\mathbb{E}g(X_1)^2 < \infty$. For integers $n, k \ge 1$ compare the full average and the thinned average

$$\bar{g}_{\text{full}} = \frac{1}{nk}\sum_{t=1}^{nk} g(X_t), \qquad \bar{g}_{\text{thin}} = \frac{1}{n}\sum_{m=1}^{n} g(X_{mk}).$$

Show that $\operatorname{Var}\bar{g}_{\text{full}} \le \operatorname{Var}\bar{g}_{\text{thin}}$.

Solution. [Supplement] For $j = 0, \dots, k-1$ let $A_j = \frac{1}{n}\sum_{m=1}^{n} g(X_{mk-j})$. The indices $mk - j$ cover $1, \dots, nk$ once, so $\bar{g}_{\text{full}} = \frac{1}{k}\sum_j A_j$. A_j is the same function of (X_{t-j}) as $A_0 = \bar{g}_{\text{thin}}$ of (X_t), so $\operatorname{Var} A_j = v := \operatorname{Var}\bar{g}_{\text{thin}}$ by stationarity, and $\operatorname{Cov}(A_i, A_j) \le v$ by Cauchy–Schwarz:

$$\operatorname{Var}\bar{g}_{\text{full}} = \frac{1}{k^2}\sum_{i,j=0}^{k-1} \operatorname{Cov}(A_i, A_j) \le \frac{1}{k^2}\cdot k^2 v = v.$$

Thinning is justified only by storage or the cost of evaluating g. ■